Where should the input enter a learned KKL observer, and when does learning beat a model-based filter?

A controlled study on the HyperKKL testbed

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Abstract

Kazantzis–Kravaris/Luenberger (KKL) observers estimate the state of a nonlinear plant through a linear, stable latent filter of the output and a learned inverse map. Under an exogenous input the inverse map must depend on the input history, but the theory does not say where that dependence should be placed in a learned observer, nor when such an observer is worth using instead of a model-based filter. We answer both questions with a pre-registered, controlled study on twelve simulated plants (six for design, six held out and run once with the frozen recipe). First, at matched parameter count and matched input history, HyperKKL, a hypernetwork that writes rank-r updates of the decoder's *weights* from a fading-memory input context, beats injecting the input into the latent dynamics and beats an additive output residual on 6/6 dev plants for unseen waveforms and 5/6 for unseen input sequences, replicated on 5/6 and 5/6 held-out plants; the stricter pre-registered rule asking for a single best location *fails*, because HyperKKL and a dense-subspace ablation of it are statistically tied. Second, against eight model-based filters (EKF, UKF, particle filter, model-based KKL, and robust and oracle-tuned variants) the learned observers lose in every one of fifteen exact-model stress conditions (sensor faults, input faults, poor initialisation, plant drift), where a tuned EKF is 5.9× more accurate; they win once the filter's model is wrong, at every tested parameter mismatch from 10% to 50%, under a missing physical term and under a realistic combination of the three (6/6 plants against the best filter that still beats a constant predictor). On real battery drive cycles with a least-squares-identified circuit model, a pre-registered test confirms both results: the weight-space observer beats the tuned EKF on 3 of 3 held-out cycles and latent injection on 3, by a small absolute margin; the advantage is mostly faster convergence on mild cycles and does not transfer to an untrained temperature. The learned observers need full-state labels from the real plant and fail under gross outliers and post-training drift; which learner wins once the model is wrong is system-dependent: on the partial-model node a grey-box EKF with a learned residual or a Koopman Kalman filter trained on the same labels beats the KKL observers on three of five plants. A fading-memory account of how much input history is needed is reported as a supporting analysis whose pre-registered tests failed. The testbed, the deployment scorecard and the mechanically applied decision rules are released with the paper.

1 Introduction

A state observer reconstructs the internal state of a dynamical system from its measured outputs. Kazantzis–Kravaris/Luenberger (KKL) observers (Kazantzis & Kravaris, 1998; Andrieu & Praly, 2006) do this with a fixed linear filter, $\dot{z} = Az + By$ with A Hurwitz, followed by the inverse of an injective map T that sends the plant state into the filter's coordinates; existence of T needs only mild observability conditions. Because T is hard to compute in closed form, a line of work learns it and its inverse from simulated data (Ramos et al., 2020; Janny et al., 2021; Peralez & Nadri, 2021; Niazi et al., 2023; Buisson-Fenet et al., 2023; Niazi et al., 2025). For plants driven by an exogenous input u, the theory says that the map must depend on the input history (Bernard & Andrieu, 2019). It does not say where that dependence should be placed in a

learned observer: in the latent dynamics, at the decoder's input, in its features, or in its weights. Nor does it say when such an observer is worth using, given that a practitioner with a model of the plant can run an extended Kalman filter (EKF) instead.

Why KKL observers. Three properties set a KKL observer apart from a Kalman filter. Its latent filter does not depend on the estimate (for the decoder-side variants of Section 2), so convergence is global and a poor initial guess cannot make it diverge, whereas the EKF's stability is local (Reif et al., 1999); at deployment it needs no model and no Jacobians, only a linear filter and a feedforward decoder, which in our evaluation harness costs less per step than the EKF (Table 12); and the map T exists under mild observability, backward distinguishability (Andrieu & Praly, 2006; Bernard & Andrieu, 2019), including strongly nonlinear and chaotic plants where EKF linearisation is poor. Whether these advantages translate into accuracy is exactly what C3 tests.

This paper answers both questions empirically, with a pre-registered protocol, and states the limits of the answers as plainly as the answers. HyperKKL, our open testbed, is the common framework: one latent filter, one pair of base maps trained once per plant, one fading-memory encoder of the input history, and five places where the resulting context can act. Our thesis is that KKL theory requires an input-dependent observer map but leaves the *location* of that dependence and the *value* of learning it open, and that both can be settled by a controlled experiment.

Contributions and verdicts. Every claim below is tied to a decision rule written before the runs (Table 13 in Appendix F) and applied mechanically; failed rules are reported as such.

1. **Where the input should enter (C1).** With identical frozen base maps, identical context encoder and a matched conditioner budget, HyperKKL, a hypernetwork that writes rank-r updates of the decoder's weights from the input-history context, beats latent injection and an additive output residual on 6/6 dev plants for unseen waveform shapes and 5/6 for unseen input sequences, and on 5/6 and 5/6 held-out plants run once with the frozen recipe. The pre-registered rule asking for one location to beat *every* other **FAIL**s: HyperKKL and a dense-subspace ablation of it are tied, and the rule reported as passed is the weaker one pre-registered for the held-out plants.

2. **When learning beats a model-based filter (C3).** With an exact model the filters win in every one of fifteen stress conditions (sensor faults, input faults, poor initialisation, plant drift; learned-wins regime: none of 15 conditions), and a tuned EKF is 5.9× more accurate (median over dev plants; range 2.7–14.1×). Once the filter's model is wrong, the learned observers win: at every mismatch level tested, 10% (4/6 plants, sequence OOD), 20% (5/6), 30% (5/6), 50% (5/6); with one physical term missing (5/6); and in a realistic deployment combining a partial model, 30% parameter error and coloured heavy-tailed noise (6/6 against the best filter that still beats a constant predictor). The pre-registered crossover hypothesis, placed between 10% and 30% mismatch, **FAIL**s in the direction favourable to learning (first winning level 10%). The learned observers themselves fail under gross outliers and 30% post-training drift on an exact model. On real battery drive cycles (Section 8) a pre-registered test confirms both orderings against a tuned EKF on an identified circuit model (3 of 3 cycles each), with limits. This is two findings: (a) learning from real-plant data beats a wrong model, which is robust; (b) *which* learner wins is system-dependent: on the partial-model node a grey-box EKF (model plus learned residual) or a Koopman Kalman filter trained on the same labels beats the KKL observers on three of five decided plants (Section 5); the other nodes' same-data comparison is still being merged.

3. **A testbed and a way of reporting (C4).** Twelve plants with partial-knowledge models, an input protocol in which out-of-distribution means a new waveform shape or event sequence inside fixed actuator bounds, eight filters including robust and oracle variants, a climatology failure line, a deployment scorecard, and an experiment tree whose nodes carry their decision rules.

4. **Supporting analysis (C2).** A fading-memory account of how much input history the decoder needs (forgetting floor, memory probe, window sweeps); all its pre-registered rules **FAIL**, and a refined rule restoring a plant-dependent constant explains part of the failures post hoc.

Requirements and limits, up front. (i) Every learned observer is trained on full-state labels from the real plant, so C3 compares "data from the true plant" with "a wrong model", not "no knowledge" with "a model". (ii) With an exact model a tuned EKF is several times better on every plant, so the learned observers' value lies in the model-wrong regime. (iii) Trained on the nominal sensor, they are fragile to gross outliers and post-training drift, where robust filters are not. (iv) The theory's counterfactual-past statements do not apply to the two chaotic plants; the real-data result holds at the training temperature only, on one cell, against a quickly identified circuit model. (v) The held-out network plant is weakly observable; the held-out C3 replication is PASS* (17/18 jobs); the same-data comparison is post hoc and, where decided, goes against the KKL observers.

Section 2 states the framework and three results of the theory informally; Section 3 gives the experimental design; Sections 4–7 report C1, C3, C2 and the scorecard; Section 9 positions the work; Sections 10–11 state limitations, broader impact and conclusions. The proofs, the full tables and the history of the study are in the appendix.

2 Framework: a learned KKL observer with an input context

Setting. The plant is $\dot{x} = f(x,u)$, $y = h(x)$, with $x \in \mathbb{R}^{n_x}$, $y \in \mathbb{R}^{n_y}$ and an input u applied through a zero-order hold with values in a fixed box (no input-affine structure is assumed; every plant in the testbed passes through a saturating actuator). *Scope and assumptions*, stated once: u is measured and bounded in a fixed box; training uses full-state labels from the real plant; the plant is observable (backward distinguishability; Appendix A, (A4)); deployment is stationary (post-training drift is a stress condition only); and training and test data come from the same plant, so the held-out test systems test the *recipe*, not transfer. The observer runs a fixed stable linear filter of the output,

$$\dot{z} = Az + By, \qquad A \text{ Hurwitz with spectral abscissa } -\alpha, \quad \|e^{At}\| \le \kappa e^{-\alpha t}, \tag{1}$$

and decodes $\hat{x}$ from z. For an autonomous plant there is a stationary map T_0 with $z \to T_0(x)$ and a left inverse T_0^+ (Andrieu & Praly, 2006). Under an input the latent equals the *finite-start map* $T^0(x(t), t; u) = \int_0^t e^{A(t-s)} B h(x(s))\, ds$, which depends on the input history (Bernard & Andrieu, 2019); the state is a function of the latent *and* of that history, so the decoder must receive an input context, and the question is where.

The HyperKKL framework. All variants share (Figure 1): the filter (1) with a Butterworth A and $n_z \ge 2n_x + 1$ (Table 5); base maps T_0, T_0^+ trained once per plant in a first phase on zero-input data with a physics-informed KKL-PDE residual (Niazi et al., 2025) using the modeller's model of f (possibly wrong, Section 3); and a causal context encoder $c_k = E(u_0, \dots, u_k)$. The default encoder is a *spectral fading-memory* (SFM) filter bank, $\xi_{k+1} = A_d^\xi \xi_k + B_d u_k$, $c_k = \psi([\xi_k, u_k])$, A^ξ Hurwitz by construction, so the context forgets old inputs at a guaranteed rate (Lemma 15); an optional *floor* forces that rate to be at least a fraction of α (the C2 intervention). Every encoder has $E(0) = 0$ exactly. In a second phase the base maps are frozen and one conditioning pathway is trained, with one objective for every variant: the deployed state error $\|(\hat{x}_k - x_k)/\sigma\|^2$ on the latent that the deployed filter actually carries. **HyperKKL** is the method: a hypernetwork (Ha et al., 2017) of the context writes a rank-r update of every decoder layer,

$$W_l \leftarrow W_l + U_l(c)\, V_l(c)/\sqrt{r}, \qquad U_l, V_l \text{ generated from } c \text{ by a shared trunk}, \tag{2}$$

with the rank r a hyperparameter (default $r = 4$): a small r is a strongly regularised, low-dimensional family of decoders, a larger r a fuller weight-space conditioning (the factorised form of LoRA, Hu et al., 2022, here generated from c). The three alternative entry points, compared at the same budget, are

$$\begin{aligned}
&\text{latent:} && \dot{\hat{z}} = A\hat{z} + By + \Phi(\hat{z}, c), && \hat{x} = T_0^+(\hat{z}) && \text{(input enters the dynamics)}\\
&\text{concat (additive):} && \hat{x} = T_0^+(\hat{z}) + R([\hat{z}, c]) - R([\hat{z}, 0]) && && \text{(output residual)}\\
&\text{FiLM:} && h_l \leftarrow h_l\,(1 + \gamma_l(c)) + \beta_l(c) && && \text{(feature modulation)}
\end{aligned}$$

where h_l, W_l are the pre-activations and weights of the frozen decoder. A *dense-subspace* generator, $W_l \leftarrow W_l + \Delta W_l(c)$ with full-shape updates confined to a k-dimensional linear subspace ($k = 4$ at the 140k

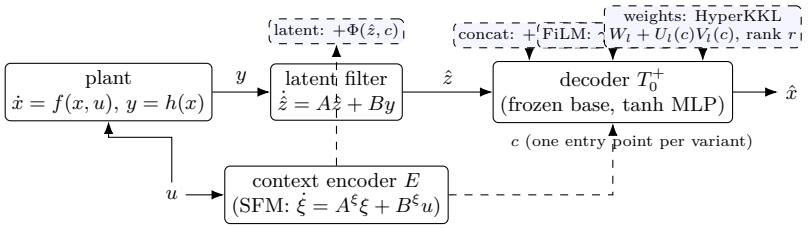


Figure 1: The HyperKKL framework. Solid: the parts shared by every variant (filter, frozen base decoder, context encoder). Dashed: the entry points of the input context c; HyperKKL is the weight-space entry (rank-r updates $U_l V_l$ written by a hypernetwork of c), the other three are the compared alternatives. The deployed latent is an input-free linear filter for every decoder-side variant; only the latent variant changes the dynamics.

budget, Table 8), is run as an ablation of the weight-space entry point, not as a competitor; "weight-space" below means HyperKKL and, where the pre-registered rules name it, this ablation. Every Δ is exactly zero at $c = 0$, so at zero input every variant is the same autonomous observer.

What the theory says, informally. The full statements and proofs are in Appendices A–C; three results frame the experiments. Each is tagged with what it proves.

C1: where the input enters is a stability–extrapolation trade-off (Proposition 7, proven). Latent injection makes the observer a recurrent system whose stability depends on the input: the error contracts only if $\mu_2(A + \partial_z \Phi(z, \hat{c})) \leq -\mu < 0$ on a set fixed in advance, the certified margin shrinks with the context (Corollary 8), and there are bounded, observable plants with an *exact* injection whose latent observer diverges at a fixed filter spectrum (Proposition 9). Decoder-side conditioning keeps the latent an input-free stable filter, so stability holds for every input, but the decoder is queried at unseen contexts and its excess error there is at best linear in the distance to the training contexts (Corollary 10). The theory does *not* rank concat, FiLM and the weight-space generators (Remark 6), which is why C1 is an experiment.

C2: a short input window suffices (Lemma 11, Theorem 12). The finite-start map and the map that sees only the last L seconds of input differ by at most $\frac{1}{2}\varepsilon(L)$, $\varepsilon(L) = 2\kappa\|B\|\bar{y}\, e^{-\alpha L}/\alpha$, so a window $L^* \leq (\ln(1/\eta) + \ln \ell + \text{const})/\alpha$ is *sufficient* for a relative error η, with ℓ the Lipschitz constant of the inverse of the windowed map (Corollary 13). Necessity is proven for linear plants only, and the counterfactual-past argument behind it does not apply to the chaotic plants. The pre-registered C2 rules dropped $\ln \ell$; Section 6 reports that they fail.

C3: a wrong model leaves a bias floor that no data lowers (Proposition 23, proven for linear plants). A fixed-gain predict–correct filter built on a model that lacks a term d has error $e_{k+1} = (I - KC)Fe_k + (I - KC)(d_k + w_k) - Kv_{k+1}$, hence a non-decaying bias and a bias–variance floor $\inf_K \text{MSE}(K) > 0$ that inflating the process covariance (what the oracle-tuned filters do) can only trade against noise; the floor depends on the model error and on no data. A decoder trained on the true plant pays its fit error, in which no model term appears (Proposition 19). Prediction: a single crossing in the relative model error. On the nonlinear benchmarks this is the reading of an *empirical* result. The ISS bound of Proposition 19 is an exact decomposition with measured constants; a priori it is vacuous on every benchmark and is not offered as a certificate.

3 Experimental design

The protocol was fixed before the reported runs, in the form of an *experiment tree*: each node is a YAML file with a question, a hypothesis, a configuration diff and a *decision rule*; the rules are applied by a script to the finished runs and the verdict (PASS, FAIL, REPORTED for descriptive rules, INCONCLUSIVE or PENDING while jobs are missing) is printed without hand editing (Table 13). Table 1 summarises the protocol; the rest of this section defines its terms.

Plants. Table 5 (Appendix F) lists the twelve plants with the state and output dimensions, the latent dimension n_z, where the input enters, the term that the *partial* model lacks, and the resulting relative model

Table 1: Protocol summary (design constants from `configs/base.yaml` and the node files; no results). A "cell" is one plant × tier × condition block; every method is evaluated on the same episodes of a cell (paired).

plants	12 (Table 5): 6 *dev* (every design decision) and 6 *test* (held out; run once with the frozen recipe, no tuning). Each set spans 2-D, 3-D chaotic and $\geq$ 4-D plants.
inputs	bounded box $v(t) \in [-1, 1]$ by construction, through a saturating actuator; times in units of a per-plant t_{ref}. Tiers: `id` (training families, fresh seeds), `shape_ood` (waveforms never trained on: chirp, filtered noise, triangle, sawtooth; same box and band), `sequence_ood` (the "drive" family with a stop-and-go event sequence; same level grid).
knowledge	what every model-based component (phase-1 PDE term, filters, oracle model-error rate) is given: *exact*; *mismatch* p% (real plant's parameters p% off nominal, $p \in \{10, 20, 30, 50\}$); *partial* (one physical term missing or simplified, Table 5); *realistic* (partial + 30% + coloured Student-t noise). The real plant keeps its full dynamics and generates every label.
conditions	`clean` (nominal sensor: white Gaussian, 3% of each output's reference std); sensor faults (noise, outliers, dropout, coloured, coloured Student-t, quantised, drift, burst outage); input faults (noise, bias, delay); poor initialisation; post-training plant drift (10%, 30%). No observer is told about a condition.
learned observers	autonomous (no context), HyperKKL and the three alternative entry points of Section 2, one frozen base per plant and seed, conditioner budget 140k (controls at 35k and 560k; hyper subspace $k = 1/4/16$, Table 8), SFM context (dim 16, 8 blocks), HyperKKL rank $r = 4$ (ablations $r = 1, 16$ and the dense-subspace generator); NLOX, a neural Luenberger observer with a learned input-affine latent term (Woelk et al., 2026), trained on the same data. All need full-state labels.
model-based filters	EKF, UKF, PF (512 particles), `kkl_mb` (the stationary-transformation controlled KKL observer of Bernard & Andrieu (2019): learned autonomous maps with the model's exact input correction $\partial_x T_0(\hat{x})[f(\hat{x}, u) - f(\hat{x}, 0)]$ in the latent dynamics); robust `ekf_huber`, `ukf_huber` (Huber update, nominal Q, R); condition-oracle `ekf_oracle`, `ukf_oracle` (rebuilt per condition with its sensor statistics and a Q that includes the model-error rate; inflation factors chosen on validation episodes). Filters hold the modeller's model and never see state labels.
reference lines	*climatology*: NRMSE of the constant mean-state predictor, same normalisation; a filter above it is FAILED and never used as a comparator. EKF on the exact model as the accuracy reference for C1.
data	per plant and seed: 400 zero-input episodes (phase 1), 600 forced training episodes, 64 validation episodes (model selection only), 64 evaluation episodes per tier; steady window = second half of each episode.
seeds, statistics	the training seed is the unit of replication: 5 seeds for `C1-where`, `C3-faults` and `C3-realistic`, 3 for every other node and for the test plants. Paired Student-t 95% CIs over per-seed relative changes; "A beats B" = CI excludes 0 in A's favour. Control nodes add a TOST equivalence test at ±10% (90% CI) (Schuirmann, 1987). System-level rules count plants (k/6) and are not corrected for multiplicity.

error median $\|f_{\text{real}} - f_{\text{model}}\|/\|f_{\text{real}}\|$ over the operating box. Dev: `battery`, `bioreactor`, `duffing`, `lorenz`, `quad_tank` (Johansson, 2000), `vdp`. Test: `cartpole`, `fhn` (FitzHugh, 1961), `highway_traffic`, `network` (five coupled neurons, three measured), `rossler` (Rössler, 1976), `two_link_arm`.

What out-of-distribution means. A deployed estimator sees inputs inside the actuator's range; what changes is the waveform's *shape* or the event *sequence* (another driver, another road). Every family in every tier realises $v(t) \in [-1, 1]$ without clipping and training covers the whole box, so no tier changes the amplitude and the evaluation states lie inside the training box (Appendix E). An earlier amplitude-OOD protocol was abandoned for driving bounded plants beyond their input range (Appendix H).

Metrics. NRMSE $= \sqrt{\text{mean}_{t \in S} \text{mean}_j (e_j(t)/\sigma_j)^2}$ on the steady window S, with σ_j the reference std of state j under in-distribution inputs of the nominal plant, fixed once per plant (one yardstick for every method, tier and condition); normalised errors are clipped at 100 so a diverged episode contributes a bounded value and is flagged. The deployment scorecard adds per block the quantities of Section 7.

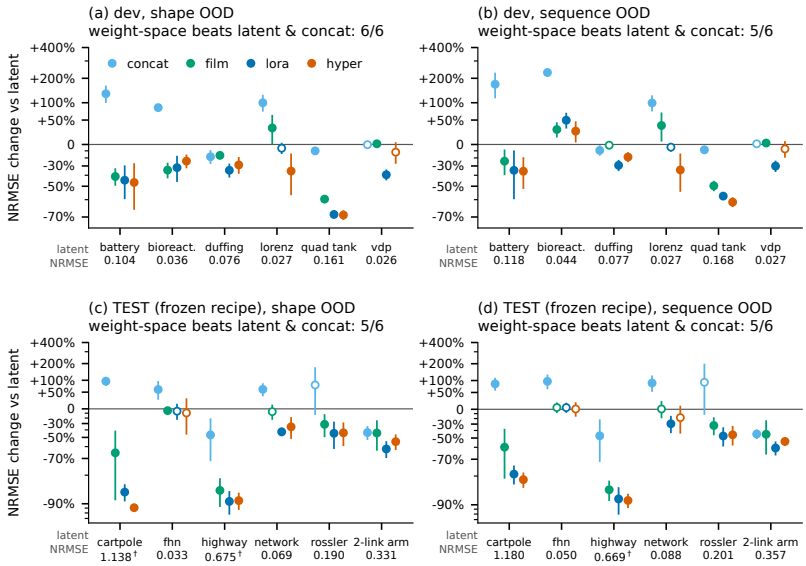


Figure 2: C1, paired change in NRMSE relative to latent injection (the reference; its absolute NRMSE is printed under each plant). (a,b) dev plants, 5 seeds; (c,d) held-out plants, frozen recipe, 3 seeds. Filled markers: 95% CI over seeds excludes 0; bars: 95% CI. †: latent injection itself above climatology. The pre-registered single-location rule fails on the dev plants (panel (e) of the generated figure, reproduced as Table 13); the weight-space-beats-latent-and-concat rule passes in all four panels.

Selection and comparison rules. "Best learned" is the lowest mean NRMSE among HyperKKL, its dense ablation and latent injection in a cell, "best filter" the lowest among the eight filters (nominal and robust/oracle reported separately); both are selected on the evaluation episodes, which favours the side with more candidates. A rule that names no tier must pass on `shape_ood` and `sequence_ood` separately; `id` is reported. The reading rules are printed in full in Appendix E.

Transparency. The held-out C1 runs come from four code snapshots (a recorded data problem, Appendix H; the recipe was frozen); control nodes have 2–3 seeds, at which "within CI" can hardly fail, hence the TOST column. Two baselines trained on the same labels (a Koopman Kalman filter on an EDMD-with-inputs lift; a grey-box EKF/UKF with a learned residual) are implemented and marked pending in Section 5. Nodes not discussed in the text carry their verdicts in Table 13 (learning curves `C3-data50/150`: REPORTED, REPORTED, descriptive, partially finished).

4 C1: where should the input enter?

Question and rule. With the same frozen base maps, the same SFM context and a 140k-parameter conditioner, which entry point generalises best to unseen input shapes and sequences? Pre-registered rule (`C1-where`): a location is *preferred* if it beats every other location with a paired 95% CI excluding 0 on $\geq 4/6$ dev plants, on `shape_ood` and `sequence_ood` separately, and is significantly worse on none. The weaker statement named by the hypothesis, "weight-space (the better of HyperKKL and its dense ablation) beats both latent and concat on $\geq 4/6$", was pre-registered as the rule for the held-out plants (`TEST-C1`, run once with the frozen recipe).

Verdict: the pre-registered rule fails; the weaker claim holds and replicates. No location is preferred (`C1-where`: FAIL; preferred location: none): HyperKKL beats every other location on 1/6 plants (`shape_ood`) and 2/6 (`sequence_ood`), its dense ablation on 1/6 and 2/6, and the two are within each other's CIs on most plants (Figure 2; absolute NRMSE in Table 6, Appendix F). The weaker claim holds: weight-space beats both latent and concat on 6/6 (`shape_ood`) and 5/6 (`sequence_ood`) plants. Read for HyperKKL alone ($r = 4$) against latent injection: better on 5/6 and worse on 0/6 plants on `shape_ood` (-32

to -69% where significant), better on 4/6 and worse on 1/6 on `sequence_ood` (the loss is `bioreactor`, where latent injection is best); FiLM is better on 4/6 and worse on 2/6, the additive residual 2/6 and 3/6. On the held-out plants (`TEST-C1`: PASS) the weight-space claim holds on 5/6 and 5/6; HyperKKL alone beats latent injection on 5/6 and 5/6 plants (-43 to -89% where significant), the exception being `fhn`, where it is within CI of latent injection; latent injection is above climatology on `cartpole` and `highway_traffic` (`shape_ood`), where HyperKKL is not. No single location passes the strict rule (none).

Controls. Each control is a pre-registered node whose counts are in Table 13. The ranking survives when the context encoder is charged to the budget (`C1-ctx`: PASS) and across budgets of 35k and 560k (`C1-budget`: PASS; an asterisk, where printed, marks a verdict decided while seeds were running). *Rank ablation.* The rank is a real knob: $r = 1$ and $r = 16$ are not equivalent to $r = 4$ at the $\pm 10\%$ margin (`C1-rank1`: FAIL, `C1-rank16`: FAIL; Table 13), with $r = 1$ losing accuracy on the chaotic and the oscillatory plant and $r = 16$ non-inferior or better on half of them, consistent with r trading regularisation against capacity; the default $r = 4$ is the pre-registered setting, not a tuned one. Base-width controls (winner not base-dependent): PASS / PASS (jobs 18/18, 18/18). The model-free phase-1 control (`C1-modelfree`; rule: the model is needed if the PDE-regularised base wins on $\geq 4/6$) reads FAIL: no evidence that the phase-1 model term is needed (Table 13).

Reference: with an exact model the EKF is far better. The best learned observer's `sequence_ood` NRMSE is 5.9× the exact-model EKF's (median over dev plants; range 2.7–14.1×) and 6.5× on the held-out plants (2.2–27.8×); C1 ranks learned observers, whether to learn at all is C3.

Caveats. (i) Vote counts are descriptive (5 variants × 2 tiers × 6 plants, no multiplicity correction) and the "better of HyperKKL/ablation" selection in the weight-space verdicts is made on the evaluation episodes. (iii) The dense-subspace ablation generates full-shape updates inside a fixed linear subspace of dimension $k = 1$, 4 and 16 at the 35k, 140k and 560k budgets, while HyperKKL's rank-4 factors are bilinear in a 11-, 44- and 170-dimensional trunk feature (Table 8); its tie with HyperKKL is a tie between these parameterisations. (iv) The latent variant is trained through 256-step rollouts at the same learning rate and steps, so location is confounded with trainability for it, and decoder-side gains could partly be repair of a weak base map; a learning-rate search and a context-swap control are not in the tree. (v) In-distribution results (Table 7) show the same ordering.

5 C3: when does a learned KKL observer beat a model-based filter?

Question and rules. The best KKL observer (HyperKKL, its dense ablation or latent injection, as pre-registered) is compared with the best of eight model-based filters, paired on the same episodes, on `id` and `sequence_ood`; "learned wins" in a scenario needs a CI excluding 0 on $\geq 4/6$ dev plants on both tiers. The filters hold the modeller's model; the learned observers are trained on labelled data from the real plant. Absolute NRMSE accompanies every change; a filter above *climatology* is FAILED, and changes are quoted against the best *working* filter with the number of failed filters. Scenarios: exact model under fifteen stress conditions (`C3-faults`, descriptive), coloured Student-t noise in training and deployment (`C3-noise-ct`), parameter mismatch at four levels (`C3-mismatch10/20/30/50`; hypothesis: the crossover sits between 10% and 30%), a missing physical term (`C3-partial`, descriptive, against the relative model error), and the realistic deployment (`C3-realistic`).

Exact model: the filters win everywhere. In none of the fifteen stress conditions does the learned side reach the learned-wins regime (`C3-faults`: none of 15 conditions), against nominal and robust/oracle filters alike (Tables 2 and 13). Under coloured heavy-tailed noise in training and deployment the learned observers beat the best filter on 0/6 plants (`C3-noise-ct`: FAIL). Two exact-model failures of the learned observers: under 2% gross sensor outliers the best learned observer is itself above climatology on `battery`, `duffing` and `quad_tank`, while the Huber and oracle filters keep their clean accuracy (Table 10); and after a 30% plant drift that occurs *after* training, it is above climatology on `bioreactor` and `quad_tank`. These limits of the method as trained here (nominal sensor, no robust loss, no adaptation) are consistent with the theory: the

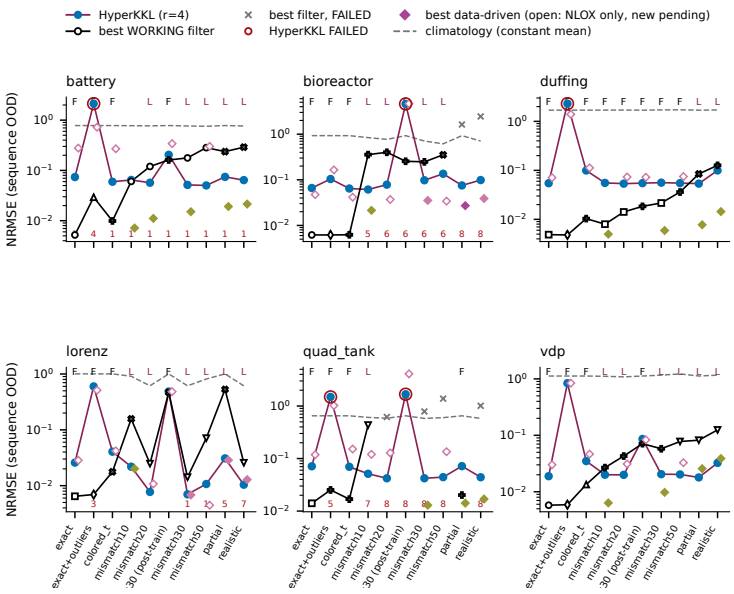


Figure 3: C3 on the dev plants, `sequence_ood`, NRMSE on a log axis: best learned observer (coloured by method) against the best working filter (black) and climatology (dashed), per scenario ordered by the median relative model error over plants (the model-error panel of the generated figure is omitted here; the values are in Table 5 and in Figure 4). Top marks: L = learned beats the best working filter (paired 95% CI), F = filter wins; red numbers = filters above climatology. Red-ringed markers: the best *learned* observer is itself above climatology. Grey crosses: the best filter is FAILED and there is no working filter. The `id` version and the held-out plants are in Appendix F.

Table 2: When does a learned KKL observer win (C3)? sequence OOD, dev systems. Counts of systems where HyperKKL ($r = 4$, the single learned method) beats the comparator with a paired 95% CI excluding 0; (kL) = comparator significantly better on k systems. (a) MODEL-BASED filters (ekf, ukf, pf, kkl_mb, Huber and condition-oracle variants); a filter is FAILED when its NRMSE exceeds climatology (constant mean-state predictor, same normalisation) and 'working' compares against the best filter that did not fail (*: no filter works on some systems). (b) DATA-DRIVEN same-data baselines (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle), trained on the same real-plant data; 'pend. m/N' = the new baselines are merged in m of N runs. Model error: median over dev plants. Per-system absolute numbers: Table 10.

		(a) vs model-based filters					(b) vs data-driven	
scenario	model err.	best	working	nominal	rob./orac.	failed	best	NLOX
exact	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	1/48	pend. 0/30	4/6
exact+outliers	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	15/48	pend. 0/30	1/6 (4L)
colored_t	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	1/48	pend. 0/18	5/6
mismatch10	0.09	4/6 (1L)	4/6 (1L)	4/6 (1L)	4/6 (1L)	13/48	0/5 (4L) [16/18]	4/6
mismatch20	0.17	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	15/48	0/1 (1L) [7/18]	5/6 (1L)
drift30 (post-train)	0.24	0/6 (5L)	0/5 (5L)*	2/6 (3L)	1/6 (5L)	15/48	pend. 0/30	3/6 (1L)
mismatch30	0.24	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	16/48	0/6 (4L)	4/6
mismatch50	0.37	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	16/48	pend. 1/18	4/6
partial	0.44	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	14/48	1/6 (4L)	4/6 (1L)
realistic	0.63	6/6	4/4*	6/6	6/6	24/48	2/6 (4L)	5/6 (1L)

fault's effect is multiplied by a decoder slope in the hundreds (Corollary 21, Remark 25), and on linear plants a Kalman filter beats any KKL observer with outliers too (Remark 22).

Wrong model: the learned observers win at every level tested. With the real plant's parameters off nominal, the learned side beats the best filter on 5/6 (`id`) and 4/6 (`sequence_ood`) plants at 10% (PASS), 5/6 and 5/6 at 20% (PASS), 5/6 and 5/6 at 30% (PASS), 6/6 and 5/6 at 50% (PASS); the same counts hold against the robust/oracle filters, which are given the model-error rate in Q. The pre-registered crossover hypothesis therefore **FAIL**s (`C3-mismatch (crossover)`: FAIL): the first winning level is the smallest one tested, and the crossing, if there is one, lies below 10%. The exception at every level is `duffing`, whose 30% parameter error moves f by only 0.02 in relative terms (Table 5) and whose best filter stays several

times better (0.0216 against 0.0561 at 30%); `battery` at 10% is the other non-win on `sequence_ood` (+7%, n.s.). With one physical term missing (`C3-partial`: REPORTED), the learned side wins on 5/6 and 5/6 plants; the exception is `quad_tank`, where linearising the orifice law moves f by 0.15 and the oracle EKF wins (0.0202 against 0.0625), whereas `bioreactor` (relative model error 2.15), `lorenz` (0.71) and `vdp` (0.66) are won at every level and `duffing` (0.13) is won here. The pattern hypothesised for this node (learned wins on `bioreactor`/`vdp`/`lorenz`, filters on `duffing`/`quad_tank`) holds on four of its five named plants (Table 13).

Realistic deployment. With partial model, 30% parameter error and coloured Student-t noise at once, the learned observers win on 6/6 (`id`) and 6/6 (`sequence_ood`) plants against the best filter of any kind and against the best working filter (`C3-realistic`: PASS). On `bioreactor` and `quad_tank` all eight filters are above climatology (8/8, 8/8), so no change is quoted: the learned observers reach 0.0702 and 0.0436 against climatology 0.708 and 0.581. On `lorenz` 7/8 filters fail and the working one, `kkl_mb`, is beaten by -58%; `battery`, `duffing` and `vdp` by -79%, -35% and -74%.

Reading the crossover. Over the 36 model-wrong cells (six plants × mismatch 10–50%, partial, realistic), every non-win sits at a small relative model error (`duffing`, `battery` at 10%, `quad_tank` partial) and every plant with a large error is won at every level (Figure 3, Table 10): the single crossing Proposition 23 predicts, at a plant- and noise-dependent value (Appendix C).

Two findings, not one. (a) *Learning from real-plant data beats model-based filters once the model is wrong.* This is the robust finding: it holds against nominal, robust and oracle-tuned filters, at every mismatch level, under a missing term and in the realistic node, and it reverses whenever the model is exact. (b) *Which learner wins is system-dependent and is not settled here.* Two further estimators trained on exactly the learned observers' data are implemented: `koopman_kf` (EDMD with inputs on a thin-plate-RBF dictionary, then a Kalman filter on the lifted state) and `greybox_ekf/ukf` (the modeller's model plus an MLP residual fitted to the one-step defect, inside the unchanged EKF/UKF). This comparison (b), best KKL observer against the best data-driven baseline (NLOX included), was added after review and is *not* pre-registered; it is merged into the runs node by node (column (b) of Table 2). Where it is complete enough to decide, `C3-partial`, the KKL observers do *not* win: FAIL (KKL beats the best data-driven baseline on 1/6 plants on `id` and 2/6 on `sequence_ood`): on `sequence_ood` the grey-box EKF/UKF is better on `battery` (0.0191 against 0.0717) and `duffing` (0.00775 vs 0.0536), the Koopman KF on `bioreactor`, and the KKL observers are better on `lorenz` (-33% vs NLOX) and `vdp` (-30% vs the grey-box EKF). For the mismatch, realistic and held-out nodes comparison (b) is FAIL at the time of writing (merge counts in Table 2). So *no claim that the KKL observer is the best learner is made.* Nor does C3 establish that the partial model helps the learned observer: the pre-registered test of the phase-1 PDE term with the partial model against a model-free phase 1 **FAIL**s (`C3-partial-modelfree`: FAIL; the differences are not significant at three seeds, including a large but non-significant one on `bioreactor`), so no claim that the partial model helps is made. Whether the KKL observers' structural advantages (global convergence, no linearisation) show against the grey-box filters where they should, under a poor initial guess, outliers, higher noise, post-training drift and a sensor outage with partial knowledge, is the question of a further node, `C3-regimes` (pre-registered rule: a KKL-wins regime needs $\geq 4/6$ dev plants per condition on `sequence_ood`), whose outcome is **PENDING** at the time of writing. The replication on the held-out plants (`TEST-C3`, the realistic scenario) is PASS* at the time of writing with 17/18 jobs finished: the learned side wins on `fhn` (-37% on `sequence_ood`) and `highway_traffic` (-21%), the UKF wins on `cartpole` (+43%) although its realistic model error (partial 0.03, 30% 0.35) is comparable to that of dev plants the learned side won; with half the plants finished, no reading of the held-out crossover is offered.

6 Supporting analysis (C2): how much input history does the decoder need?

The theory says that a window of $\ln(1/\eta)/\alpha$ plus a plant-dependent $\ln \ell/\alpha$ is sufficient. Three pre-registered tests were run on the dev plants; all three **FAIL** and are reported as supporting analysis.

Forgetting floor. The SFM context was forced to forget at a rate $\geq c\alpha$, $c \in \{0.25, 0.5, 1\}$ (`C2-floor-025/05/1`). Rule (decided at $c = 0.5$): no significant loss relative to the un-floored context

for HyperKKL and its dense ablation, on `id` and `sequence_ood`, on $\geq 5/6$ plants, and the memory probe must decay at least at the floor rate. Outcome: FAIL: no-loss on 4/6 plants (TOST at ±10%: 0/6; three seeds make "no significant loss" weak evidence), probe at or above the floor rate on 0/6 (Figure 6, Appendix F). The dose–response is monotone: at $c = 0.25$, 6/6 (FAIL on the TOST and probe parts); at $c = 1$, 4/6 (FAIL). The one plant that clearly loses at $c = 0.5$ is `battery` (HyperKKL +33% on `id`, +36% on `sequence_ood`; +106% at $c = 1$), the plant with the largest decoder-Lipschitz proxy ($\ell_{\mathrm{dec,q95}} = 806$ against 10–295 elsewhere): its state of charge enters the voltage only through the open-circuit-voltage slope, so inverting the windowed map costs a large ℓ and a longer window, as Corollary 13 says. The `bioreactor` loss has no such explanation. Without a floor the learned SFM's slowest pole sits at 0.14–$0.18\,\alpha$ (HyperKKL), between the two slowest *initial* poles ($\alpha/1.5^5 = 0.13\alpha$, 0.20α): the un-floored memory is inherited from the initialisation, not learned.

Memory probe. The last L input samples are kept and everything older is swapped for another episode's input; the decoded change is fitted to a rate (the quantity Lemma 15(c) bounds). Under a 0.5α floor the fitted rate is 0.73–0.96 of the floor on five plants (no contradiction of an upper envelope, but the strict check fails) and `lorenz`'s probe does not decay at all (-0.02; Figure 7), which is unexplained and reported as such.

Window sweeps. With HyperKKL on a raw window of L samples in place of the SFM context (`C2-window-a1`), the rule "$e(L)$ within 5% of $e(512)$ for every $L \geq L^* = \ln(1/\eta)/\alpha$, $\eta = 0.05$, on $\geq 5/6$ plants" gives FAIL: 2/5 (`id`) and 3/5 (`sequence_ood`) among the evaluable plants (`quad_tank`'s $L^* = 599$ exceeds the grid). The ℓ-free rule fails where ℓ is largest: on `battery` ($L^* = 135$) the error at $L = 256$ is still +73% above $e(512)$; restoring $\ln \ell$ ($L^* = 435$) moves the rule to 2/3 and 2/3 on the three plants where it stays on the grid, a *post hoc* explanation, not a passed test. Measured knees are far below either rule where they exist (`vdp` 32, `duffing` 32 samples), as a sufficient-side bound allows; `bioreactor` is non-monotone. The second-bandwidth test of the $1/\alpha$ scaling (`C2-window-a2`, three plants at 2α) gives FAIL: the knee in samples moved by $\leq 2\times$ on one of two evaluable plants, so the scaling is not supported by the current data.

What survives. The sufficient-side statement is consistent with every sweep and the floor costs nothing significant on four of six plants; not established are that memory beyond the theory window is unnecessary, the $1/\alpha$ scaling, and the probe rate. The intervention and the probe remain cheap, reusable diagnostics.

7 C4: the testbed and the deployment scorecard

Every evaluation block stores a *deployment scorecard*: NRMSE per tier and on the *unmeasured* states, worst sensor-fault NRMSE, p99 and divergence rate, convergence from a far-corner prior and the paired "forget the prior" time, recovery after a burst outage, long-horizon drift, the AUROC of an output-residual alarm for the method's own large errors, and time per step (Appendix E). Figure 9 (Appendix F) ranks every method on every column over the dev plants for the exact-model stress node and the realistic node (values per plant: Table 12).

With an exact model the nominal and oracle filters take the top ranks on accuracy, tails, unmeasured states, convergence and recovery; the learned observers rank better only on the divergence rate (they never diverge; the EKF and UKF sometimes do) and, with the particle filter, on the residual alarm. In the realistic deployment the learned KKL observers take the top ranks on every accuracy column, every filter is above climatology on three of six plants, and the learned observers' alarm is no better than the filters'; time per step is the batched-harness cost, not an embedded number.

The released testbed comprises the twelve plants with their partial-knowledge models, the input protocol, the eight filters with their tuning rules, the climatology reference, the memory probe, the experiment tree with its rules and the script that applies them (Table 13), and the regeneration path from raw runs to every number here.

8 Real-data validation: battery SOC on held-out drive cycles

C1 and C3 were tested once on real hardware with a design pre-registered before any training (Appendix G): the Panasonic 18650PF drive cycles (Kollmeyer, 2018) at 25°C, split by cycle: Cycles 1–3, UDDS and LA92 for training, Cycle 4 for validation (model selection, rank-4-vs-dense-ablation choice, filter tuning), and US06, HWFET and NN as test cycles reported once. The target is the laboratory coulomb count ($Q = 2.995\,\mathrm{Ah}$); the modeller's model is a constant-parameter 2-RC circuit identified by least squares on the training cycles (open-loop voltage RMSE 24–37 mV there, 23–72 mV on validation/test), used for phase 1 and inside the EKF/UKF (Q, R tuned on the validation cycle). Episodes are 30-min windows through each discharge from a prior SOC 20% off; the metric is settled SOC RMSE; a cycle counts only if the 95% block-bootstrap CI of the paired difference excludes 0 and all 3 seeds agree.

Both pre-registered rules are met (Table 14, Appendix G); they were decided on "WS-best", the per-seed validation choice between HyperKKL ($r = 4$) and its dense ablation. H1 (C3): WS-best beats the tuned EKF on 3 of 3 test cycles (US06 -6.50, HWFET -0.41, NN -1.27 % SOC; likewise against the UKF), and HyperKKL at $r = 4$ alone on 3 of 3. H2 (C1): WS-best beats latent injection on 3 of 3 (by 0.24–0.63 % SOC, a small margin); HyperKKL at $r = 4$ alone passes the strict rule on 1 of 3 cycles with the same sign everywhere, and the simulated ordering weight-space > latent > concat $\gg$ autonomous holds. The mechanism is that of C3: on the aggressive US06 cycle the filters settle on a biased SOC (8.85 % against 1.76 %) because the constant-parameter circuit cannot explain the voltage error; the decoder trained on real data can. The limits matter as much: on whole discharges the filters have hours to converge and the gap closes except on US06 (EKF 0.66 vs learned 0.69 % on NN), so on mild cycles the advantage is mostly faster convergence; at 0°C, where nothing was trained or identified, the Arrhenius-scaled EKF is better on 8 of 9 cycles (learned 9.9–16.0 %); one cell, one campaign; a coulomb-counting reference; a quickly identified circuit model. The same-data baselines (Koopman KF, grey-box EKF) on the real cycles are **pending**.

9 Related work

KKL observers. The construction goes back to Kazantzis & Kravaris (1998); Andrieu & Praly (2006) proved existence of the injective map under backward distinguishability, Bernard et al. (2018) addressed the inverse, Bernard & Andrieu (2019) extended the theory to non-autonomous plants (history-dependent map), and Brivadis et al. (2023) settled the generic latent dimension ($2n_x + 1$ autonomous, $2n_x + 2$ time-varying). `kkl_mb` is that line's stationary-transformation controlled observer; Remark 1 states what is cited and what is assumed.

Learned KKL and neural Luenberger observers. Learning T and its inverse from simulated data was proposed by Ramos et al. (2020) and developed by Janny et al. (2021); Peralez & Nadri (2021); Peralez et al. (2022); Niazi et al. (2023); Buisson-Fenet et al. (2023) and Niazi et al. (2025) (physics-informed training with robustness guarantees, whose phase-1 recipe we use). Peralez & Nadri (2024) learn KKL observers model-free from data with a switching scheme, for autonomous systems; for driven plants, Woelk et al. (2026) learn a Luenberger observer with an input-affine latent term (our NLOX baseline). Our preprint (Anonymous, 2026) introduced hypernetwork conditioning of the KKL maps; this paper replaces its comparisons by a matched, pre-registered study.

Conditioning mechanisms. FiLM (Perez et al., 2018), hypernetworks (Ha et al., 2017; Chauhan et al., 2024) and low-rank adaptation (Hu et al., 2022), whose factorised form HyperKKL's generated updates take, are the conditioning families compared here; Zeng & Lee (2024) analyse the expressive power of low-rank adaptation. Context-generated parameters for dynamical systems appear in CoDA (Kirchmeyer et al., 2022) and HyperPINN (de Avila Belbute-Peres et al., 2021).

Koopman operators with inputs. The KKL map is the resolvent of the Koopman generator at the filter poles (Remark 2); latent injection with a stationary map is a lifted Koopman model with inputs and NLOX its bilinear form. DMD with control (Proctor et al., 2016), EDMD (Williams et al., 2015), Koopman linear predictors with inputs (Korda & Mezić, 2018), learned linear embeddings (Lusch et al., 2018) and Koopman

observers (Surana, 2016) are the model class the decoder-side route must be compared with on the same data; our EDMD-with-inputs Kalman filter is that comparison (Section 5).

Sequence models and learned filtering. Neural CDEs (Kidger et al., 2020), ODE-RNNs and latent ODEs (Rubanova et al., 2019; Chen et al., 2018) are universal encoders of input histories; the SFM context is a fixed fading-memory filter bank in the spirit of Boyd & Chua (1985) with a guaranteed forgetting rate (Lemma 15); a matched neural-CDE encoder is not in the tree. Learned Kalman filters such as KalmanNet (Revach et al., 2022) and Backprop KF (Haarnoja et al., 2016) learn the gain or the model inside a Kalman structure from labels; our grey-box EKF is the simplest member of that family (Section 5). The classical comparators are the EKF, UKF (Julier & Uhlmann, 2004), bootstrap particle filter (Gordon et al., 1993) and Huber-type robust updates (Huber, 1964).

10 Limitations

Labels. Every learned observer, NLOX included, is trained on full-state trajectories of the real plant; C3 is about a practitioner with a wrong model *and* labelled data (a virtual sensor with laboratory ground truth), not model-free estimation. *Exact models.* With an exact model a tuned EKF is 5.9× better on the dev plants and the filters win every stress condition: if the model is right, do not learn. *Fragility.* Trained on the nominal sensor, the learned observers are above climatology under gross outliers and 30% post-training drift, where robust filters are not; robust losses, fault augmentation and adaptation are not studied. *Same-data baselines.* The Koopman-KF and grey-box comparison is post hoc and complete on one node, where they beat the KKL observers on three of five plants; C3 is about learning from real-plant data, not the KKL structure. *Capacity.* HyperKKL's tie with its dense-subspace ablation is at $r = 4$ and $k = 4$; $r = 1$ and 16 are not equivalent (open controls: Section 4). *Statistics.* Control nodes have 2–3 seeds (most TOST cells underpowered); vote counts are uncorrected for multiplicity; best-of selections use the evaluation episodes. *Chaotic plants.* The theory's counterfactual-past statements do not apply to `lorenz` and `rossler`; `lorenz`'s probe is flat and unexplained; `rossler` is weakly chaotic; whether the forced plants stay chaotic was not verified. The held-out `network` plant is weakly observable. *Real data.* One cell, one campaign, one temperature; coulomb-counting reference; a quickly identified circuit model; on whole discharges the filters match the learned observers on mild cycles. *Theory.* The ISS bound is vacuous a priori; the window lower bound and the C3 proposition are linear results whose nonlinear reading is empirical; predictions no node measures are stated as untested.

Broader impact. A learned observer that beats a filter when the model is wrong can make soft sensing cheaper where laboratory ground truth exists; the same results caution against deploying one where the model is good, sensors produce gross outliers or the plant drifts after training, and its residual alarm is no better than a filter's, so none of these observers should run in a safety-critical loop without a fallback filter. No personal data is used; the compute is a few hundred GPU-hours.

11 Conclusion

KKL theory requires an input-dependent observer map for a forced plant; where a learned observer should carry it, and when learning one is worth it, were open. On twelve plants with pre-registered rules, the input should enter the decoder's *weights*, as HyperKKL's rank-r updates, rather than the latent dynamics or an additive residual (6/6/5/6 dev, 5/6/5/6 held-out plants), while the single-best-location rule failed because HyperKKL ties with its dense-subspace ablation. Learned observers lose to model-based filters in every exact-model condition (about 5.9× against a tuned EKF) and win once the model is wrong, at every mismatch level, under a missing term and in a realistic deployment; on real battery drive cycles both orderings replicate (small margin over latent injection; no transfer to an untrained temperature); which learner wins once the model is wrong is system-dependent and, where decided, favours a grey-box EKF. The fading-memory analysis failed its pre-registered tests.

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Appendix overview

Appendices A–C are the theory, included unchanged from the maintained source (`theory/m1_existence_location.tex`, `m2_fading_memory.tex`, `m3_error_bounds.tex`; numerical checks in `theory/checks/*.py`, all of which pass). They are self-contained: Appendix A states the one assumption block (A1)–(A4) used everywhere, the decoder target, the Koopman reading and the stability–extrapolation trade-off (C1); Appendix B the fading-memory lemma, the window theorem, the practical rule and the context encoders (C2); Appendix C the ISS decomposition of the deployed sampled-data observer, the sensor-fault corollary, the Kalman-filter remark, the model-incompleteness proposition that anchors C3, and the single sampled-data lemma (Lemma D). Every statement carries a status tag ([proven], [proven, linear plants], [conjecture], [empirical]) and names the experiment node that tests it or says that none does. Empirical numbers quoted inside the theory are the same generated macros as in the main text.

Appendix E gives the protocol details (signal families, training recipe, filter tuning, scorecard definitions, the reading rules of the results layer). Appendix F holds the generated tables and figures not shown in the main text (in-distribution C1 table, per-plant C3 tables for both tiers, the `id` and held-out versions of the C3 figure, the memory probe and window sweeps, the realistic-deployment scorecard, every decision rule with its verdict, and the capacity table of the conditioners). Appendix H records what was tried, superseded or abandoned, and the data problems recorded by the results layer.

Reading guide for the theory (status table).

statement	status	tested by
Lemma 4, Prop. 5 (decoder target, sufficient contexts, history necessary)	proven	–
Prop. 7 (latent injection: contraction margin; decoder side: stable, pays extrapolation)	proven	no node measures the margin or the realised gains (eval-only)
Cor. 8, Prop. 9 (margin sufficient only; cannot be removed at a fixed spectrum)	proven	numerical checks
Cor. 10 (OOD excess $\leq$ Lipschitz extrapolation in context distance)	proven	context distance per tier not reported
Lemma 11 (fading-memory tail; (b) not for chaotic plants)	proven	–
Thm. 12 (window rule; (b) linear plants)	(a) proven; (b) linear; nonlinear (b) conjecture	`C2-window-a1` FAIL, `C2-window-a2` FAIL
Cor. 13 (rule with $\ln \ell$)	proven; its use on the sweep is post hoc	explains the failures post hoc
Prop. 14 (current input not enough)	proven	$L = 1$ row of `C2-window-a1`
Lemma 15 (SFM forgets at a guaranteed rate; (c) = the memory probe)	proven	probe rate $\geq$ floor: 0/6
Prop. 19 (ISS relative to the nominal observer)	(i),(ii),(iv) exact; (iii) decomposition	realised gains not measured by any node
Cor. 21 (sensor faults bounded and forgotten)	proven	`C3-faults`: filters win in none of 15 conditions
Prop. 23 (bias floor vs fit error)	proven, linear plants	`C3-mismatch10-50` PASS, `C3-partial`, `C3-realistic` PASS; `TEST-C3` PASS*
Rem. 22 (KF beats KKL on linear plants, also with outliers)	proven	–
Lemma 26 (sampled-data observer)	proven	checks

A Where the input enters a learned KKL observer

Setting and conventions. Plant $\dot{x} = f(x, u)$, $y = h(x)$, $x \in \mathbb{R}^{n_x}$, $y \in \mathbb{R}^{n_y}$; no input-affine structure. Inputs are applied through a zero-order hold at the sample rate with values in a compact $U \ni 0$, $\bar{u} := \max_U |u|$; u_t is the window $u|_{[t-L,t]}$ shifted to $[-L, 0]$, a point of $W_L = U^{N_L}$ at sample instants when $L/dt = N_L \in \mathbb{N}$ (the decoder runs only there), with the sup norm. $X(x, t; s; u)$ is the value at s of the solution through x at t. The latent filter is $\dot{z} = Az + By$. *Latent initial condition (one convention for Sections A–C).* The *label* and training latents start from $z(0) = 0$, so $z(t) = T^0(x(t), t; u) := \int_0^t e^{A(t-s)} Bh(X(x(t), t; s; u))\, ds$ exactly; this finite-start map is the only time-varying map used. The *deployed* observer (decoder-side and latent-injection

alike, `evaluate.py z0_map`) starts from $\hat{z}(0) = \hat{T}_0(\hat{x}_0)$, which adds the transient $e^{At}\hat{T}_0(\hat{x}_0)$ to the latent; it is carried as the Δz_0 term of Proposition 19 and never absorbed into a map. The decoder target is the windowed map

$$T_L(x, w) := \int_{-L}^{0} e^{-A\tau} B\, h\big(X(x, 0; \tau; w)\big)\, d\tau, \tag{3}$$

which by Lemma 11(a) is within $\frac{1}{2}\varepsilon(L)$ of $T^0(x(t), t; u)$ at $w = u_t$. $\mathcal{P}_L$ is the set of pairs $(x(t), u_t)$, $t \geq L$, generated by admissible trajectories.

Assumption 1 (standing; shared by all sections)**.** Regularity: $f(\cdot, u) \in C^1$ uniformly in $u \in U$, $h \in C^2$.

(A1) *Forward invariance.* A compact $\mathcal{X}$ contains every solution from the initial set $\mathcal{X}_0 \subset \mathcal{X}$ under every admissible input, for all $t \geq 0$.

(A2) *Bounded output.* $\|h(x)\| \leq \bar{y}$ on $\mathcal{X}$.

(A3) *Hurwitz filter.* $-\alpha < 0$ is the spectral abscissa of A and $\kappa := \sup_{t \geq 0} e^{\alpha t}\|e^{At}\| < \infty$ ($\kappa = 1$ for the normal Butterworth A of the code).

(A4) *Uniform Lipschitz injectivity of the windowed map.* There are L_0 and $\ell < \infty$ with $|x_1 - x_2| \leq \ell\, |T_L(x_1, w) - T_L(x_2, w)|$ for all $(x_1, w), (x_2, w) \in \overline{\mathcal{P}_L}$ and all $L \geq L_0$ (i.e. $\ell := \sup_{L \geq L_0} \ell(L) < \infty$; Theorem 12(a) varies L at fixed ℓ).

Chaotic plants. (A1)–(A3) hold for the dissipative chaotic benchmarks (Lorenz, Rössler), (A4) is assumed for them as for every other plant, but its constant is not under control when α is below the *backward* expansion rate λ_b of the flow (for Lorenz $\lambda_b \approx 14.6$ against $\alpha = 2$): $\partial_x T^0$ contains $\int_0^t e^{A(t-s)} B\, \partial h\, \partial_x X(x, t; s; u)\, ds$ and $\|\partial_x X\|$ grows like $e^{\lambda_b (t-s)}$ along the backward flow, so the forward Lipschitz constant of $T^0(\cdot, t)$ grows with t when $\alpha < \lambda_b$; this is the eigenvalue condition of Andrieu & Praly (2006) (the spectrum of A must dominate the Lipschitz constant of f for a C^1 stationary map; a Hölder map needs less), and it is a plausible mechanism (not tested: no node sweeps α or measures λ_b) behind the phase-1 PDE residual $\varepsilon_{\text{pde,rms}} = 74$ on Lorenz against ≤ 0.3 elsewhere. Which results survive on such plants is stated at each result: everything that uses only the true past and (A1)–(A2) does (Lemma 11(a), Theorem 12(a), Proposition 19); the counterfactual-past statements (Lemma 11(b)) and the linear lower bounds (Theorem 12(b), Proposition 14(b)) do not apply to them.

Remark 1 (existence and injectivity: cited, and what is assumed)**.** T^0 is a C^1 solution of $\partial_x T\, f(x, u(t)) + \partial_t T = AT + Bh(x)$ whenever backward solutions exist on $[0, t]$ (Bernard & Andrieu, 2019, Lemma 1), and an observer $\hat{x} = T^*(\hat{z}, t)$ converges if there are $\bar{t}$ and a concave class-$\mathcal{K}_\infty$ ρ_T with $|x_1 - x_2| \leq \rho_T(|T(x_1, t) - T(x_2, t)|)$ for all $t \geq \bar{t}$ (Bernard & Andrieu, 2019, Theorem 1). For a fixed input that is backward distinguishable in finite time, $T^0(\cdot, t)$ is injective for almost every complex diagonal A with $n_y(n_x + 1)$ eigenvalues (Bernard & Andrieu, 2019, Theorem 3), but the exceptional set depends on the input (Bernard & Andrieu, 2019, Remark 5); the Lipschitz, uniform-in-time version (Bernard & Andrieu, 2019, Theorem 2) needs strong differential observability, inputs of class C^{m+1} (their time derivatives enter) and a gain k above a threshold, which excludes held, switching and PRBS inputs and a fixed (A, B). For autonomous systems, $2n_x + 1$ generic real dimensions suffice (Brivadis et al., 2023, Theorem 3.4), $2n_x + 2$ for time-varying ones (Brivadis et al., 2023, Remark 3.6). (A4) is therefore *assumed*: it is the quantitative, uniform-in-input version of what these results give for each regular input. The code's $n_z = 5$ for 2-D plants is the autonomous count; the time-varying count $2n_x + 2$ is not met, so (A4) at $n_z = 5$ is an assumption whose constant must be reported, not a consequence of the cited results; the phase-1 proxy for ℓ (`l_dec_q95`, Remark 25) is 24 (vdp), 44 (bioreactor), 299 (duffing), 10 (lorenz), 14 (quad_tank), 816 (battery) for the C1-where phase 1 (used in `check_apriori_vacuity`); the C2-window-a1 phase 1 of the results layer gives $23, 44, 295, 10, 14, 806$ (Table 3).

Remark 2 (Koopman reading; which model classes the variants are)**.** For the autonomous plant and a diagonalised filter $A = \mathrm{diag}(\lambda_i)$, $B = (b_i)$, the KKL PDE $\partial_x T_i\, f = \lambda_i T_i + b_i h$ reads $(L_f - \lambda_i) T_i = b_i h$ with $L_f = f \cdot \nabla$ the Koopman generator on observables, so

$$T_i = b_i\, (L_f - \lambda_i)^{-1} h = b_i \int_0^\infty e^{\lambda_i \tau}\, h\big(X(x, 0; -\tau; 0)\big)\, d\tau,$$

the resolvent of the generator at the filter pole, i.e. the Laplace transform of the backward Koopman semigroup applied to the output observable; if $h = \sum_j c_j \varphi_j$ with $L_f \varphi_j = \mu_j \varphi_j$ then $T_i = \sum_j b_i c_j \varphi_j / (\mu_j - \lambda_i)$. For a linear plant this is the row identity $p_i = b_i C (F - \lambda_i I)^{-1}$ of the Sylvester solution (check [7]); for a nonlinear plant it gives the spectral reading of the "A fast enough" condition above ($\mathrm{Re}\,\lambda_i$ must clear the backward spectrum) and of the zero-input floor ε_0 of Remark 6(i) as a spectral truncation. Under an input the same object is the resolvent for the skew-product (state $\times$ input-shift) system, and the windowed map T_L is its finite-horizon version with the window as a truncated coordinate on the shift space; for a linear plant the context $\dot{q} = Aq - PGu$ of Proposition 5(a) has the structure of a linear predictor with inputs (Proctor et al., 2016; Korda & Mezić, 2018). In these terms, *latent injection* with a stationary T_0 is a lifted Koopman model with inputs (its input-affine case $\Omega(z)u$, and the NLOX baseline $\omega(z)u$, are bilinear lifted observers with a fixed output-injection gain B), whose stability requires the input-dependent generator $A + \partial_z \Phi$ to stay contracting (Proposition 7(a)); *decoder-side conditioning* keeps a fixed generator and parametrises the left inverse of the lifting by the input context. Proposition 9 is then the statement that the Koopman spectrum moves with u (eigenvalue $u - a$; rotation rate $\omega_0 + u$) while a fixed A does not. We state this as positioning only; no result below depends on it.

A.1 The decoder target and sufficient contexts

Definition 3 (sufficient context)**.** A context map $E : W_L \to \mathbb{R}^{d_c}$ is *sufficient* if the latent and the context determine the state: for $(x_i, w_i) \in \overline{\mathcal{P}_L}$, $E(w_1) = E(w_2) \Rightarrow |x_1 - x_2| \le \ell\, |T_L(x_1, w_1) - T_L(x_2, w_2)|$. It is ϵ_E-sufficient if the same holds with an additive ϵ_E on the right.

The window itself ($E = \mathrm{id}$) is sufficient by (A4). A lower-dimensional E cannot be injective on W_L (invariance of domain), but it need not be: it only has to resolve what the latent leaves open.

Lemma 4 (decoder target)**.** *Let E be continuous and ϵ_E-sufficient, and $S_L := \{(T_L(x,w), E(w)) : (x,w) \in \overline{\mathcal{P}_L}\}$. If $\epsilon_E = 0$ there is a continuous $G^* : S_L \to \mathcal{X}$ with $G^*(T_L(x,w), E(w)) = x$, ℓ-Lipschitz in z for each c; extend $G^*(\cdot, c)$ to $\mathbb{R}^{n_z}$ with the same constant (Kirszbraun). Then along every admissible trajectory, for $t \ge L$, $|G^*(z(t), E(u_t)) - x(t)| \le \frac{\ell}{2}\varepsilon(L)$. For $\epsilon_E > 0$, any decoder D satisfies $\sup |D(T_L(x,w), E(w)) - x| \ge \frac{1}{2}\delta_E$, where δ_E is the largest $|x_1 - x_2|$ over pairs with equal latent and equal context.*

Intuition. Over a fixed window the plant has no clock, so the state is a fixed function of (latent, context) as soon as the context resolves the latent's ambiguity.

Proof. $\Psi(x,w) = (T_L(x,w), E(w))$ is continuous on the compact $\overline{\mathcal{P}_L}$ (under the hold, the backward solution is a composition of N_L C^1 flows with constant parameters). By sufficiency, $\Psi(x_1, w_1) = \Psi(x_2, w_2)$ implies $x_1 = x_2$, so π_x factors as $G^* \circ \Psi$; G^* is continuous because Ψ is a closed map (continuous on a compact set into a Hausdorff space), and ℓ-Lipschitz in z by sufficiency. Then $|G^*(z(t), c) - x(t)| = |G^*(z(t), c) - G^*(T_L(x(t), u_t), c)| \le \ell\, |z(t) - T_L(x(t), u_t)|$ and Lemma 11(a). The lower bound: D returns one value for two states δ_E apart. □

Proposition 5 (sufficient contexts exist, and must carry history)**.** *(a)* Linear plants. *For $\dot{x} = Fx + Gu$, $y = Cx$, the windowed map is affine in x with a w-independent slope, $T_L(x,w) = P_L x + Q_L w$, $P_L = P - e^{AL} P e^{-FL}$, $PF = AP + BC$. Hence the n_z-dimensional linear context $c = Q_L w$ is exactly sufficient whenever P_L is injective ($\ell = \|P_L^+\|$), and as $L \to \infty$ (under the washout $e^{AL} P e^{-FL} \to 0$, (H1) of Theorem 12) it is the output q of the stable filter $\dot{q} = Aq - PGu$: a spectral filter bank with identity readout.* *(b)* Nonlinear plants. *The window ($d_c = n_u N_L$) is sufficient, and by (a) of Lemma 11 any context that determines u_t for some L is $\ell\varepsilon(L)$-sufficient for the infinite past: $|x_1 - x_2| \le \ell\, |T_1^0 - T_2^0| + \ell\, \varepsilon(L)$ (each T_i^0 is within $\frac{1}{2}\varepsilon(L)$ of T_L).* *(c)* History is necessary. *The current input alone is not sufficient in general (Proposition 14).*

Proof. (a) $X(x, 0; \tau; w) = e^{F\tau} x + (\text{linear in } w)$; substitute in (3) and use $\int_{-L}^{0} e^{-A\tau} B C e^{F\tau} d\tau = P - e^{AL} P e^{-FL}$ (integrate $\frac{d}{d\tau}(e^{-A\tau} P e^{F\tau}) = e^{-A\tau} B C e^{F\tau}$). If $Q_L w_1 = Q_L w_2$ then $T_L(x_1, w_1) - T_L(x_2, w_2) = P_L(x_1 - x_2)$. The limit: $z - Px$ obeys $\dot{(z - Px)} = A(z - Px) - PGu$. (b) (A4) and Lemma 11(a). □

For nonlinear plants and contexts of fixed dimension below $n_u N_L$, finite stable filter banks with a static readout uniformly approximate every time-invariant fading-memory operator on slew-limited bounded inputs

(Boyd & Chua, 1985, Theorem 2); applying it to the observer operator $(y,u)\mapsto x$ (a fading-memory operator under (A1)–(A4), on slew-limited signals, and for the filters the theorem chooses rather than ours) motivates the spectral context but is not a guarantee for the trained encoder.

Remark 6 (expressivity at a fixed base; what C1 compares)**.** (i) *Shared floor.* Every conditioner with exact zero-input collapse, $D(\cdot,0)=T_0^+$, errs at $c=0$ by $\varepsilon_0:=\sup_z|G^*(z,0)-T_0^+(z)|$: the zero-input target is the inverse of the zero-input windowed map, not the learned stationary base. (ii) *Additive (concat).* $T_0^+(z)+R(z,c)-R(z,0)$ approximates on S_L, as the width of R grows, every continuous target that equals T_0^+ on the zero slice: SiLU satisfies $\mathrm{SiLU}(s)-\mathrm{SiLU}(-s)=s$, so a two-hidden-layer SiLU network contains every one-hidden-layer one, and the universal approximation theorem for non-polynomial activations (Leshno et al., 1993) applies (with a Tietze extension of the target). (iii) *Base-capped (FiLM, LoRA, hypernetwork).* For each fixed c the decoder is a network $N_{\theta(c)}$ of the frozen base's architecture, so for every conditioner budget

$$\inf_{D}\sup_{S_L}|D-G^*|\;\geq\;\sup_{c}\inf_{\theta\in\Theta}\sup_{z}|N_\theta(z)-G^*(z,c)|,$$

with Θ the reachable parameter set. At the matched budget Θ is much smaller than the base's weight space: the hypernetwork's linear bias-free heads place $\theta(c)$ in an affine subspace of dimension equal to its trunk width; LoRA updates have the configured rank and leave biases fixed; FiLM never modulates the readout layer, so $D(\cdot,c)=W_L\phi_c(z)+b_L$ with (W_L,b_L) frozen. In the ReLU setting, a LoRA rank of order width times a depth ratio suffices to represent a smaller target network exactly (Zeng & Lee, 2024); that result does not cover tanh bases or small fixed ranks. (iv) *Latent injection.* Φ is a universal class, but as an observer it represents only targets of the stationary route: a stationary T_0 with an exact correction. The theory therefore does not rank the variants by expressivity; at a matched budget on one base, C1 measures inductive bias and optimisation within these classes.

A.2 Where the input enters: a stability–extrapolation trade-off

Both observers run the filter on $y+\nu$, $\|\nu\|_\infty\le\delta_y$, with a context $\hat c=E(\hat u)$ of the input $\hat u$ they are given; $c=E(u)$ and $\delta_c:=\sup_t|\hat c-c|\le\ell_E\|\hat u-u\|_\infty$ (Lemma 15).

$$\begin{aligned}&\text{(latent)}\quad \dot{\hat z}=A\hat z+B(y+\nu)+\Phi(\hat z,\hat c), && \hat x=T_0^+(\hat z),\quad \hat z(0)=T_0(\hat x_0),\\ &\text{(decoder side)}\quad \dot{\hat z}=A\hat z+B(y+\nu), && \hat x=D(\hat z,\hat c),\quad \hat z(0)=\hat T_0(\hat x_0).\end{aligned}$$

For the latent variant, $T_0\in C^1$ is a stationary map with left inverse T_0^+ (ℓ_{dec}-Lipschitz, $\varepsilon_{\mathrm{dec}}=\sup_{\mathcal X}|T_0^+T_0(x)-x|$), the residual along trajectories is $r=\partial_xT_0(x)f(x,u)-AT_0(x)-Bh(x)-\Phi(T_0(x),c)$ with $|r|\le\varepsilon_{\mathrm{res}}$, and $\ell_{\Phi,c}$ is the Lipschitz constant of $\Phi(z,\cdot)$ *uniformly over* $z\in T_0(\mathcal X)+B(r)$ (the set used in (a)). $\mu_2(M)=\lambda_{\max}(\frac12(M+M^\top))$ is the logarithmic norm.

Proposition 7 (where the input enters)**.** *Under Assumption 1:*

(a) Latent injection: stability depends on the input *[proven]. Let* $\bar d:=\varepsilon_{\mathrm{res}}+\|B\|\delta_y+\ell_{\Phi,c}\,\delta_c$ *and let* $r\ge|e(0)|+\bar d/\mu$. *If* $\mu_2\big(A+\partial_z\Phi(z,\hat c)\big)\le-\mu<0$ *for all* z *in the set* $\mathcal Z_r:=T_0(\mathcal X)+B(r)$ *and all contexts* $\hat c$ *the observer meets, then* $e=\hat z-T_0(x)$ *stays in* $B(r)$ *for all* t *(so the hypothesis is checked on a set fixed in advance, not on the solution) and*

$$|e(t)|\le e^{-\mu t}|e(0)|+\tfrac1\mu\big(\varepsilon_{\mathrm{res}}+\|B\|\delta_y+\ell_{\Phi,c}\,\delta_c\big),\qquad |\hat x-x|\le\ell_{\mathrm{dec}}|e|+\varepsilon_{\mathrm{dec}}.$$

(b) Decoder side: stability is free, extrapolation is not *[proven]. The latent is an input-free LTI filter, so* $\hat z-z$ *is driven only by* ν *and the initial transient, and* $|\hat x-x|$ *obeys the bound of Proposition 19 for every input, with the decoder's Jacobian norms* $\ell_z(\hat c),\ell_c(\hat c)$ *taken at the contexts actually queried. Both conditioner families have c-uniform bounds on a bounded context set, but with very different constants: for an additive conditioner* $\ell_z(c)\le\ell_z(T_0^+)+2\sup\|\partial_zR\|$*; for multiplicative ones (FiLM, LoRA, hypernetwork) the available bound is the product form* $\ell_z(c)\le\prod_l\|W_l+\Delta W_l(c)\|\sup|\sigma'|^d$*, finite because* $\theta(c)$ *is bounded (tanh trunk, bounded c), but* 10^1*–*10^5 *times larger at random initialisation (check [5]) before the tanh base saturates.*

Intuition. Injecting the input makes the observer a recurrent network whose stability depends on how strongly the injection reacts to the latent, but its decoder never sees an unfamiliar context; conditioning the decoder keeps the recurrent part a fixed stable filter, but the network is queried at every unseen context directly.

Proof. (a) $\dot e = Ae + [\Phi(\hat z,\hat c) - \Phi(T_0x,\hat c)] + d$, $d = B\nu + [\Phi(T_0x,\hat c) - \Phi(T_0x,c)] - r$, $|d| \le \bar d$. While $e \in B(r)$ the segment $[T_0x, \hat z]$ lies in the ball $T_0x + B(r) \subseteq \mathcal{Z}_r$ (convexity of the ball, not of $\mathcal{Z}_r$, is what is used); write the bracket as $\bar J e$, $\bar J = \int_0^1 \partial_z\Phi(T_0x + se, \hat c)ds$, so $\mu_2(A + \bar J) \le -\mu$ by subadditivity and convexity of μ_2, and $D^+|e| \le -\mu|e| + \bar d$. The comparison function $\phi(t) = e^{-\mu t}|e(0)| + \bar d(1 - e^{-\mu t})/\mu \le r$ bounds $|e|$ as long as $e \in B(r)$, and $\phi < r$ whenever $|e| = r$ would be reached, so $B(r)$ is forward invariant and the bound holds for all t (contraction, Lohmiller & Slotine, 1998; equivalently the perturbation lemmas of Khalil, 2002, Ch. 9). The state bound is the triangle inequality through $T_0^+(T_0(x))$. (b) The latent statement is linearity; the bound is Proposition 19; the Jacobian bounds are the chain rule (R's z-Jacobian does not involve c multiplicatively, whereas modulated layers multiply z-features by c-dependent weights), and $\theta(c)$ is bounded because the trunk is tanh and c is bounded (Lemma 15). □

Corollary 8 (the certified margin shrinks with the context)**.** *$\mu_2(A + \partial_z\Phi) \le \mu_2(A) + \ell_\Phi$, and $\mu_2(A) = -\alpha$ for normal A, so $\mu \ge \alpha - \ell_\Phi$ (for general A, the Gronwall version gives $\alpha - \kappa\ell_\Phi$). For an injection with exact zero-input collapse, $\Phi(z,c) = M(z,c) - M(z,0)$, $M \in C^2$: $\ell_\Phi(c) \le \min\big(|c|\, L_{zc}, 2\sup\|\partial_z M\|\big)$ with $L_{zc} = \sup\|\partial_c\partial_z M\|$, and $|c| \le \ell_E\|u\|_\infty$ (sup norm) since $E(0) = 0$; for an input-affine injection $\Omega(z)u$, $\ell_\Phi \le \ell_\Omega\|u\|_\infty$. The margin is evaluated at the* observer's *context $\hat c$, so a corrupted input shrinks it. The margin saturates: if $2\kappa\sup\|\partial_z M\| < \alpha$ the latent observer is certified at every amplitude, and when the plant input saturates (as through a saturating actuator) the exact correction's z-Lipschitz constant stays bounded too. The condition is* sufficient only*: for the code's trained injection the certified margin is already negative at moderate gain (-5.47, log-norm $+8.1$ in check [3]) while two observer copies still contract, so a negative margin predicts nothing, and only a measured incremental rate does.*

Proposition 9 (the margin cannot be removed at a fixed spectrum)**.** *[proven] There are bounded, observable plants with an* exact *injection ($\varepsilon_{\text{res}} = 0$) whose latent observer diverges while a decoder-side observer with a bounded sufficient context is within $\frac{\ell}{2}\varepsilon(L)$ of exact. For $\dot x = (u - a)x + b$, $y = x$, $a > \alpha > 0$, scalar latent $\dot z = -\alpha z + y$ and constant $u \in (\alpha, a)$: the plant converges to $b/(a - u)$, the stationary map is $T_0(x) = Px + q$, $P = 1/(\alpha - a)$, $q = -Pb/\alpha$, the exact injection is $\Phi^*(z,u) = u(z - q)$ ($\ell_\Phi = |u|$), and $\dot e = (u - \alpha)e$. The windowed map is $T_L(x,w) = P_L(u)\,x + Q_L(u)$ with $P_L = \int_{-L}^0 e^{(\alpha+u-a)\tau}d\tau > 0$ and $Q_L = \int_{-L}^0 e^{\alpha\tau}\, b\, \frac{e^{(u-a)\tau}-1}{u-a}\, d\tau$ (both bounded functions of the window input), so the decoder $\hat x = (\hat z - Q_L)/P_L$ errs by at most $\frac{\ell}{2}\varepsilon(L)$ with $\ell = 1/P_L$ (Lemma 4), for every $u \in (\alpha, a)$.*

Proof. $T_0'(x)f(x,0) = P(b - ax)$ and $-\alpha T_0 + x = (1 - \alpha P)x - \alpha q = -aPx + Pb$; so T_0 solves the zero-input PDE and $\partial_x T_0[f(x,u) - f(x,0)] = Pux = u(T_0(x) - q)$. Then $\dot e = -\alpha\hat z + x + u(\hat z - q) - P((u - a)x + b) = (u - \alpha)e$. For the window: $X(x, 0; \tau; u) = e^{(u-a)\tau}x + b(e^{(u-a)\tau} - 1)/(u - a)$ in (3). □

Here the small-gain threshold $u = \alpha$ is attained exactly. The *exact* time-varying map $z = m(t)x + n(t)$, $\dot m = 1 - (\alpha + u - a)m$, $\dot n = -\alpha n - bm$, is also a sufficient context, but for $u < a - \alpha$ its coefficient $-(\alpha + u - a)$ is positive and $m, n \to \infty$ (check: $m(30) = 2 \cdot 10^{19}$ at $u = 0.5$); it is outside the bounded class of Proposition 7(b), which is why the windowed context is the one used. The obstruction is intrinsic to the stationary route *at a fixed spectrum*: a designer removes it by choosing $\alpha > \bar u$ (here) or $\alpha > \bar u\ell_\Omega\kappa$ (Corollary 8), i.e. by a gain that grows with the input bound, which is also what Bernard & Andrieu (2019, Theorem 4) require. A second example, the rotating oscillator $\dot x = (\omega_0 + u)Jx$, $y = x_1$ ($|x|$ is conserved for every input), with the drift map P and the exact correction $\Omega(z) = PJP^+z$, has error rate $\max \mathrm{Re}\,\lambda(A + uPJP^+)$: stable for all tested $u > -\omega_0$, but unstable ($+2.65$, $+5.16$ at $u = -2, -4$, $\omega_0 = 1$, $n_z = 5$, $\alpha = 1.5$) when the rotation reverses, although the plant is bounded and observable there; this is the input-independence required by the stationary route (Bernard & Andrieu, 2019, Theorem 4), which also needs an input-affine plant and a gain growing with $\bar u$. The decoder-side route needs only Assumption 1 and a sufficient context.

What the bounds predict, and what the current nodes test. Under protocol v2 no tier changes the input amplitude (every family lives in the same box), so the quantity that varies between tiers is the

distance in context space, $d(\hat{c}, \mathcal{C}_{\text{train}})$, not $|c|$. (P1) The latent variant's excess error on `shape_ood` and `sequence_ood` is the extrapolation error of Φ at the new contexts, filtered at the measured incremental rate; it degrades sharply only if that rate approaches zero. Decoder-side variants pay $\ell_z(\hat{c})\,|\Delta z|$ and $\ell_c(\hat{c})\,|\Delta c|$ at the new contexts (Corollary 10), with the additive conditioner's ℓ_z small and c-uniform. (P2) Input noise moves the latent variant's operating point to a corrupted context $\hat{c}$ (its margin is evaluated there), whereas decoder-side variants stay stable for every input and pay $\ell_c \ell_E \delta_u$. *Status: untested.* No current node measures μ_2 or the incremental rate on visited segments, $\ell_z(\hat{c})$, $\ell_c(\hat{c})$, or the context distance per tier; under C3-faults `input_noise` every variant, latent included, degrades by about the same factor ($\approx 10\times$ on vdp, read from the C3-faults node report; not a number of the generated results layer), which neither supports nor falsifies (P2). The quantities to record are eval-only on saved checkpoints: $\min_{c' \in \mathcal{C}_{\text{train}}} \|\hat{c} - c'\|$ per tier (C1-where, TEST-C1), the latent margin $\alpha - \kappa \ell_\Phi$ already computed by `metrics.py`, and the realised gains of Definition 24. Measured Lipschitz constants and log-norms are estimates on the visited set (lower bounds of suprema), not certificates.

Corollary 10 (out of distribution: Lipschitz extrapolation, and nothing more)**.** *[proven] Let* $\varepsilon_{\text{train}} := \sup |D(z', E(w')) - x'|$ *over the training support* S_{train} *of triples (label latent* $z' = T^0$*, window* w'*, state* x'*), and* assume *the target* $x = G^*(z, w)$ *of Lemma 4 has, on its domain* S_L*, Lipschitz constants* ℓ *in* z *(this is (A4)) and* ℓ_w *in* w *(sup norm; a new assumption, not implied by sufficiency), and assume its extension beyond* S_L *keeps these constants (a componentwise McShane extension achieves them only up to a factor* $\sqrt{n_x}$*, which then multiplies* ℓ *and* ℓ_w *in (a)). (a) For a query* (z, w) *and any* $(z', w') \in S_{\text{train}}$*, with* D*'s Jacobian norms* ℓ_z^D, ℓ_c^D *on the segment,*

$$|D(z, E(w)) - G^*(z, w)| \le \varepsilon_{\text{train}} + \tfrac{\ell}{2}\varepsilon(L) + (\ell_z^D + \ell)|z - z'| + (\ell_c^D\, \ell_E + \ell_w)\|w - w'\|_\infty,$$

the $\frac{\ell}{2}\varepsilon(L)$ *because* $G^*(z', w') = x'$ *only at* $z' = T_L$*, whereas the training latent is* T^0*; and* $|G^*(z(t), u_t) - x(t)| \le \frac{\ell}{2}\varepsilon(L)$ *(Lemma 4). (b) The linear rate cannot be improved without structure. Among targets that are* 2λ*-Lipschitz and agree with a* λ*-Lipschitz one on* S_{train}*,* $G_\pm = G_0 \pm \lambda\, d(\cdot, S_{\text{train}})\, e$ *(*$|e| = 1$*) both fit the data, so every learner errs by at least* $\lambda\, d(q, S_{\text{train}})$ *at some query* q*. At the plant level, if* φ *is a diffeomorphism of* $\mathcal{X}$ *with* $h \circ \varphi = h$*, equal to the identity on an open neighbourhood of the states reached under the training inputs (so that* $D\varphi = I$ *there), then the plants* f *and* $\varphi_* f$ *produce identical training data, and under an input that drives the state where* $\varphi \ne \mathrm{id}$ *they produce identical* (y, u) *from* x_0 *and* $\varphi(x_0)$*, so any observer errs by at least* $\frac{1}{2}|\varphi(x(t)) - x(t)|$ *on one of them.*

Proof. (a) Triangle inequality through $D(z', E(w'))$, $G^*(z', w')$ and $G^*(T_L(x', w'), w') = x'$, with $|E(w) - E(w')| \le \ell_E \|w - w'\|_\infty$ and $|z' - T_L(x', w')| \le \frac{1}{2}\varepsilon(L)$. (b) $d(\cdot, S)$ is 1-Lipschitz and vanishes on S; $|G_+ - G_-| = 2\lambda d$ and one learner output is at distance $\ge \lambda d$ from one of them. For the plants: φ maps solutions of f to solutions of $\varphi_* f$ and preserves y; on the training set it is the identity. □

What it predicts under protocol v2. Distances are measured in input (window) space so that they are comparable across encoders. In v2 at most 0.1% of evaluation states lie outside the training box on any dev system and tier (C1-where node report; not in the generated results layer), and the input amplitude box is shared by all tiers, so $|z - z'|$ and the state-level construction of (b) are inert: the relevant distance is $\|w - w'\|_\infty$ (equivalently $d(\hat{c}, \mathcal{C}_{\text{train}})$ in context space), which `shape_ood` and `sequence_ood` change through the waveform and the event timing at the same amplitude. The corollary therefore predicts a shape/sequence-OOD excess that is at most linear in that context distance and, by (b), can be no better without structure; it does not predict its size, and the context distance per tier is not yet reported (eval-only; C1-where, TEST-C1). A model-based filter with the correct model extrapolates by structure; a learned observer cannot, beyond this Lipschitz rate.

Zero input. With $E(0) = 0$, $\Delta(\cdot, 0) = 0$ in every decoder conditioner and $\Phi(\cdot, 0) = 0$, every conditioned observer is the autonomous KKL observer at $u \equiv 0$, so the autonomous theory (Andrieu & Praly, 2006, Theorems 2.2, 2.4, 2.7) and the robustness bound of Niazi et al. (2025, Theorem 19) (their cond(V) is our κ) are the special case $c \equiv 0$ of Proposition 19.

Numerical checks. `theory/checks/m1_location.py`: [1] zero-input collapse of all encoders and conditioners (errors $\le 5\cdot 10^{-16}$); [2] at fixed c, FiLM, LoRA and hypernetwork decoders equal base-architecture networks with explicit weights ($\le 3\cdot 10^{-13}$); [3] the injection's z-Lipschitz constant is $0, 0.61, 1.22, 2.33, 4.26, 6.82, 7.29$ at $|c| = 0, 0.25, 0.5, 1, 2, 8, 32$ (linear, then saturating), and for two observer copies the measured incremental rate is below the sup log-norm on the visited segments ($-1.52 \le -1.23$; $-1.05 \le -0.29$, where the small-gain margin is -5.47 and certifies nothing; at the largest gain the log-norm is $+8.1$ and also certifies nothing while the copies still contract, so the log-norm condition is sufficient only); [4] the rotating oscillator; [5] *at random initialisation* (trained decoders are not tested here) the z-Jacobian of concat stays below its uniform-in-c bound (≤ 1.25 against 12 for $|c| \le 256$) while FiLM, LoRA and hypernetwork reach 81, $3.7\cdot 10^5$, $2.3\cdot 10^6$ and then fall (LoRA and hyper to 0 and 2.9 at $|c| = 256$): c-uniformly bounded, with a far larger constant, as stated in Proposition 7(b); [6] the linear windowed map is affine with w-independent slope to 10^{-15}; [7] the Koopman resolvent identity $p_i = b_i C(F - \lambda_i I)^{-1}$ (to 10^{-16}) and its Laplace-transform form (to 10^{-8}); [8] the a-priori form of Proposition 7(a): with $\mu = 0.46$, $|e(0)| = 0.5$, $\sup|d| = 0.3$ the error never leaves $B(r)$, $r = 1.15$. `theory/checks/check_theory.py`: `where_input_enters` (Proposition 9: measured rate $u - \alpha$ to 10^{-3}; windowed-context error equals $\frac{\ell}{2}\varepsilon(L)$ to three digits at $L = 6$, i.e. the bound is attained for a constant input; the exact context's $m(t)$ reaches 10^{11}–10^{19}).

B Why a short input window suffices

Setting. Assumptions (A1)–(A4) of Section A: forward invariance of a compact $\mathcal{X}$, $\|h\| \le \bar{y}$ on $\mathcal{X}$, $\|e^{At}\| \le \kappa e^{-\alpha t}$, and Lipschitz injectivity (modulus ℓ, uniform in $L \ge L_0$) of the windowed map. Here we also use that $-\alpha$ is *exactly* the spectral abscissa of A and that the eigenvalues on $\mathrm{Re}\,\lambda = -\alpha$ are semisimple (otherwise no finite κ exists for that α). The Butterworth real-Jordan A is normal: each block is $sI + \omega J$, so $\|e^{At}\| = e^{-\alpha t}$ and $\kappa = 1$; its real parts are $\alpha_j = \alpha \sin\frac{(2j-1)\pi}{2n_z} / \sin\frac{\pi}{2n_z}$, $j = 1, \dots, \lceil n_z/2 \rceil$, so $\alpha = \alpha_1 \le \alpha_j \le \alpha_{\max} = \alpha / \sin\frac{\pi}{2n_z}$. The *label* latent starts at $z(0) = 0$ (the convention of Section A), hence $z(t) = T^0(x(t), t; u)$ exactly; the deployed observer's initial transient is a separate term (Proposition 19) and plays no role in this section. $T_L(x, w)$ is the windowed map (3) (zero history before the window), and

$$\varepsilon(L) := \frac{2\kappa\|B\|\bar{y}}{\alpha}\, e^{-\alpha L}.$$

Lemma 11 (fading-memory tail)**.** *(a) [proven] Along every admissible trajectory and $t \ge L$, $\|T^0(x(t), t; u) - T_L(x(t), u_t)\| \le \frac{1}{2}\varepsilon(L)$. (b) [proven under its hypothesis] If $u = v$ on $[t - L, t]$ and* both *backward solutions from x stay in $\mathcal{X}$ on $[0, t]$, then*

$$T^0(x, t; u) - T^0(x, t; v) = e^{AL}\big[T^0(\xi, t - L; u) - T^0(\xi, t - L; v)\big], \qquad \|T^0(x, t; u) - T^0(x, t; v)\| \le \varepsilon(L),$$

with ξ the common backward state at $t - L$. In particular the map forgets at exactly rate α: for a unit left eigenvector p of an eigenvalue with real part $-\alpha$, $|p^*[T^0(x, t; u) - T^0(x, t; v)]| = e^{-\alpha L}\,|p^*[T^0(\xi, t - L; u) - T^0(\xi, t - L; v)]|$.

Intuition. Inside the window the two backward solutions are one curve; everything older reaches the latent through e^{AL}.

Proof. (a) $T^0 - T_L$ is the integral over $[0, t - L]$ of $e^{A(t-s)}Bh(x(s))$ along the true past, which lies in $\mathcal{X}$ by (A1); bound the integrand by $\kappa\|B\|\bar{y}e^{-\alpha(t-s)}$. (b) On $[t - L, t]$ both backward solutions solve the same ODE from x, so they coincide; split (3)-type integrals at $t - L$ and use the flow property: the window parts cancel and the rest is $e^{AL}T^0(\xi, t - L; \cdot)$. Bound each map's remainder as in (a). Finally $p^*e^{AL} = e^{\lambda L}p^*$, $|e^{\lambda L}| = e^{-\alpha L}$. □

No Lipschitz property of f in u is used; the plant enters only through $\bar{y}$. *Where (b) does not apply.* Part (a) uses only the true past and (A1), so it holds for every benchmark. Part (b) needs the *counterfactual* backward solution (from x under v) to stay in $\mathcal{X}$; for a dissipative chaotic flow this fails generically, because backward time expands at the rate λ_b of the most contracting forward direction (Lorenz: $\lambda_b \approx 14.6$, so a counterfactual past leaves any compact set within a fraction of a second). For Lorenz and Rössler, (b) and

everything that rests on it (Proposition 5(b)'s $\ell\varepsilon(L)$-sufficiency of a context that determines the window; the lower bounds below) are not available; the sufficient direction (a) is. The zero-order hold changes nothing: if the held samples agree for $j = k-M, \dots, k-1$, apply the lemma at t_k with $L = M\,dt$ (the sample u_k has not acted yet); the deployed discrete filter has the same bounds with $e^{-\alpha L} \mapsto e^{-\alpha M dt}$ and constants no larger than the continuous ones, plus e_{disc} (Lemma 26).

Theorem 12 (window rule for the observer)**.** *Let $L^*(\eta)$ be the shortest window for which some decoder $D(z, u_t)$ has worst-case error at most η.*

*(a) (*sufficient; every plant under (A1)–(A4)*) [proven] $D = T_L^*$ achieves $\frac{\ell}{2}\varepsilon(L)$, so*

$$L^*(\eta) \;\le\; \frac{1}{\alpha}\ln\frac{\kappa\ell\|B\|\bar{y}}{\alpha\eta} \;=\; \frac{\ln(1/\eta) + \ln\ell + \ln(\kappa\|B\|\bar{y}/\alpha)}{\alpha}.$$

*(b) (*necessary; proven for linear plants*) For every decoder and every plant,* $\sup\|D - x\| \ge \frac{1}{2}\delta(L)$*, where the fibre diameter $\delta(L)$ is the largest $\|x - x'\|$ over* admissible *pairs with $T^0(x, t; u) = T^0(x', t; v)$ and $u = v$ on $[t-L, t]$ (each state reached at t under its own past). For a linear plant $\dot{x} = Fx + Gu$, $y = Cx$, normal A, (A, PG) controllable, symmetric inputs $|u| \le \bar{u}$, under*

(H1) washout*: $e^{At}Pe^{-Ft} \to 0$, so that after a transient $z = Px + q$ with $PF = AP + BC$, P injective, $q(t) = -\int_{-\infty}^{t} e^{A(t-r)}PGu(r)dr$ (this holds iff α exceeds the fastest* contraction *rate of F; automatically for neutrally stable F);*

(H2) overlapping reachable sets*: for any two admissible input histories and every d with $\|d\| \le 2\kappa R_q e^{-\alpha L}/\sigma_{\min}(P)$ there is a state x reachable at t under the first history with $x + d$ reachable under the second. (Every $x \in \mathcal{X}$ being reachable under every history is* not *required and is false under (A1) whenever the forced response is nonzero: the reachable set under one history is the translate $e^{Ft}\mathcal{X}_0 +$ (forced response), a strict subset of $\mathcal{X}$. (H2) holds for F skew-symmetric or $F = 0$ when $\mathcal{X}_0$ contains a ball whose radius exceeds the forced-response bound plus $\|d\|/2$, as in the oscillator and integrator examples; it fails for contracting F, where the reachable set under a fixed history shrinks to a point and the fibre, hence the necessary window, is* smaller*);*

the fibre diameter satisfies

$$\frac{r_q}{\|P\|}\,e^{-\alpha_k L} \;\le\; \delta(L) \;\le\; \frac{2\kappa R_q}{\sigma_{\min}(P)}\,e^{-\alpha L},$$

where $R_q = \kappa\|PG\|\bar{u}/\alpha$, $r_q > 0$ is the radius of a ball contained in the set Q of reachable q, and α_k is the real part of the first mode group such that the slowest groups up to it span a dimension $\ge n_z - n_x + 1$. Hence $\alpha \le \alpha_k \le \alpha_{\max}$ and

$$\frac{1}{\alpha_k}\ln\frac{r_q}{2\|P\|\eta} \;\le\; L^*(\eta) \;\le\; \frac{1}{\alpha}\ln\frac{\kappa\ell\|B\|\bar{y}}{\alpha\eta}, \qquad L^*(\eta) = \Theta\Big(\frac{1}{\alpha}\ln\frac{1}{\eta}\Big) \;\; \textit{(linear plants)},$$

with constants that depend on n_z only through $\alpha_k/\alpha \le 1/\sin\frac{\pi}{2n_z}$ ($\alpha_k/\alpha = 2.618$ at $n_z = 5$, $n_x = 2$; 4.049 at $n_z = 7$, $n_x = 3$).

Intuition. The decoder cannot recover what the window does not contain, and what it does not contain reaches the latent through e^{AL}; a decoder may lean on the faster modes, but never faster than $\alpha_{\max}$.

Proof. (a) Lemma 4 with Lemma 11(a). (b) Two points of the fibre receive the same input (z, u_t), so D errs by $\ge \frac{1}{2}\|x - x'\|$ on one of them. Linear case: by (H1) the window parts of q coincide, so $Px + q = Px' + q'$ iff $P(x - x') = e^{AL}(m' - m)$ with $m, m' \in Q$ the history terms at $t - L$; every such $d = x' - x$ obeys the upper bound below, so by (H2) the pair can be realised by admissible states, and $\delta(L) = \sup\{\|d\| : e^{-AL}Pd \in Q - Q\}$. Upper: $\|d\| \le \|Pd\|/\sigma_{\min}(P) \le \|e^{AL}\|\,2R_q/\sigma_{\min}(P)$. Lower: $\dim\operatorname{range} P = n_x$ and the slow span S_k has dimension $\ge n_z - n_x + 1$, so some unit d has $Pd \in S_k$; for normal A, $\|e^{-AL}Pd\| \le e^{\alpha_k L}\|P\|$, and $Q - Q \supseteq Q \supseteq B(r_q)$ ($0 \in Q$, controllability), so $d\,r_q e^{-\alpha_k L}/\|P\|$ is feasible. □

What is proven and what is not. The two-sided $\Theta(\ln(1/\eta)/\alpha)$ is a statement about linear plants (with the hidden factor α_k/α in the lower constant at $n_z = 5$). For nonlinear plants only (a) is proven; "(b) with rate between α and $\alpha_{\max}$" is a **conjecture**, and for the chaotic benchmarks even the counterfactual-past step is unavailable (Lemma 11). *The fibre is a needle.* Writing $Q = \{\int_0^\infty e^{As} PGu\, ds : |u| \le \bar{u}\}$, a symmetric convex body with support function $h_Q(w) = \bar{u} \int_0^\infty |w^\top e^{As} PG|\, ds$, the fibre diameter is computed exactly as $\delta(L) = \max_{|d|=1} 2/g_Q(e^{-AL}Pd)$ with g_Q the gauge of Q (a linear programme on a Riemann discretisation of Q). For the oscillator with $n_x = 2$ the maximiser is the single direction d with $Pd \in S_k$: $\delta(L)$ decays at rate 1.000α ($n_z = 3$) and 2.617α ($n_z = 5$; $\alpha_k = 2.618\alpha$), every other direction decays at $\approx \alpha_{\max}$, and the angular half-width of the needle shrinks from 3° to 1° over $L = 2$–6 (Table 4). An RMS error curve averages over directions, so a sweep's knee on a log scale is expected between the slopes $-\alpha_k$ and $-\alpha_{\max}$, with the slow needle visible only at the largest L. *Tested by* C2-window-a1 (six dev systems, one bandwidth; pre-registered rule "$e(L \ge \ln(1/\eta)/\alpha)$ within 5% of $e(512)$ on $\ge 5/6$": **FAIL**, `id` 2/5, `sequence_ood` 3/5) and C2-window-a2 (battery, duffing, vdp at 2α, the only test of the $1/\alpha$ scaling; rule "knee in samples changes by $\le 2\times$ on $\ge 2/3$": **FAIL**, 1/2). The ℓ-free rule that was pre-registered is the sufficient bound with ℓ dropped; the corollary below restores ℓ and is *post hoc* consistent with the largest failure (battery), not with lorenz (Table 3). Source: `paper/generated/RESULTS.md`.

Corollary 13 (practical window rule, and why one plant needs more)**.** *[proven: (a) of the theorem, rewritten; its use as an explanation of the sweep is* post hoc*, not pre-registered] For a target relative error η the window that is* sufficient *is*

$$L^*(\eta) \;\approx\; \frac{\ln(1/\eta) + \ln \ell}{\alpha} \qquad \textit{(plus the design constant } \ln(\kappa \|B\| \bar{y}/\alpha)/\alpha\textit{)},$$

where ℓ is the Lipschitz constant of the inverse of the windowed map (A4). The $\ln(1/\eta)/\alpha$ part is plant-independent; the plant enters through $\ln \ell$ and, in the design constant, through $\bar{y}$. The lower bound of the theorem (linear plants) can only shorten *the necessary window (rate $\alpha_k \ge \alpha$), so a plant whose sufficient window exceeds $\ln(1/\eta)/\alpha$ does so through ℓ (or $\bar{y}$), not through the fading-memory rate.*

Integrator-type states (coulomb counting; a heuristic reading, not a proven statement). If a state enters the output only weakly, as the battery's state of charge enters the terminal voltage through the open-circuit-voltage slope $\partial V_{\mathrm{oc}}/\partial \mathrm{SOC}$, then the latent carries that state at gain $\propto$ slope and inverting it costs $\ell \propto 1/\text{slope}$: large ℓ, long L^*. The state itself is the integral of the input (coulomb counting), so a decoder must integrate the window, which is possible only over the window's length; the memory needed is $\ln \ell/\alpha$ longer than for a well-observed state. This is the sufficient side of the theorem; the digest's earlier attribution of the battery to "the Lemma 1′ worst case" is withdrawn: Lemma 11(b) is plant-independent (rate exactly α for every plant) and the lower bound at $n_z - n_x + 1 = n_z$ (scalar integrator) gives $\alpha_k = \alpha_{\max}$, the *shortest* necessary window. *Proxy for ℓ.* No node measures ℓ itself; the phase-1 diagnostic `l_dec_q95` (random-direction quotient of the *stationary* base decoder, $\approx \|J\|_F/\sqrt{n_z}$, Remark 25) is used as its proxy in Table 3. It is an estimate on data of a related constant, not the constant of the bound.

Table 3: Post-hoc reading of the C2-window-a1 sweep (lora, `sequence_ood`, NRMSE at $L \in \{1, \dots, 512\}$ samples, $\eta = 0.05$). $L_0 = \ln(1/\eta)/(\alpha dt)$ is the pre-registered ℓ-free rule and $L^* = (\ln(1/\eta) + \ln \ell)/(\alpha dt)$ the rule of Corollary 13 with $\ell =$ `l_dec_q95` of the node's own phase 1 (both in samples; the design constant is omitted; not pre-registered). "ratio" is $\max_{L \ge L_\cdot} e(L)/e(512)$ on the dyadic grid; the node's criterion is ≤ 1.05; n.t. = not testable (no grid window in $[L_\cdot, 512)$, so the ratio is $e(512)/e(512)$ and says nothing; the results layer leaves these out). ℓ and L^* are the results-layer values. Monotone: no window worse than a shorter one by $> 10\%$.

system	n_z	α	αdt	ℓ	L_0	L^*	ratio, $L \ge L_0$	ratio, $L \ge L^*$	monotone
battery	7	1.5	0.0223	806	135	435	1.73 (fails)	n.t.	yes
bioreactor	5	0.5	0.0309	44	97	219	1.00	1.00	no (min at 128; 2.4× worse at 512)
duffing	5	3.0	0.0309	295	97	281	1.00	n.t.	yes
lorenz	7	2.0	0.0223	10	135	239	1.07 (fails)	1.07 (fails)	yes
quad_tank	6	0.1	0.0050	14	599	1121	n.t.	n.t.	yes (falling at 512)
vdp	5	1.5	0.0309	23	97	199	1.00	1.00	yes

Reading the table (post hoc). The pre-registered ℓ-free rule fails where ℓ is largest (battery, $1.73\times$ the $L = 512$ error at $L = 256$) and marginally on lorenz (1.07). With $\ln \ell$ restored the rule can be evaluated only on bioreactor, lorenz and vdp: on battery, duffing and quad_tank the refined L^* leaves no grid window below

512, so the refined rule is *untested* there (it does not "hold"); it holds on bioreactor and vdp and still fails by 7% on lorenz (2/3 on `id` and 2/3 on `sequence_ood` in the results layer). For battery the refined rule only says that its failure at $L \geq L_0$ is not in contradiction with the theory. This is what a sufficient-side statement can do: the measured knee sits at or below it (duffing's knee is at 32 samples against $L^* = 281$; the bound is far from tight there), and it does not predict the knee. Bioreactor is non-monotone (longer windows are *worse*: an optimisation or over-fitting effect that no truncation bound describes) and quad_tank's fixed $dt = 0.05$ puts both rules beyond the grid, as predicted below. The refined rule is offered as the theory's explanation of why the pre-registered test failed, not as a passed test; a pre-registered test of it needs a new node. The $L = 1$ row of the same sweep (current input only, Proposition 14(a)) is +143% to +1522% worse than $L = 512$ on every dev system (`sequence_ood`, lora; paired changes from the C2-window-a1 node report, not a number of the generated results layer).

Proposition 14 (the current input is not enough)**.** *(a) [proven] Under the zero-order hold, $T^0(\cdot, t_k; u)$ does not depend on u_k. (b) [proven, linear SISO plants] For the linear plant of Theorem 12 with $n_y = n_u = 1$ under (H1)–(H2), but without assuming (A, PG) controllable, (z, u_t) determines x iff* range $P \cap \langle A \,|\, PG\rangle = \{0\}$*, for every $L \geq 0$. If (A, B) is controllable, the eigenvalues of A are distinct and none is a transmission zero of $H(s) = C(sI - F)^{-1}G$, then $\langle A|PG\rangle = \mathbb{R}^{n_z}$ and the condition fails* for every *n_z: no latent dimension makes the latent store the history. (For $n_y > 1$ the condition $p_i^* BH(\lambda_i) \neq 0$ is stronger than "not a transmission zero".) (c) [proven, local] Parametrise the input before $t - L$ by $\theta \in \mathbb{R}^p$ (e.g. held levels in the interior of U), keep the window fixed, and let $\Psi(x, \theta)$ be the resulting latent at t. If Ψ is C^1 near $(\bar{x}, \bar{\theta})$ with trajectories in the interior of $\mathcal{X}$ and* rank $\partial_\theta \Psi(\bar{x}, \bar{\theta}) = n_z$*, then every x in a neighbourhood of $\bar{x}$ has a past $\theta(x)$ with the same latent and the same window; any decoder $D(z, u_t)$ errs by at least half that neighbourhood's radius.*

Intuition. With a time-varying map the past input is part of what must be inverted, and the latent alone cannot hold it.

Proof. (a) The backward solution on $(-\infty, t_k]$ uses u on $[t_j, t_{j+1})$, $j \leq k - 1$, only. (b) By Theorem 12(b), ambiguity exists iff $Pd = e^{AL}\Delta m$ for some $d \neq 0$ and admissible Δm; the reachable differences form a neighbourhood of 0 in $\langle A|PG\rangle$, which e^{AL} maps onto itself. For distinct eigenvalues, controllability of (A, PG) is $p_i^* PG \neq 0$ for every left eigenvector, and $p_i^* PG = -p_i^* B\, H(\lambda_i)$ (from $p_i^* P(F - \lambda_i I) = p_i^* BC$). (c) Implicit function theorem: choose n_z components of θ with invertible $\partial\Psi$-block and solve $\Psi(x, \theta) = \Psi(\bar{x}, \bar{\theta})$ for them as a C^1 function of x. □

The rank condition in (c) is checkable on a simulator: for Van der Pol with the saturating actuator, twelve held levels before the window give rank $\partial_\theta \Psi = 5 = n_z$ (smallest singular value $1.6 \cdot 10^{-4}$: the check shows existence of the ambiguity, not its size, which is $\sigma_{\min}\times$ the admissible range of θ). Example: the oscillator $\dot{x}_1 = \omega_0 x_2$, $\dot{x}_2 = -\omega_0 x_1 + u$, $y = x_1$, $n_z = n_x = 2$, with $v \equiv 0$, $u = a\mathbf{1}_{(-\infty, t-L)}$: x and $x' = x - P^{-1}e^{AL}A^{-1}PGa$ share latent, window and current input. On the stationary route the current input does suffice (the injection $\Phi(z, u)$ depends on $u(t)$ only; Section A); the price there is a small-gain margin (Proposition 7). *Tested by* the $L = 1$ row of C2-window-a1 (above). No current node compares the context encoders (current, window, GRU, SFM) at a matched budget; C1-ctx only moves the SFM's parameters into the budget and does not test this proposition.

B.1 Predictions for the window sweeps

With $dt = 0.1 \sin(\pi/2n_z)/\alpha$, $\alpha\, dt$ does not depend on the bandwidth scale s, and a window of L samples covers $(L - 1)\, dt$ seconds of relevant history (Proposition 14a).

1. The knee in seconds scales as $1/(s\alpha)$; in *samples* it is the same for all s (vdp, duffing, battery in C2-window-a2). One e-fold of the rate-α bound is $1/(\alpha dt) = 32.4$ samples ($n_z = 5$) or 44.9 ($n_z = 7$); at rate α_k it is α_k/α times fewer (2.6 at $n_z = 5$, $n_x = 2$; 4.0 for lorenz, $n_z = 7$, $n_x = 3$; the `a2/a` column of `c2_predictions` is the second-slowest mode group, which equals α_k only when $n_z - n_x + 1 \leq 4$). The worst-case window of Theorem 12(a) at $\eta = 10^{-3} z_{\text{scale}}$ (ℓ omitted) is $\approx 311, 313, 320, 467$ samples; measured tails are 10^2–$5 \cdot 10^2$ times below the bound, i.e. the bound overstates the window by $\ln(485)/(\alpha\, dt) \approx 200$ samples. At one bandwidth (C2-window-a1) the knees within 5% of $e(512)$

are 32 (vdp, `id` and `sequence_ood`; duffing and bioreactor on `sequence_ood`), 64 (bioreactor `id`), 128 (quad_tank `id`) and not reached on the grid for battery and lorenz (both tiers), duffing (`id`) and quad_tank (`sequence_ood`); see Table 3. At 2α (C2-window-a2) the knee in samples moved by $\leq 2\times$ on only 1/2 evaluable systems: the $1/\alpha$ scaling is *not* supported by the current data.

2. quad_tank has a fixed $dt = 0.05$: $\alpha dt = 0.0025, 0.005, 0.01$ for $s = \frac{1}{2}, 1, 2$; the knee in samples scales as $1/s$ and the grid ($L \leq 512$) spans only $1.3, 2.6, 5.1$ e-folds, so the curve should still fall at $L = 512$ for $s \leq 1$ (it does at $s = 1$): this plant cannot test the scaling.

3. On a dyadic grid "knee$\times\alpha$ varies by less than $2\times$" holds only if the knee sits at the same grid point for every s (neighbours differ by exactly 2); use "$\leq 2\times$" or a knee interpolated in $\log L$.

B.2 Context encoders: forgetting by construction

Lemma 15 (SFM context)**.** *[proven] Let $\xi_{k+1} = A_d^{\xi}\xi_k + B_d u_k/u_{\mathrm{ref}}$, $c_k = \psi([\xi_k, u_k/u_{\mathrm{ref}}])$, with A^{ξ} block-normal, $s_b = -\mathrm{softplus}(\cdot) < 0$, $\alpha_\xi = \min_b |s_b|$, and ψ ℓ_ψ-Lipschitz. Then for $l \geq 1$*

$$(a)\ \Big\|\frac{\partial c_k}{\partial u_{k-l}}\Big\| \leq \frac{\ell_\psi}{u_{\mathrm{ref}}}\|B_d\|e^{-\alpha_\xi(l-1)dt}, \qquad (b)\ \ell_E \leq \frac{\ell_\psi}{u_{\mathrm{ref}}}\Big(1 + \frac{\|B_d\|}{1-e^{-\alpha_\xi dt}}\Big),$$

*and the per-lag bound is attained (ψ isometric on the slowest block, B_d in that block). (c) (*the memory probe*) If two input records agree on the last L samples and differ arbitrarily before, with $|u| \leq \bar{u}$,*

$$\|c_k - c_k'\| \leq \frac{\ell_\psi\|B_d\|\,2\bar{u}}{u_{\mathrm{ref}}}\,\frac{e^{-\alpha_\xi L\,dt}}{1-e^{-\alpha_\xi dt}}.$$

Proof. $\xi_k = \sum_{l\geq 1}(A_d^{\xi})^{l-1}B_d u_{k-l}/u_{\mathrm{ref}}$, $\|(A_d^{\xi})^{l-1}\| = e^{-\alpha_\xi(l-1)dt}$; chain rule. For ℓ_E add the direct path u_k and sum the geometric series. (c) Only lags $l \geq L+1$ differ; sum (a) over them. □

The rate exists for every parameter value but is not bounded below uniformly (softplus$\to 0$); a floor $s_b \leq -\alpha_{\min}$ makes it uniform (the C2-floor nodes set $\alpha_\xi \geq c\,\alpha$, $c \in \{0.25, 0.5, 1\}$).

What the probe measures. The C1-where and C2-floor memory probes keep the last L inputs, swap everything older, and fit a rate to the decoded change $|\Delta\hat{x}| \leq \ell_c\|c - c'\|$. This is the quantity of (c): a test of the *encoder's* memory, not of Lemma 11 (the latent z is unchanged by the probe). Since (c) is an upper envelope, a fitted rate below α_ξ over a finite L range does not contradict it unless the curve is in its exponential regime over several decades; the probe tests the rate only there.

At initialisation the SFM's fill-in poles are $\alpha/1.5^b$, $b = 1, \ldots, 5$, so its slowest mode is $\alpha/1.5^5 = 0.132\alpha$ and the next 0.198α: at initialisation the SFM forgets $\approx 7.6\times$ more slowly than the latent. *Measured (un-floored, C1-where, lora, median over seeds; `paper/generated/RESULTS.md`):* the learned SFM's slowest pole is 0.14–0.18 α (battery 0.18, bioreactor 0.14, duffing 0.14, lorenz 0.18, quad_tank 0.17, vdp 0.15), i.e. between the two slowest *initial* poles: training moves the slowest pole only slightly (by at most $\approx 40\%$, never past the second initial pole), so the "5–7$\times$ extra memory" of the un-floored SFM is largely inherited from initialisation rather than learned. *The floor intervention (C2-floor-025/05/1) fails its pre-registered rule.* At 0.5α: no significant loss on 4/6 dev systems (rule: $\geq 5/6$; battery +23–36% and bioreactor lose), TOST non-inferiority at $\pm 10\%$ on 0/6, and the strict probe check "fitted rate $\geq$ floor" on 0/6 (ratios 0.73–0.96: battery 0.78, bioreactor 0.77, duffing 0.73, quad_tank 0.96, vdp 0.82; lorenz -0.02, flat). At 0.25α: no significant loss on 6/6 but TOST 3/6 and probe 1/6; at 1α: 4/6, 0/6, 0/6 (battery $\approx 2\times$ worse). So the data are *consistent with* the memory being unused on four systems but do not establish it (three seeds; see the TOST counts), and battery, the large-ℓ plant of Corollary 13, is the one system where a floor at 0.5α clearly costs accuracy, as the corollary says it should (its sufficient window $\ln(1/\eta)/\alpha + \ln\ell/\alpha$ is attenuated by $e^{-0.5\alpha L^*}$ under the floor); the bioreactor loss has no such explanation. The probe ratios below one do not contradict Lemma 15(c), which is an upper envelope, but they mean the pre-registered check of its *rate* fails on every system, and the flat lorenz curve ($d(L)$ within $\times 2$–3 over $L \leq 5.7$ s) is unexplained: a probe outside its exponential regime, to be reported as such. For the window prediction the SFM has no truncation loss on the

latent's time scale (it should match the window encoder's plateau beyond the knee, not its knee), but an input glitch persists 7.6× longer in c and ℓ_E is correspondingly larger ($\|B_d\|/(1-e^{-\alpha_\xi dt}) \approx 7.6/(\alpha dt) \approx 246\|B_d\|$ for $n_z = 5$).

Remark 16 (SFM is an exact context for linear plants)**.** If $\mathrm{spec}(A^\xi) \supseteq \mathrm{spec}(A)$ with nonzero input weights (true at initialisation), then $q(t_k) = -\sum_{i<k} A_d^{k-1-i} A^{-1}(A_d - I)PGu_i$ is a fixed linear function of ξ_k (each 2×2 block commutes with J): no truncation, whereas a window of L samples truncates at $e^{-\alpha(L-1)dt}$. Once the poles are trained this holds only approximately.

Remark 17 (a GRU has no such guarantee)**.** (1) An update-gate bias b gives $h_k = \sigma(b)h_{k-1} + \ldots$, forgetting at rate $-\ln\sigma(b) \to 0$ as $b \to \infty$. (2) No fading memory in the sense of Boyd & Chua (1985): with reset gate $\equiv 1$ and update gate $\equiv 0$, $h_{k+1} = \tanh(3h_k + 2u_k)$ is bistable; one pulse flips the state and the context difference stays 1.99 for all later k, although the two inputs agree after the pulse. Contractivity of recurrent networks needs weight constraints (Miller & Hardt, 2019); the SFM has it by construction. *Tested by* no current node (an encoder ablation SFM vs GRU at a matched budget is requested; C1-ctx does not contain a GRU variant).

Table 4: Numerical checks of this section (CPU).

result	check	outcome
Lemma 11	integrator; oscillator $n_z = 5$; Van der Pol	rate $1.5000 = \alpha$; $1.0000 = \alpha$ (both); bound loose 135–485×
Thm. 12(b)	$\delta(L)$ by gauge/LP, oscillator $n_x = 2$	rate $1.000, 2.617$ at $n_z = 3, 5$ ($\alpha_k = 1, 2.618$); needle at predicted angle
	bounds $r_q e^{-\alpha_k L}/\|P\| \le \delta \le 2R_q e^{-\alpha L}/\sigma_{\min}$	hold at every L (slack 21×, 1.5–106×)
	$\sigma_{\min}(e^{-AL}P)$ (arithmetic)	same rates $1.000, 2.618$
Cor. 13	Table 3 from the C2-window-a1 report	ℓ-free rule fails 1.73× (battery); with $\ln\ell$: holds on 2/3 testable systems (lorenz 1.07), untestable on battery, duffing, quad_tank
Prop. 14(c)	Van der Pol, 12 past levels	$\mathrm{rank}\,\partial_\theta\Psi = 5$
sampled form	ZOH step identity; discrete constant	exact to 10^{-15}; $c_d = 0.985 \le 1/\alpha$
Lemma 15(a,b)	tight SFM; random SFM	ratio 1.0000000000; ≤ 0.13
Lemma 15(c)	history swap, floor 0 and 0.5; initial poles	observed/bound ≤ 0.01; $\alpha_\xi = \alpha/1.5^5$ at init, $\ge 0.5\alpha$ with floor
GRU remark	latch in the code's GRU encoder	difference 1.990 at $k = 100, 1000, 4999$

C Error bound for the deployed observer

Setting. Plant $\dot{x} = f(x,u)$, $y = h(x)$, input held on $[t_k, t_{k+1})$, $t_k = k\,dt$. The decoder-side observer, exactly as deployed, is

$$\hat{z}_{k+1} = A_d\hat{z}_k + W_0\tilde{y}_k + W_1\tilde{y}_{k+1}, \qquad \hat{x}_k = \hat{T}^+(\hat{z}_k;\hat{c}_k), \qquad \hat{c}_k = E(\tilde{u}_0,\ldots,\tilde{u}_k), \tag{4}$$

with A_d, W_0, W_1 the exact step of Lemma 26(a), the context held between samples, and $\hat{z}_0$ a fixed function of the prior $\hat{x}_0$. Its response to a sensor sequence (v_k) is the *latent noise filter*, the sampled form of $(sI - A)^{-1}B$:

$$H_0 = W_1, \qquad H_m = A_d^{m-1}W_0 + A_d^m W_1 \ (m \ge 1), \qquad \gamma_y := \sum_{m\ge 0}\|H_m\| \le \frac{\kappa\|B\|}{\alpha}, \qquad \|H\|_2^2 := \sum_{m\ge 0}\|H_m\|_F^2, \tag{5}$$

where the inequality holds at every dt (Lemma 26(c)).

Reference condition. Every bound is stated relative to the *nominal* deployed observer $\hat{x}_k^{\mathrm{nom}}$: the same episode, the nominal sensor model (the one used in training, `data.sensor`), the true input record and the nominal prior. Deviations are $v_k := \tilde{y}_k - \tilde{y}_k^{\mathrm{nom}}$ (sensor faults, extra noise), $\hat{c}_k - c_k^{\mathrm{nom}}$ (input corruption) and $\hat{z}_0 - \hat{z}_0^{\mathrm{nom}}$ (initialisation). This is the condition the decoder was trained in: phase 2 regresses x_k on the deployed latent of the nominal-noise record, so its validation error measures the nominal error directly, and no noise-free quantity is needed.

Assumptions. (A1)–(A4) and the regularity clause ($f(\cdot,u)\in C^1$, $h\in C^2$, hence $M_2:=\sup\|\ddot y\|$, $L_y:=\sup\|\dot y\|$ on each hold interval) of Assumption 1, and

(L) $\hat T^+\in C^1$, and the context encoder satisfies $\|E(\tilde u_{0:k})-E(u_{0:k})\|\le \ell_E\sup_{j\le k}\|\tilde u_j-u_j\|$ (for SFM, $\ell_E\le(\ell_\psi/u_{\mathrm{ref}})(1+\|B_d\|/(1-e^{-\alpha_\xi dt}))$, Lemma 15).

Write ℓ_z,ℓ_c for the sup of $\|\partial_z\hat T^+\|$, $\|\partial_c\hat T^+\|$ over the segments a given episode traverses (defined in Proposition 19), and $\ell_z^{\mathrm g},\ell_c^{\mathrm g}$ for the global constants (finite for tanh/SiLU networks on a bounded context set, for every conditioner; for FiLM, LoRA and hyper the available bound is a product of layer norms that is orders of magnitude larger, Proposition 7(b)). The latent deviations are linear, input-free and decoder-independent, so they are fixed before any Lipschitz constant is used.

Definition 18 (nominal error, held-out quantile)**.** For an episode $\omega\sim\mathcal D$ (initial state and input path from the training episode distribution) let $e_{\mathrm{nom}}(\omega):=\sup_{k\in S}\|\hat x_k^{\mathrm{nom}}-x(t_k)\|$, S the steady window, and let ε^β be its $(1-\beta)$-quantile. For n i.i.d. held-out episodes, $\hat\varepsilon:=\max_i e_{\mathrm{nom}}(\omega_i)$ satisfies

$$\mathbb P\big(\hat\varepsilon\ge\varepsilon^\beta\big)\ge 1-(1-\beta)^n\qquad(n=64,\ \beta=0.05:\ 0.96),\tag{6}$$

because all n draws fall below the quantile with probability at most $(1-\beta)^n$.

ε^β contains everything the nominal observer gets wrong: approximation, window truncation (by Lemma 11(a) and Theorem 12(a) at most $\frac{\ell}{2}\varepsilon(L)$ for a sufficient context), nominal noise and the discretisation (Lemma 26(b) only relates the deployed latent to the continuous map, $\|\bar z_k-z(t_k)\|\le e_{\mathrm{disc}}=O(dt^2)$). It is a quantile with distribution-free confidence, not a supremum, and only for $\mathcal D$. It is defined on the *true* plant's episodes (phase 2 regresses true-plant state labels on the deployed latent): no model of f enters it, which is the point of Proposition 23.

Proposition 19 (decoder-side observer: ISS relative to the nominal observer)**.** *[(i), (ii), (iv) proven and exact; (iii) proven as a decomposition, see the paragraph after the proof] Assume (A1)–(A4), (L). Let* $\Delta z_k:=\hat z_k-\hat z_k^{\mathrm{nom}}$, $\Delta c_k:=\hat c_k-c_k$.

(i) (latent deviation, exact) $\Delta z_k=A_d^k\Delta z_0+\sum_{i\le k}G_i^{(k)}v_i$ *with* $G_i^{(k)}=H_{k-i}$ *($i\ge1$),* $G_0^{(k)}=A_d^{k-1}W_0$*; hence* $\|\Delta z_k\|\le\kappa e^{-\alpha t_k}\|\Delta z_0\|+\frac{\kappa\|B\|}{\alpha}\sup_{i\le k}\|v_i\|$ *for every k, and* $\limsup_k\|\Delta z_k\|\le\gamma_y\limsup_k\|v_k\|$.

(ii) (path identity, exact)

$$\hat x_k-x(t_k)=\big(\hat x_k^{\mathrm{nom}}-x(t_k)\big)+\bar J_{z,k}\Delta z_k+\bar J_{c,k}\Delta c_k,\tag{7}$$

$\bar J_{z,k}=\int_0^1\partial_z\hat T^+(\hat z_k^{\mathrm{nom}}+s\Delta z_k;\hat c_k)ds$, $\bar J_{c,k}=\int_0^1\partial_c\hat T^+(\hat z_k^{\mathrm{nom}};c_k+s\Delta c_k)ds$.

(iii) (ISS bound) If $\|v_k\|\le\delta_y$ *and* $\|\tilde u_k-u_k\|\le\delta_u$*, then for every k*

$$\|\hat x_k-x(t_k)\|\le\|\hat x_k^{\mathrm{nom}}-x(t_k)\|+\ell_z\Big(\kappa e^{-\alpha t_k}\|\Delta z_0\|+\frac{\kappa\|B\|}{\alpha}\delta_y\Big)+\ell_c\,\ell_E\,\delta_u,\tag{8}$$

with ℓ_z,ℓ_c the sup of the Jacobian norms on the traversed segments $[\hat z_k^{\mathrm{nom}},\hat z_k]\times\{\hat c_k\}$ and $\{\hat z_k^{\mathrm{nom}}\}\times[c_k,\hat c_k]$; on the steady window the first term is $\le\varepsilon^\beta$ for a $(1-\beta)$ fraction of episodes, and in the limsup $\kappa\|B\|/\alpha$ may be replaced by the exact γ_y.

(iv) (stochastic deviations) For zero-mean i.i.d. v_k with covariance σ_v^2I (e.g. extra Gaussian noise), $\limsup_k(\mathbb E\|\hat x_k-x(t_k)\|^2)^{1/2}\le\limsup_k(\mathbb E\|\hat x_k^{\mathrm{nom}}-x(t_k)\|^2)^{1/2}+\ell_z^{\mathrm g}\sigma_v\|H\|_2+\ell_c\ell_E\delta_u$, $\|H\|_2^2=dt\,\mathrm{tr}W_c+o(dt)$, $AW_c+W_cA^\top+BB^\top=0$. *The global constant is needed because unbounded noise leaves every neighbourhood of the data with positive probability.*

Intuition. The learned map never sees raw measurements: sensor deviations and initialisation reach it through a known, input-free stable filter, a wrong input record moves only the context, and each deviation is multiplied by the decoder's slope along the path it actually travels.

Proof. (i) $\hat{z}$ and $\hat{z}^{\mathrm{nom}}$ obey the same linear recursion with forcing difference $W_0 v_k + W_1 v_{k+1}$; unroll it, and use $\|A_d^k\| \le \kappa e^{-\alpha t_k}$ and $\sum_i \|G_i^{(k)}\| \le \kappa\|B\|/\alpha$ (Lemma 26(c)). (ii) Split $\hat{T}^+(\hat{z};\hat{c}) - \hat{T}^+(\hat{z}^{\mathrm{nom}};c)$ at $\hat{T}^+(\hat{z}^{\mathrm{nom}};\hat{c})$ and apply the fundamental theorem of calculus on each segment. (iii) Norms in (7), (i), (L), Definition 18. (iv) $L^2(\mathbb{P})$ norms of the pointwise inequality (Minkowski) with $\|\bar{J}_z\| \le \ell_z^{\mathrm{g}}$; $\mathbb{E}\|\sum_i G_i^{(k)} v_i\|^2 = \sigma_v^2 \sum_i \|G_i^{(k)}\|_F^2 \to \sigma_v^2 \|H\|_2^2$, a Riemann sum ($H_m = e^{At_m} B\, dt + O(dt^2)$) of $dt \int_0^\infty \|e^{As}B\|_F^2 ds$. □

When the nominal sensor is noise-free, (8) is the classical decomposition: fit error plus decoder slope times the ISS gain $\kappa\|B\|/\alpha$ of the latent filter (cf. Niazi et al., 2025, Lemma 17, Theorem 19), here for the sampled observer and with an input-corruption term. The latent error equation of Anonymous (2026, Proposition 1) assumes no injection; with injection $\Phi \neq 0$ the latent deviation is no longer input-free and the margin of Proposition 7 is needed (sampled form: Lemma 26(d)). The bound has no amplitude-dependent condition, but its constants ℓ_z, ℓ_c may depend on the context.

(iii) is a decomposition with measured constants, not a certificate. With ℓ_z, ℓ_c the suprema over the traversed segments, (iii) is (ii) with norms taken, and the constants are only known after the fact (lower bounds of the suprema when estimated from samples). Evaluated *a priori*, with the phase-1 proxy `l_dec_q95` for the global constant ℓ_z^{g} and the nominal sensor level $\delta_y = 0.03\, y_{\mathrm{std}}$, the second term alone is $\ell_z \kappa\|B\|\delta_y/\alpha = 66$ (battery; $\kappa\|B\|/\alpha = 2.68$, $\ell = 816$), 15 (quad_tank; 37.4, 13.5), 6.7 (duffing), 5.9 (bioreactor), 1.1 (vdp), 0.40 (lorenz), against measured nominal NRMSE 0.016–0.052 (C1-where, lora, `id`): 15× to 1400× too large (`check_apriori_vacuity`). The bound is therefore vacuous as a guarantee on every benchmark and is *not* presented as one. What it offers is the exact attribution (ii) and the exact stochastic form (iv), both of which are measurable (Definition 24).

Remark 20 (reading the bound)**.** Three quantities set the error: the nominal error ε^β (capacity, data, window), the decoder slope ℓ_z times a design-only latent deviation ($\propto 1/\alpha$), and the input sensitivity $\ell_c \ell_E$, which is zero for the autonomous observer. The architecture enters only through these constants: the bound says which one a conditioning mechanism must keep small, not which mechanism wins.

What would test it (eval-only; implemented by no current node). Under `input_noise/bias/delay` the excess error is $\bar{J}_c \Delta c$ exactly: the autonomous observer is unaffected, and variants would order by their realised input gain $\|\bar{J}_c \Delta c\|/\|\Delta c\|$ if that gain were measured on saved checkpoints (Definition 24; C3-faults has the conditions, no node records the gains). Under `poor_init` the excess is $\bar{J}_z A_d^k \Delta z_0$, gone after a few $1/\alpha$ (C3-faults records the convergence time, which is the observable consequence). Upper bounds do not rank errors, so none of this is implied by the bound.

Check (`m3_checks.py::check_noise_iss`, `check_input_corruption`, `check_identity`; linear plant with exact maps $z = Px + \nu$, decoder $P^+(z - c)$, $\ell_z = \ell_c = 12.48$, $\ell_E = 0.048$, $n_z = 5$, $\alpha = 1.5$, $dt = 0.0206$ as for vdp). Worst-case bounded deviation: latent $0.0372 \le \gamma_y \delta_y = 0.0537 \le \kappa\|B\|\delta_y/\alpha = 0.0745$. White deviation, RMS latent 0.00427 (Monte Carlo) vs $\sigma_v\|H\|_2 = 0.00430$; the sup form at $\delta_y = 4\sigma_v$ is 30× looser, so only (iv) is informative for Gaussian conditions. Input bias 0.1 / delay 5 / bounded noise: state error $0.022 \le 0.061$, $0.197 \le 1.21$, $0.024 \le 0.18$. Identity (7) for a tanh-MLP decoder holds to 4×10^{-10} (finite-difference Jacobians); its realised gain differs from the on-data $\|\partial_z \hat{T}^+\|_2$ by a factor 0.01–1.38 (median 0.32), so on-data slopes neither bound nor estimate it. Coverage of (6) at $n = 64$, $\beta = 0.05$: 0.961 vs 0.963.

Corollary 21 (sensor faults: bounded and forgotten, uniformly in the state)**.** *For a decoder-side observer and any sensor deviation v, pathwise $\|\hat{x}_k - x(t_k)\| \le \|\hat{x}_k^{\mathrm{nom}} - x(t_k)\| + \ell_z^{\mathrm{g}} \|\sum_i G_i^{(k)} v_i\|$. In particular: an outlier of size M at k_0 adds at most $\ell_z^{\mathrm{g}}\kappa\|B\|M\left(w_1 + e^{\alpha dt} w_0\right) e^{-\alpha(t_k - t_{k_0})} \le \ell_z^{\mathrm{g}}\kappa\|B\|M\, dt\, e^{\alpha dt} e^{-\alpha(t_k - t_{k_0})}$ (w_0, w_1 of Lemma 26(c), $w_0 + w_1 \le dt$), with time integral at most $\ell_z^{\mathrm{g}}\kappa\|B\|M\, dt/\alpha$, for every state, input and M; an outlier process (each sample with probability p, size M, random sign) adds at most $\ell_z^{\mathrm{g}}\gamma_y p M$ in mean and $\ell_z^{\mathrm{g}}\sqrt{p} M \|H\|_2$ in RMS; sample-and-hold dropout adds at most $\ell_z^{\mathrm{g}}\gamma_y L_y\, dt\, p/(1-p)$ in mean; a drift b adds at most $\ell_z^{\mathrm{g}}\gamma_y \sup\|b\|$, and for slow drift $\Delta z_k \approx -A^{-1}B b_k$, since the discrete DC gain is $(I - A_d)^{-1}(W_0 + W_1) = -A^{-1}B$ exactly; coloured deviations obey $\mathbb{E}\|\Delta z\|^2 = \sum_{m,m'} \mathrm{tr}(H_m R_v (m' - m) H_{m'}^\top)$.*

Intuition. There is no feedback from the estimate to the latent: a fault is a forcing of a fixed stable filter, so the observer's own error can never amplify it.

Proof. Proposition 19(ii) with $\Delta c = 0$, $\Delta z_0 = 0$, $\|\bar{J}_z\| \le \ell_z^{\mathrm{g}}$. Outlier: $\Delta z_{k_0} = W_1 M e$ and $\Delta z_k = (A_d^{k-k_0-1} W_0 + A_d^{k-k_0} W_1) M e$, bounded with the merged kernels of Lemma 26(c); the integral is $dt \sum_m \|H_m\| M \le dt\, \kappa \|B\| M/\alpha$. Process: $\mathbb{E}\|v_i\| \le pM$ in (i), and $R_v(m) = pM^2 \delta_m$. Dropout: a held sample differs from the current one by at most $L_y\, dt\, N_k$, N_k geometric with $\mathbb{E}N_k \le p/(1-p)$. DC gain: $W_0 + W_1 = A^{-1}(A_d - I)B$ and A, A_d commute; the quasi-static error is bounded by summation by parts, $\|\Delta z_k + A^{-1} B b_k\| \le \sum_{m \ge 1} \|\sum_{j \ge m} H_j\| \sup_j \|b_j - b_{j-1}\|$. □

What this proves about filters, and what it does not. The corollary is a *qualitative* ISS guarantee: the excess is bounded and decays at rate α for every state, input and fault size, given ℓ_z^{g}, with no feedback from the estimate to the latent. The EKF's stochastic stability is local (small initial error and noise, Reif et al., 1999) and an outlier moves its linearisation point. A qualitative guarantee is not a comparative advantage, and the data say so: on *exact* models no C3-faults condition (sensor, input, plant-drift and outage) reaches the learned-wins regime of $\ge 4/6$ systems (learned-wins regime: none of 15 conditions; the learned observer beats the best filter on 0/6 under every sensor-side condition and on at most 2/6 against the best filter of either kind under input bias, input delay or 10% plant drift), and it wins on 0/6 under colored Student-t noise (C3-noise-ct), as Remark 22 predicts for linear plants and as the EKF's 2.7–14.1× (median 5.9×) smaller exact-model `sequence_ood` error on the dev systems (C1-where) makes expected. Worse, the best *learned* observer is itself above climatology (a failed estimator) under exact-model `outliers` (battery, duffing, quad_tank) and under 30% post-training plant drift (bioreactor, quad_tank): the corollary bounds the excess by ℓ_z^{g}×(filter response), and with ℓ_z^{g} in the hundreds that bound is consistent with failure. Where learned observers win is model incompleteness (C3-mismatch10/20/30/50, C3-partial, C3-realistic), and the theoretical lever there is not robustness to sensor faults but the model term of Proposition 23.

Exact filter-level ratios (design-invariant). For the Butterworth design with $dt = 0.1/\|A\|$, the *latent* deviations of the stress conditions, relative to the nominal noise ($0.03\, y_{\mathrm{std}}$), do not depend on the system: RMS ×3.4 for `colored` ($\varrho = 0.95$), ×57 for `outliers` ($p = 0.02$, $M = 12\, y_{\mathrm{std}}$), and ×30–32 at the quasi-static peak for `sensor_drift` ($0.3\, y_{\mathrm{std}}$); `dropout` costs $O(L_y dt)$ per held sample. These are exact properties of the known filter; the *state*-level excess multiplies them by the variant's realised $\|\bar{J}_z\|$, which no current node records (Definition 24). *Check* `m3_checks.py::check_fault_ratios` computes these ratios for vdp, duffing, bioreactor and lorenz (all agree to two digits; drift $\times 32.1, 32.1, 32.1, 30.5$). `check_outliers`, `check_dropout_drift`: outlier peak $0.519 \le 0.536$, envelope never violated, integral $0.266 \le 0.369$; process mean $0.174 \le 0.258$, RMS 0.241 vs 0.243; dropout ($p = 0.1$) mean $0.00105 \le 0.00737$; DC identity to 10^{-16}; random-walk drift peak $0.137 \le 0.322$, quasi-static deviation $0.0052 \le 0.049$.

Remark 22 (Kalman filter vs. learned KKL on linear plants)**.** For $\dot{x} = Fx + Gu$, $y = Cx$ the exact maps are $z = Px + \nu$ with $PF = AP + BC$, $\dot{\nu} = A\nu - PGu$ and decoder $P^+(z - \nu)$, so the KKL estimate is an LTI filter of (y, u) with sensor-to-error response $\Gamma_m = P^+ H_m$; for $n_z = n_x$ it is the Luenberger observer with gain $P^{-1}B$. The steady KF has the same structure with gain K. (1) Gaussian noise with correct (Q, R): the KF is the MMSE estimator, so no KKL design beats it. (2) White non-Gaussian noise (Gaussian plus outliers): every LTI filter has steady MSE $\mathrm{MSE}_w + (\sigma_v^2 + pM^2)\|\Gamma\|_{H_2}^2$, the KF with $R = \sigma_v^2 + pM^2$ is the best linear filter (Anderson & Moore, 1979), and the KKL observer against a KF tuned to the nominal R is decided by $\|\Gamma\|_{H_2}$ alone. *Check* (`check_kf_comparison`; oscillator, $\sigma_v = 0.03$, $p = 0.02$, $M = 12$): Gaussian MSE KF 1.98×10^{-3} < KKL 6.75×10^{-3}; with outliers KF(nominal R) 0.514, KF(R_{eff}) 0.0115, KKL 2.40, each within 2% of the formula. The KKL loses to both here ($\|P^+\| = 12.5$): outlier robustness is not a general property of KKL observers, and on the benchmarks any advantage over the EKF must be attributed to its mis-tuning, local stability and linearisation, or, as the data show, to the filter's model being wrong (next).

C.1 Model incompleteness: the term that decides C3

Proposition 23 (persistent model error: filter bias floor vs. learned fit error; linear plants)**.** *[proven for linear plants] Let the true plant be $x_{k+1} = Fx_k + Gu_k + d_k + w_k$, $y_k = Cx_k + v_k$, where d_k is the part of the dynamics the modeller's model lacks (a wrong F or a missing term: $d_k = \Delta F\, x_k + \Delta G\, u_k$), w_k, v_k white with covariances Q, R. Any fixed-gain predict–correct observer built on the model (F, G, C), in particular the steady-state Kalman filter and every Luenberger observer, $\hat{x}_{k+1} = \hat{x}_{k+1}^- + K(y_{k+1} - C\hat{x}_{k+1}^-)$, $\hat{x}_{k+1}^- = F\hat{x}_k + Gu_k$, has*

error $e_k = x_k - \hat{x}_k$ *obeying*

$$e_{k+1} = (I - KC)F\, e_k + (I - KC)(d_k + w_k) - K v_{k+1}.$$

*(i) (*bias*) For a constant* $d_k \equiv \bar{d}$ *(e.g. a constant input and* $\Delta F \bar{x}$ *at the steady state), the error converges to the non-decaying bias* $\bar{e} = \big(I - (I - KC)F\big)^{-1}(I - KC)\,\bar{d}$*; in continuous time, for* $\dot{e} = (F - KC)e + \Delta F\, x$*,* $\bar{e} = -(F - KC)^{-1}\Delta F\, \bar{x}$*, the DC value of* $(sI - (F - KC))^{-1}\Delta F\, x$*.*

*(ii) (*bias–variance floor*) Let* $Q \succ 0$*,* $R \succ 0$*. For a stationary* d_k *independent of* (w, v) *(e.g. d evaluated along the noise-free trajectory, as in the check; with* $d_k = \Delta F\, x_k$ *along the noisy trajectory the part* $\Delta F\, x_k^{\mathrm{sto}}$ *is an extra process noise correlated with* w*, and* Σ *must then be taken from the joint Lyapunov equation of* (x^{sto}, e)*) the steady mean-square error splits exactly as* $\mathrm{MSE}(K) = b^2(K) + \operatorname{tr}\Sigma(K)$*, with* b^2 *the mean square of the deterministic response to d through* $\Phi_K = (I - KC)F$ *and* $\Sigma = \Phi_K \Sigma \Phi_K^\top + (I - KC)Q(I - KC)^\top + KRK^\top$*. Inflating the process covariance to* $Q + \lambda I$ *(the oracle filters'* `q_inflate`*) moves* $K(\lambda)$ *along a one-parameter curve that trades* b^2 *against* $\operatorname{tr}\Sigma$ *(in the check:* $\|K\|$ *and* $\operatorname{tr}\Sigma$ *increase,* b^2 *decreases, monotonically); the best achievable error* $\mathrm{MSE}^\star := \inf_K \mathrm{MSE}(K) \geq \inf_K \operatorname{tr}\Sigma(K) > 0$ *depends on* $(F, G, C, \Delta F, \Delta G, Q, R)$ *and the input, and on no data. Only* n_y *directions of d can be cancelled by K, so* b^2 *keeps a positive limit as* $\lambda \to \infty$ *whenever d has a component K cannot cancel (e.g.* $C\bar{d} = 0$*, as for the missing stiffness term of the check).*

*(iii) (*learned decoder-side observer*) Its error obeys Proposition 19 with* ε^β *measured on the true plant:* $\|\hat{x}_k - x_k\| \leq \varepsilon^\beta + \ell_z\|\Delta z_k\| + \ell_c\|\Delta c_k\|$ *on a* $(1-\beta)$ *fraction of episodes, and no term depends on the model. In the linear case, a decoder fitted on true-plant data is* $P^+_{\mathrm{true}}(z - \nu_{\mathrm{true}})$ *with* $P_{\mathrm{true}}F_{\mathrm{true}} = AP_{\mathrm{true}} + BC$ *(continuous-time Sylvester equation of the true plant; for the sampled plant its discrete counterpart): its error is driven by* w, v *only (no bias), with a mean-square error* $\mathrm{MSE}_{\mathrm{KKL}}$ *that is a property of the true plant and the noise; at a fixed true plant it does not depend on the modeller's model at all. The model enters the learned observer only through the phase-1 PDE regulariser of* T_0 *(and through* `kkl_mb`*, which is a model-based filter and belongs to the other class).*

Prediction. *The learned observer beats the best (oracle-tuned) filter iff* $\mathrm{MSE}^\star > \varepsilon^2$*, where* ε *is its fit error on the true plant. For a missing term* $s\,d$ *of fixed shape and scale s (at a fixed true plant),* $\mathrm{MSE}^\star(s) = \inf_K[s^2 b_1^2(K) + \operatorname{tr}\Sigma(K)]$ *(*b_1^2 *the bias term at* $s = 1$*; d independent of* (w, v) *as in (ii)) is nondecreasing in s and* ε *does not depend on s, so there is at most one crossing: filters win below a threshold in the model error (possibly* 0*) and learned observers above it. The threshold depends on the plant and the noise.*

Intuition. A model-based filter corrects its wrong model only through the n_y measured directions and only at the cost of trusting the noisy measurement more; a decoder trained on the true plant has no model to be wrong about, it pays its fit error instead.

Proof. Subtract the predict–correct recursion from the true one: $x_{k+1} - \hat{x}^-_{k+1} = Fe_k + d_k + w_k$ and $e_{k+1} = (I - KC)(x_{k+1} - \hat{x}^-_{k+1}) - Kv_{k+1}$. (i) Set $e_{k+1} = e_k$ with $w = v = 0$; $(I - (I - KC)F)$ is invertible when the filter is stable. Continuous time: set $\dot{e} = 0$. (ii) Linearity: the response to d (deterministic) and to (w, v) (zero mean, independent of d) are orthogonal in L^2, so the mean squares add; Σ is the stationary covariance of the noise response (discrete Lyapunov equation). $\operatorname{tr}\Sigma(K)$ is the error of the linear filter K under the nominal model and the true noises, so it is bounded below by the nominal-model MMSE, which is positive for $Q \succ 0$, $R \succ 0$; hence $\mathrm{MSE}^\star > 0$ for every d. As $\lambda \to \infty$ the gain can cancel d only on the n_y directions that C measures, so b^2 has a positive limit when $(I - KC)\bar{d}$ cannot be made zero. (iii) Proposition 19 with Definition 18; in the linear case $z - P_{\mathrm{true}}x - \nu_{\mathrm{true}}$ obeys a stable linear recursion forced by $W_0 v_k + W_1 v_{k+1} - P_{\mathrm{true}} w_k$ only. □

Check (`m3_checks.py::check_model_mismatch`; damped oscillator, missing stiffness term $\Delta F = \epsilon\,[0\ 0; -1\ 0]$, $q = 0.05$, $\sigma_v = 0.03$, PRBS input). (i) the steady bias agrees with the formula to 10^{-8} (discrete, $\lambda = 0$ and 10^{-3}) and 10^{-6} (continuous Luenberger). (ii) at $\epsilon = 0.3$ (relative model error $|\Delta Fx|/|Fx| = 0.29$),

over $\lambda \in [0, 10^{-1}]$: $\|K\|$ rises $0.25 \to 1.19$, b^2 falls $5.24 \to 4.80 \cdot 10^{-2}$, $\operatorname{tr}\Sigma$ rises $2.0 \to 3.0 \cdot 10^{-3}$, MSE is U-shaped with $\mathrm{MSE}^\star = 5.09 \cdot 10^{-2}$; Monte Carlo agrees with $b^2 + \operatorname{tr}\Sigma$ within 1%. (iii) the true-plant KKL has $\mathrm{MSE} = 6.6 \cdot 10^{-3}$ and mean error 10^{-3} at every ϵ. Threshold: filters win at relative model error 0 and 0.05 ($\mathrm{MSE}^\star = 2.0, 4.2 \cdot 10^{-3}$), learned wins from 0.10 on ($10.0 \cdot 10^{-3}$ and above), one crossing.

What it predicts, and where it is tested (verdicts from `paper/generated/RESULTS.md`, the mechanical application of each node's pre-registered rule; model errors are the generated macros). With model error the C3 nodes measure, the proposition predicts a single crossing in the relative model error: filters win below it, learned observers above, per plant. On the exact-model side no condition reaches a learned-wins regime (C3-faults: none of 15 conditions; C3-noise-ct 0/6). On the incomplete-model side the learned observer beats the best *working* filter (a filter above climatology counts as failed) at every mismatch level, 10% (5/6 `id`, 4/6 `sequence_ood`), 20% (5/6, 5/6), 30% (5/6, 5/6), 50% (6/6, 5/6), under a missing term (C3-partial 5/6, 5/6) and in the realistic node (6/6, 6/6); the pre-registered crossover hypothesis "between 10% and 30%" therefore **fails** in the direction of the learned observer (first winning level 10%; the crossing lies below the smallest tested mismatch for the $\geq 4/6$ level rule, although duffing and battery are non-wins at 10%). Across the 36 model-wrong cells (`sequence_ood`, vs. the best working filter) the learned observer wins in 30/36. *Within* each system the outcome is ordered by model error: the six non-wins are duffing under every parameter mismatch (model error 0.02 at 30%), while duffing under the missing term (0.13) is a win; battery at 10% mismatch, while battery wins from 20% (model error 0.064); and quad_tank under the missing term (0.15), while quad_tank wins under parameter mismatch (0.69 at 30%); the systems with the largest errors (bioreactor 2.15, lorenz 0.71, vdp 0.66) are won at every level. *Across* systems the ordering does *not* hold: the smallest model error with a win (0.064, mismatch20/battery) is below the largest without one (0.152, partial/quad_tank). This is what the proposition allows (the threshold depends on the plant and the noise; the toy crosses at 0.05–0.10): the data are consistent with one crossing per plant, at plant-dependent thresholds that overlap in the band 0.06–0.15, not with a single cross-system threshold. This reading is post hoc, not a pre-registered test; TEST-C3 (frozen recipe, held-out plants) is **inconclusive** at the time of writing (6/18 jobs). The proposition is proven for linear plants; on the nonlinear benchmarks "learned wins iff the floor exceeds the fit error" is the reading of an **empirical** result. The "no model term" half of (iii) has one caveat: the phase-1 PDE term does use the (wrong) model; whether that helps or hurts is *not* established (C3-partial-modelfree: the partial model beats a model-free phase 1 with a CI excluding zero on 0/6 systems, so no claim that the partial model helps is made). Two further comparisons are needed before C3 can be read as a statement about KKL rather than about "data beats a wrong model": a Koopman/EDMDc Kalman observer and a grey-box (partial model plus learned residual) EKF trained on the same data (`baselines_data.py`; not yet in the reported nodes).

C.2 Measurement and certification

Definition 24 (measured decomposition and predicted bound)**.** For a trained model, a stress condition and the steady window, in the σ-normalised norm of `metrics.py`, compute exactly: Δz_k by running the known filter on $v = \tilde{y} - \tilde{y}^{\mathrm{nom}}$ (and $A_d^k \Delta z_0$ for the prior); Δc_k by running E on $\tilde{u}$ and u; and $\bar{J}_{z,k}, \bar{J}_{c,k}$ in (7) by Gauss quadrature along the traversed segments. Report the three terms of (7) (an exact attribution of the excess error) and the *realised effective gains* $g_z = \mathrm{RMS}\|\bar{J}_z \Delta z\| / \mathrm{RMS}\|\Delta z\|$, $g_c = \mathrm{RMS}\|\bar{J}_c \Delta c\| / \mathrm{RMS}\|\Delta c\|$. The *predicted bound* is $\widehat{\mathcal{B}} = \mathrm{NRMSE}_{\mathrm{nom}} + \hat{g}_z \mathrm{RMS}\|\Delta z\| + \hat{g}_c \mathrm{RMS}\|\Delta c\|$ with $\hat{g}_z, \hat{g}_c$ measured on *other* conditions (or the a priori $\kappa\|B\|\delta_y/\alpha$, $\sigma_v\|H\|_2$, $\ell_E \delta_u$ in place of the RMS terms).

With the gains of the same condition, $\widehat{\mathcal{B}} \geq \mathrm{NRMSE}$ is the triangle inequality and tests nothing. With transferred gains it is a prediction, and the test is one-sided: a point above the diagonal falsifies the transferred gain (the decoder is steeper along this condition's paths); a point below the diagonal confirms nothing. *Status.* This procedure is evaluation-only on saved checkpoints (the filter, the encoder and the decoder are all available) and has not been run by any node in the current tree; until it is, the transferred-gain prediction is untested, and this section claims only (i), (ii), (iv) and the exact filter-level ratios above.

Remark 25 (what can be certified)**.** The result is a decomposition into measurable terms, not a certificate. *Exact* (no learning): $\kappa, \alpha, \gamma_y, \|H\|_2$, the latent deviation of any recorded fault, the path Jacobians on recorded data, and an upper bound on the SFM gain ℓ_E (product of layer norms, Lemma 15). *Lower bounds only*: any

supremum of Jacobian norms estimated from samples. Exact Lipschitz constants are NP-hard to compute even for two-layer ReLU networks (Virmaux & Scaman, 2018); certified upper bounds (layer-norm products, LipSDP, Fazlyab et al., 2019) are valid but loose. On-data Jacobians are the wrong set for sensor faults: with $n_z > n_x$ the nominal latents lie near an n_x-dimensional manifold (times the context fibre) and $\hat{z} = \hat{z}^{\mathrm{nom}} + \Delta z$ leaves it, where the decoder's slope is inductive bias; hence the traversed-segment Jacobians of Definition 24. The random-direction quotient l_{dec} of `train.phase1_diagnostics` estimates about $\|J\|_F/\sqrt{n_z}$ on the base decoder's data (check: $q_{0.95} = 0.81\|J\|_2$ for a linear decoder) and is not the constant of the bound. *Statistical*: the nominal error only through (6).

D Lemma D: the sampled-data observer

Lemma 26 (Lemma D). *Let $\|e^{At}\| \le \kappa e^{-\alpha t}$, $\rho_0 := e^{-\alpha dt}$, and let input and context be held on each $[t_k, t_{k+1})$. Let $w_0 := \int_0^{dt} \frac{\tau}{dt} e^{-\alpha\tau} d\tau = \frac{1-\rho_0(1+\alpha dt)}{\alpha^2 dt}$, $w_1 := \int_0^{dt} (1 - \frac{\tau}{dt}) e^{-\alpha\tau} d\tau = \frac{\alpha dt - 1 + \rho_0}{\alpha^2 dt}$, so $w_0 + w_1 = (1-\rho_0)/\alpha$.*

(a) Exact step. *For y affine and g constant on the step, $\dot{z} = Az + By + g$ gives exactly $z_{k+1} = A_d z_k + W_0 y_k + W_1 y_{k+1} + W_g g$, with $A_d = e^{Adt}$, $W_0 = \int_0^{dt} (1 - \frac{s}{dt}) e^{A(dt-s)} B\, ds$, $W_1 = \int_0^{dt} \frac{s}{dt} e^{A(dt-s)} B\, ds$, $W_g = A^{-1}(A_d - I)$ (`design.latent_step`).*

(b) Interpolation. *With (A3) and the regularity clause, the clean iterate $\bar{z}_k$ from $\bar{z}_0 = z(0)$ satisfies $\sup_k \|\bar{z}_k - z(t_k)\| \le e_{\mathrm{disc}} := \frac{\kappa\|B\| M_2 dt^3}{12(1-\rho_0)} \le \frac{\kappa\|B\| M_2 dt^2 e^{\alpha dt}}{12\alpha}$, also for switching inputs, since switches sit on the grid.*

(c) Merged kernels. *$\|A_d^m W_0\| \le \kappa\|B\|\rho_0^m w_0$ and $\|A_d^m W_1\| \le \kappa\|B\|\rho_0^m w_1$. Hence $\sum_i \|G_i^{(k)}\| \le \kappa\|B\|/\alpha$ for every k and $\gamma_y \le \kappa\|B\|/\alpha$ at every dt, and $\gamma_y \to \int_0^\infty \|e^{As} B\| ds$ as $dt \to 0$. More generally every sampled constant is bounded by its continuous counterpart: $\rho_0^M = e^{-\alpha L}$, $c_d \le 1/\alpha$ in (e), $\gamma_y \le \kappa\|B\|/\alpha$.*

(d) Injection: step-size condition. *For the deployed exponential midpoint step (an exponential integrator, Hochbruck & Ostermann, 2010, with an explicit Euler predictor) $z_{k+1} = A_d z_k + W_0 y_k + W_1 y_{k+1} + W_g \Phi(z_k^h, c_k)$, $z_k^h = z_k + \frac{dt}{2}(Az_k + By_k + \Phi(z_k, c_k))$, with $\Phi(\cdot, c)$ ℓ_Φ-Lipschitz, two solutions driven by the same (y, c) satisfy $\|z_k - z_k'\| \le \kappa\rho^k \|z_0 - z_0'\|$,*

$$\rho := \rho_0 + \kappa(1-\rho_0)\frac{\ell_\Phi \beta}{\alpha}, \qquad \beta := 1 + \tfrac{dt}{2}(\|A\| + \ell_\Phi).$$

So $\kappa\ell_\Phi\beta < \alpha$ suffices; it tends to $\kappa\ell_\Phi < \alpha$ (Corollary 8) as $dt \to 0$ with $-\ln\rho/dt \to \alpha - \kappa\ell_\Phi$, and $\beta \le 1.1$ under $\|A\| dt \le 0.1$ (`design.choose_dt`). If the stationary latent $\zeta_k = \hat{T}_0(x(t_k))$ satisfies $\zeta_{k+1} =$ (clean step at ζ_k) $+ \int_0^{dt} e^{A(dt-s)} R\, ds + q_k$, with $\|R\| \le \varepsilon_{\mathrm{res}}$ the PDE residual and q_k the interpolation/quadrature defect with $\|A_d^m q_k\| \le \kappa\rho_0^m \bar{q}$, and the step is fed samples with $\|v_k\| \le \delta_y$, then

$$\|\hat{z}_k - \zeta_k\| \le \kappa\rho^k \|\hat{z}_0 - \zeta_0\| + \frac{\kappa\big(\|B\|(1 + \ell_\Phi \frac{dt}{2})\delta_y + \varepsilon_{\mathrm{res}}\big)}{\alpha - \kappa\ell_\Phi\beta} + \frac{\kappa\,\alpha\,\bar{q}}{(1-\rho_0)(\alpha - \kappa\ell_\Phi\beta)},$$

a single factor κ, with $\bar{q} = O(dt^3)$ so the last term is $O(dt^2)$.

(e) Deployed fading memory. *If $u_j = u_j'$ for $j = k-M, \dots, k-1$, the deployed map $T_k^d(x; u) = \sum_{j \ge 0} A_d^j (W_0 y_{k-1-j} + W_1 y_{k-j})$ satisfies $\|T_k^d(x; u) - T_k^d(x; u')\| \le 2\kappa\|B\|\bar{y}\,\rho_0^M c_d$, $c_d := (w_0 + \rho_0 w_1)/(1-\rho_0) \le 1/\alpha$.*

Proof. (a) Variation of constants over one step with the affine y inserted. (b) $\bar{z}_k - z(t_k) = -\int_0^{t_k} e^{A(t_k - s)} B(y - \ell)(s) ds$ with ℓ the piecewise-linear interpolant. On $[t_j, t_{j+1}]$ the input is constant, so $y \in C^2$ there, and the Peano form gives $\|y(t_j + \sigma) - \ell(t_j + \sigma)\| \le \frac{\sigma(dt-\sigma)}{2} M_2$ in any norm. Use $\|e^{A(t_k - s)}\| \le \kappa\rho_0^{k-1-j}$ on that interval and $\int_0^{dt} \frac{\sigma(dt-\sigma)}{2} d\sigma = \frac{dt^3}{12}$, sum the geometric series, and use $1 - \rho_0 \ge \alpha dt\, \rho_0$. (c) $A_d^m W_0 = \int_0^{dt} (1 - \frac{s}{dt}) e^{A(mdt + dt - s)} B\, ds$, so $\|A_d^m W_0\| \le \kappa\|B\|\rho_0^m \int_0^{dt} (1 - \frac{s}{dt}) e^{-\alpha(dt-s)} ds = \kappa\|B\|\rho_0^m w_0$, and likewise for W_1.

Then $\|H_0\| + \sum_{m\geq 1}\|H_m\| \leq \kappa\|B\|\big(w_1 + \sum_{m\geq 1}(\rho_0^{m-1}w_0 + \rho_0^m w_1)\big) = \kappa\|B\|\frac{w_0+w_1}{1-\rho_0} = \frac{\kappa\|B\|}{\alpha}$, and the finite-$k$ coefficients are a sub-sum of the same terms. The limit holds because $H_m = e^{At_m}B\,dt + O(dt^2)$ with summable tails. (d) $\Delta_k = A_d^k\Delta_0 + \sum_{j<k}A_d^{k-1-j}W_g(\Phi(z_j^h) - \Phi(z_j'^h))$ with $\|z_j^h - z_j'^h\| \leq \beta\|\Delta_j\|$ and the merged kernel $\|A_d^{k-1-j}W_g\| = \|\int_0^{dt} e^{A((k-j)dt-s)}ds\| \leq \kappa\rho_0^{k-1-j}(1-\rho_0)/\alpha$. With $a_k = e^{\alpha t_k}\|\Delta_k\|$ and $g = \kappa(e^{\alpha dt}-1)\ell_\Phi\beta/\alpha$, $a_k \leq \kappa a_0 + \sum_{j<k} g a_j$; the right side S_k obeys $S_{k+1} \leq (1+g)S_k$, so $\|\Delta_k\| \leq \kappa(\rho_0(1+g))^k\|\Delta_0\| = \kappa\rho^k\|\Delta_0\|$. If $\kappa\ell_\Phi < \alpha$ and $\|A\|dt \leq 0.1$, then $\ell_\Phi dt < \alpha dt \leq \|A\|dt$, so $\beta \leq 1.1$. For the forced bound, unroll $e_k = \hat{z}_k - \zeta_k$; every forcing term is merged with the kernel that propagates it: $\|A_d^m(W_0 v_j + W_1 v_{j+1})\| \leq \kappa\rho_0^m\|B\|(w_0+w_1)\delta_y$, $\|A_d^m\int_0^{dt} e^{A(dt-s)}R\,ds\| \leq \kappa\rho_0^m(1-\rho_0)\varepsilon_{\text{res}}/\alpha$, the predictor mismatch $\leq \beta\|e_j\| + \frac{dt}{2}\|B\|\delta_y$ enters through $A_d^m W_g$, and $\|A_d^m q_j\| \leq \kappa\rho_0^m\bar{q}$. So $a_k \leq \kappa a_0 + \sum_{j<k}\big(g a_j + \kappa e^{\alpha(j+1)dt}D\big)$ with $D = \frac{1-\rho_0}{\alpha}\big(\|B\|(1+\ell_\Phi\frac{dt}{2})\delta_y + \varepsilon_{\text{res}}\big) + \bar{q}$, hence $S_{k+1} \leq (1+g)S_k + \kappa e^{\alpha(k+1)dt}D$ and $\|e_k\| \leq \kappa\rho^k\|e_0\| + \kappa D/(1-\rho)$; insert $1-\rho = (1-\rho_0)(\alpha - \kappa\ell_\Phi\beta)/\alpha$. (e) The samples y_i coincide for $i \geq k-M$; bound by (c), summing from $j = M$ (W_0) and $j = M+1$ (W_1), with $\|y_i - y_i'\| \leq 2\bar{y}$. □

Continuous-time interpretation. The latent step is exact for the held input, the context and any generated decoder weights are constant on each interval by construction, and the decoder runs at t_k only. The bounds are about the implemented observer, and the continuous formulas are their $dt \to 0$ limits; the sampled observer adds only e_{disc} and, with injection, the factor β.

Check (`m3_checks.py`). (b) `check_interpolation`: PRBS input, four step sizes, order 2.00, bound/error 3.6 (`check_theory.lemma_d` uses the same constant, bound/error 4.4–4.5). (c) `check_design_constants`: $\gamma_y = 1.074 \leq \kappa\|B\|/\alpha = 1.491$ ($\int\|e^{As}B\| = 1.075$); $\gamma_y \leq \kappa\|B\|/\alpha$ at $\alpha dt \in \{0.03, 0.3, 1, 3\}$ and for a non-normal A with $\kappa = 1.46$. The κ of a non-normal A is estimated by `spectral_bounds` on a 400-point grid, so those rows are consistency checks, not certificates; the deployed Butterworth A is normal and has $\kappa = 1$ exactly. (d) `check_midpoint`: for $\Phi = \ell z$ the per-step gain is below ρ (e.g. $0.99591 \leq 0.99762$ at $\ell = 1.3$); 200 random linear injections: $\max\|\text{step}\|/\rho = 0.999$; for a non-normal A ($\kappa = 1.99$) $\max_k\|\text{step}^k\|/(\kappa\rho^k) = 0.92$, so a single κ suffices; $\Phi = -400z$, more stable in continuous time, gives discrete spectral radius 26.4 at $\ell dt = 8.2$, so the step-size part is needed; the numpy step equals `LatentStep` to 1.5×10^{-7}. (e) `m2_checks.py::lemma_d_constants`.

E Protocol details

Design of the latent filter. A is the real-Jordan form of a Butterworth spectrum with abscissa α (per plant, `configs/systems/*.yaml`), hence normal with $\|e^{At}\| = e^{-\alpha t}$ ($\kappa = 1$); dt is chosen so that $\max|\lambda|\,dt = 0.1$; B is the all-ones input matrix. The latent step is the exact zero-order-hold step with linear interpolation of y inside a sample (Lemma 26(a)), and the context and any generated weights are held between samples.

Signal families. Training families (used for `id` with fresh seeds): constant, step, staircase, PRBS, square, sinusoid, multisine, ramp, amplitude-modulated and "drive"; zero input is a training-only family. `shape_ood`: chirp, filtered noise (Ornstein–Uhlenbeck), triangle, sawtooth, inside the training frequency band. `sequence_ood`: the drive family (a level on a 9-level grid in $[-1, 1]$ held for a dwell, then a linear ramp to the next level) with a stop-and-go event sequence: level jumps of 1.0–2.0 instead of 0.25–0.5, dwells of 0.5–2 t_{ref} instead of 2–6, ramps of 0.2–0.5 t_{ref} instead of 0.5–1.5. All families realise $v(t) \in [-1, 1]$ without clipping (periodic families carry an offset inside $[-(1-A), 1-A]$); the applied input is $u_{\text{scale}}\,v(t)$; plants that declare input bounds are rescaled, never clipped. Switch times and knots are snapped to the integration grid so that no RK4 step straddles a switch. A per-tier state-coverage diagnostic (fraction of evaluation states outside the training-state box and beyond the training k-NN 99% quantile) is stored with every run; under this protocol it is 0% outside the box on every dev plant and tier.

Training. Phase 1 (base maps, once per plant and seed): T_0 ($x \to z$, SiLU MLP, width 192, depth 3) from latent labels of washed-out zero-input episodes plus the PDE residual $\partial_x T_0(x)f(x,0) - AT_0(x) - Bh(x)$ on a Sobol collocation set, weighted by $-A^{-1}$ and divided by $\|f(x,0)\|$; T_0^+ ($z \to x$, tanh MLP, width 128, depth 3) on fresh states encoded by the frozen T_0; 6000 steps each, batch 4096, learning rate $2 \cdot 10^{-3}$. f here is the *modeller's* model (nominal parameters; `f_known` when partial); `phase1.pde: false` gives the

model-free variant. Phase 2 (conditioning): base maps frozen; one width knob per variant solved so that the conditioner's trainable parameters hit the budget (counts within 2%); every generator is bias-free with zero-preserving SiLU activations (not odd, so that FiLM and hyper can represent input effects that do not flip sign with u); 3000 steps, learning rate $2 \cdot 10^{-3}$, 32 episodes $\times$ 256 time points per batch, model selection on the validation loss. Decoder-side variants minimise the deployed error on the latent that the deployed filter carries (`z_obs`); the latent variant minimises the error of 256-step rollouts from $T_0(x_{k_0})$ through the injected dynamics, decoded by the frozen T_0^+. NLOX: $\omega(z)$ and $T^\dagger$ (widths 64 and 128) trained jointly from rollouts started at $z_0 = 0$, 3000 steps, learning rate 10^{-3}, model-free.

Filters. EKF and UKF propagate with RK4 on the modeller's model; the particle filter uses 512 particles with a proposal inflation chosen per job on validation episodes. Nominal Q is the process-noise rate of the data generator (0.5% of each state's reference std) and nominal R the nominal sensor variance (3%); the prior covariance is set from the initial-condition box. `ekf_huber`/`ukf_huber`: Huber M-estimation update (threshold 1.345, three IRLS iterations), nominal Q, R, not told the condition. `ekf_oracle`/`ukf_oracle`: rebuilt per condition with that condition's sensor statistics (outlier mixture variance, coloured-noise variance), a Q that includes the measured model-error rate between the filter's model and the block's real plant, and inflation factors $q_{\text{inflate}} \in \{0, 0.3, 1, 3, 10\}$ (searched only when the model is inexact), $r_{\text{inflate}} \in \{1, 2\}$ chosen per job on at most 32 validation episodes; they skip held samples (dropout, burst outage) unless the sensor is quantised. `kkl_mb`: Section 3.

Evaluation. 64 episodes per tier, paired across methods and conditions; the steady window is the second half of each episode; every observer starts from the same prior ($\hat{z}_0 = \hat{T}_0(\hat{x}_0)$ for the KKL observers). Conditions (`configs/base.yaml`): `noise10` (10% noise), `outliers` (rate 2%, scale 12), `dropout` (10%), `colored` (AR(1), $\rho = 0.95$), `colored_t` (AR(1), $\rho = 0.9$, driven by Student-t with $\nu = 3$, unit stationary variance), `quantized` (step 0.25), `sensor_drift` (0.3), `burst_outage` (sample-and-hold for 2 t_{ref} from 60% of the horizon), `poor_init` (prior in a far corner of the IC box), `input_noise` (10%), `input_bias` (0.1), `input_delay` (5 samples), `plant_drift10/30` (plant parameters shifted after training), `long` (16 episodes of five horizons, `id` only). Clean-condition climatology is used for every stress condition of `C3-faults` (sensor and input faults do not change the true state; plant drift does slightly, flagged). For `id` the climatology is analytic (equal to 1 by the normalisation) for the nominal plant and a proxy (mean of the two OOD tiers) for a perturbed plant.

Scorecard definitions. With $ne(t) = \sqrt{\text{mean}_j(e_j(t)/\sigma_j)^2}$ (clipped at 100): `nrmse_p95/p99` are percentiles of the episode NRMSE; `div_rate` the fraction of episodes with a non-finite estimate or $|e_j/\sigma_j| > 100$; `t_conv` the first time after which the centred moving RMS of ne over one t_{ref} stays below $\max(2\,\text{median}\,ne,\, 0.05)$ (censored at the horizon); `t_forget` the paired first time after which the moving RMS of the normalised difference to the same observer's estimate from the nominal prior stays below 0.05; `recovery` the first time after the first fresh sample from which the RMS of ne over the next $0.5\,t_{\text{ref}}$ is at most 1.5 times its pre-outage level; `drift` the RMS of ne over the last 20% of a long episode divided by the RMS over the 20% after settling; `alarm_auroc` the Mann–Whitney AUROC of $r_k = \text{EMA}_{t_{\text{ref}}}\|(y_k - h(\hat{x}_k))/y_{\text{std}}\|$ as a detector of steps with $ne > 3\times$ the median settled ne.

Reading rules of the results layer (printed with every generated report). I1: "beats" = paired 95% CI of the relative change (Student-t over seeds; episode bootstrap for a single seed) excludes 0 in the better direction. I2: a rule that names no tier is applied on `shape_ood` and `sequence_ood` separately and passes only on both; `id` is reported. I3: "best learned" = lowest mean NRMSE among latent injection, HyperKKL ($r = 4$) and its dense ablation in the block; "best filter" = lowest among `ekf, ukf, pf, kkl_mb` (nominal) and `ekf_huber, ukf_huber, ekf_oracle, ukf_oracle` (robust/oracle); NLOX is reported apart; the selection is made on the episodes it is tested on, which favours the side with more candidates. I4: "within CI" = the 95% CI includes 0; "no loss" = not significantly worse; control nodes add TOST at $\pm 10\%$ (90% CI) with an "underpowered" flag (fewer than 3 seeds or CI half-width $\geq 10\%$). I5: climatology is the failure line; a filter above it is FAILED and changes are quoted against the best WORKING filter with the count of failed filters. I6: clean-condition climatology for every stress condition of `C3-faults`.

Compute and reproducibility. Every artefact (design, scales, data, base maps) is cached by a hash of the configuration sections it depends on, so sibling nodes share data and base maps; each run stores its configuration, code snapshot hash, metrics, scorecard, memory probe and timing in `results.json`; the results layer (`analysis/paper_results.py`) regenerates every figure, table and macro of this paper from those files and prints missing data as PENDING or INCONCLUSIVE rather than dropping it. A job (all methods on one plant and seed) takes 30–120 minutes on one GPU; the tree reported here is 519 jobs.

F Generated tables and figures

All tables and figures in this appendix are produced by `analysis/paper_results.py` from the runs' `results.json` and are not edited by hand.

F.1 The plants

Table 5: The 12 systems. Design decisions use dev systems only; test systems are run once with the frozen recipe. Model error: median $\|f_{\text{real}} - f_{\text{model}}\|/\|f_{\text{real}}\|$ over a Sobol sample of the IC box and the input box, for the partial-knowledge model (nominal parameters) and for a 30% parameter error (full model).

system	role	n_x	n_y	n_z	t_{ref}	input enters	unmodelled term (partial model)	err. partial	err. 30%
battery	dev	5	2	7	1	current (SOC + 2-RC + thermal; 1-min time unit; Vm and...	the slow RC branch R2 ‖ C2 (model: the textbook 1-RC equivalent circuit; V2 is not driven,...	0.22	0.09
bioreactor	dev	2	1	5	1	bilinear (dilution rate)	Monod saturation of the growth rate (model: first-order kinetics mu_max s / Ks)	2.15	0.50
duffing	dev	2	1	5	1	state-dependent gain	the cubic stiffness -x1^3 (model: linear spring)	0.13	0.02
lorenz	dev	3	1	7	1	additive, chaotic	the coupling -x1 x3 in the measured equation	0.71	0.29
quad_tank	dev	4	2	6	15	distributed, non-minimum phase	the square-root orifice law (model: flows linearised at the nominal equilibrium)	0.15	0.69
vdp	dev	2	1	5	1	additive + saturating actuator	the nonlinear damping law mu (1 - x1^2) x2 (model: undamped oscillator)	0.66	0.20
cartpole	test	4	2	6	1	matrix-valued G(x), balancing loop	large-angle pendulum terms (model: small-angle sin th = th, cos th = 1, no om^2)	0.03	0.35
fhn	test	2	1	5	2	additive	the recovery leak -eps b w	0.04	0.16
highway_traffic	test	5	3	12	10	ramp inflow	congestion in the fundamental diagram (model: free-flow q = v_free rho)	0.82	0.40
network	test	10	3	12	1.5	node 0 of a ring of 5 FHN neurons (nodes 0, 1, 3 measured)	the coupling edge 2-3 of the ring (model: chain 3-4-0-1-2, graph partly unknown)	0.22	0.21
rossler	test	3	1	7	8	additive, chaotic	the folding nonlinearity x1 x3	0.30	0.25
two_link_arm	test	4	2	6	2	M(q)^{-1} e1	Coriolis and centrifugal terms (model: M(q) qdd = tau - g(q) - kd qd)	0.04	0.29

Table 6: Where the input enters (C1). Absolute NRMSE (mean over seeds) and paired change vs latent injection (mean of per-seed relative changes). **Bold**: 95% CI (Student-t over seeds) excludes 0, better; underlined: excludes 0, worse. Dev systems: C1-where (5 seeds); TEST systems: TEST-C1, frozen recipe (3 seeds). †: NRMSE above climatology (worse than the constant mean-state predictor). Last rows per block: pre-registered vote counts.

tier	system	latent	concat	film	lora	hyper
dev systems (C1-where)						
shape OOD	battery	0.104	0.238 (+132%)	0.0601 (**-41%**)	0.0559 (**-45%**)	0.0534 (**-47%**)
	bioreactor	0.0364	0.0671 (+84%)	0.0237 (**-35%**)	0.0248 (**-32%**)	0.0276 (**-24%**)
	duffing	0.0758	0.0616 (**-18%**)	0.0632 (**-17%**)	0.0493 (**-35%**)	0.0539 (**-29%**)
	lorenz	0.0269	0.0537 (+100%)	0.0345 (+32%)	0.0251 (-6%)	0.0168 (**-36%**)
	quad_tank	0.161	0.145 (**-10%**)	0.0649 (**-60%**)	0.0503 (**-69%**)	0.0498 (**-69%**)
	vdp	0.0259	0.0258 (-0%)	0.0262 (+1%)	0.0156 (**-40%**)	0.0228 (-12%)
	better / worse vs latent	–	2 / 3 of 6	4 / 2 of 6	5 / 0 of 6	5 / 0 of 6
	beats every other location	0/6	0/6	0/6	1/6	1/6
	weight-space (better of lora/hyper) beats latent and concat: 6/6 (PASS)					
sequence OOD	battery	0.118	0.312 (+172%)	0.0869 (**-24%**)	0.0738 (**-35%**)	0.0733 (**-36%**)
	bioreactor	0.0442	0.146 (+230%)	0.0565 (+28%)	0.0661 (+50%)	0.055 (+25%)
	duffing	0.0773	0.07 (**-9%**)	0.0765 (-1%)	0.0547 (**-29%**)	0.0628 (**-19%**)
	lorenz	0.0272	0.0542 (+99%)	0.0364 (+37%)	0.0259 (-4%)	0.0173 (**-34%**)
	quad_tank	0.168	0.154 (**-8%**)	0.0844 (**-50%**)	0.0712 (**-58%**)	0.0644 (**-62%**)
	vdp	0.0271	0.0274 (+1%)	0.0277 (+2%)	0.0189 (**-30%**)	0.0252 (-7%)
	better / worse vs latent	–	2 / 3 of 6	2 / 3 of 6	4 / 1 of 6	4 / 1 of 6
	beats every other location	1/6	0/6	0/6	2/6	2/6
	weight-space (better of lora/hyper) beats latent and concat: 5/6 (PASS)					
TEST systems (TEST-C1, frozen recipe, run once)						
shape OOD	cartpole	1.14†	2.23† (+96%)	0.396 (**-66%**)	0.152 (**-87%**)	0.103 (**-91%**)
	fhn	0.0328	0.0528 (+60%)	0.0314 (**-4%**)	0.0309 (-5%)	0.0294 (-9%)
	highway_traffic	0.675†	0.357 (**-47%**)	0.0932 (**-86%**)	0.0714 (**-89%**)	0.073 (**-89%**)
	network	0.0691	0.111 (+61%)	0.0645 (-7%)	0.0396 (**-43%**)	0.0446 (**-35%**)
	rossler	0.19	0.344 (+79%)	0.13 (**-31%**)	0.104 (**-45%**)	0.106 (**-44%**)
	two_link_arm	0.331	0.186 (**-44%**)	0.185 (**-44%**)	0.125 (**-62%**)	0.149 (**-55%**)
	better / worse vs latent	–	2 / 3 of 6	5 / 0 of 6	5 / 0 of 6	5 / 0 of 6
	beats every other location	0/6	0/6	0/6	0/6	1/6
	weight-space (better of lora/hyper) beats latent and concat: 5/6 (PASS)					
sequence OOD	cartpole	1.18	2.18† (+85%)	0.478 (**-60%**)	0.246 (**-79%**)	0.216 (**-82%**)
	fhn	0.05	0.0984 (+96%)	0.0523 (+4%)	0.0521 (+4%)	0.0503 (+1%)
	highway_traffic	0.669†	0.351 (**-47%**)	0.096 (**-86%**)	0.0769 (**-88%**)	0.0744 (**-89%**)
	network	0.088	0.166 (+88%)	0.0885 (+1%)	0.062 (**-29%**)	0.0717 (-18%)
	rossler	0.201	0.389 (+92%)	0.135 (**-32%**)	0.105 (**-47%**)	0.108 (**-46%**)
	two_link_arm	0.357	0.197 (**-45%**)	0.196 (**-45%**)	0.14 (**-61%**)	0.165 (**-54%**)
	better / worse vs latent	–	2 / 3 of 6	4 / 0 of 6	5 / 0 of 6	4 / 0 of 6
	beats every other location	0/6	0/6	0/6	1/6	1/6
	weight-space (better of lora/hyper) beats latent and concat: 5/6 (PASS)					

F.2 C1: all tiers and capacity

Table 7: Where the input enters (C1). Absolute NRMSE (mean over seeds) and paired change vs latent injection (mean of per-seed relative changes). **Bold**: 95% CI (Student-t over seeds) excludes 0, better; underlined: excludes 0, worse. Dev systems: C1-where (5 seeds); TEST systems: TEST-C1, frozen recipe (3 seeds). †: NRMSE above climatology (worse than the constant mean-state predictor). Last rows per block: pre-registered vote counts.

tier	system	latent	concat	film	lora	hyper
dev systems (C1-where)						
id	battery	0.114	0.221 (+98%)	0.0609 (**-45%**)	0.0483 (**-56%**)	0.0512 (**-53%**)
	bioreactor	0.035	0.0516 (+48%)	0.0196 (**-44%**)	0.0193 (**-45%**)	0.0229 (**-34%**)
	duffing	0.0761	0.0587 (**-23%**)	0.0614 (**-19%**)	0.0469 (**-38%**)	0.0509 (**-33%**)
	lorenz	0.0302	0.0565 (+87%)	0.0384 (+30%)	0.027 (**-10%**)	0.0203 (**-32%**)
	quad_tank	0.164	0.146 (**-11%**)	0.0677 (**-59%**)	0.0524 (**-68%**)	0.0533 (**-67%**)
	vdp	0.0258	0.0258 (-0%)	0.0262 (+1%)	0.0164 (**-36%**)	0.0226 (**-13%**)
	better / worse vs latent	–	2 / 3 of 6	4 / 2 of 6	6 / 0 of 6	6 / 0 of 6
	beats every other location	0/6	0/6	0/6	1/6	1/6
	weight-space (better of lora/hyper) beats latent and concat: 6/6 (PASS)					
TEST systems (TEST-C1, frozen recipe, run once)						
id	cartpole	1.17†	2.11† (+80%)	0.406 (**-66%**)	0.15 (**-87%**)	0.0944 (**-92%**)
	fhn	0.0298	0.0442 (+48%)	0.0291 (-2%)	0.0281 (-5%)	0.0275 (-7%)
	highway_traffic	0.672	0.354 (**-47%**)	0.0941 (**-86%**)	0.0694 (**-90%**)	0.0727 (**-89%**)
	network	0.0773	0.11 (+42%)	0.0655 (**-15%**)	0.0422 (**-45%**)	0.0485 (**-37%**)
	rossler	0.187	0.328 (+73%)	0.126 (**-32%**)	0.103 (**-44%**)	0.103 (**-44%**)
	two_link_arm	0.336	0.193 (**-42%**)	0.19 (**-44%**)	0.127 (**-62%**)	0.153 (**-54%**)
	better / worse vs latent	–	2 / 3 of 6	5 / 0 of 6	5 / 0 of 6	5 / 0 of 6
	beats every other location	0/6	0/6	0/6	1/6	1/6
	weight-space (better of lora/hyper) beats latent and concat: 5/6 (PASS)					

Table 8: What each conditioner can express at each budget (hyperkkl.conditioning.match_width on the phase-1 decoder shape, dec. width 128, depth 3, context dim 16). The dense-subspace ablation ('hyper') generates every weight update inside a FIXED k-dimensional linear subspace (k = trunk output width); HyperKKL ('lora') has rank $\leq r$ per layer with factors bilinear in a w-dim trunk feature. Observed: widths in the runs' results.json (– = no run yet).

system (n_z)	budget	kind	width	params	effective parameterisation	observed
vdp (5)	35k	HyperKKL r=4	11	35953	rank $\leq$ 4 per layer (max ranks [4, 4, 4, 2]), U(t)V(t) bilinear in a 11-dim trunk feature	11
vdp (5)	35k	dense (abl.)	1	35138	Delta W in a fixed 1-dim linear subspace (trunk 16→64→1); heads 34050 x 1	1
vdp (5)	140k	HyperKKL r=1	149	140449	rank $\leq$ 1 per layer (max ranks [1, 1, 1, 1]), U(t)V(t) bilinear in a 149-dim trunk feature	–
vdp (5)	140k	HyperKKL r=4	44	140596	rank $\leq$ 4 per layer (max ranks [4, 4, 4, 2]), U(t)V(t) bilinear in a 44-dim trunk feature	44
vdp (5)	140k	HyperKKL r=16	11	142921	rank $\leq$ 16 per layer (max ranks [5, 16, 16, 2]), U(t)V(t) bilinear in a 11-dim trunk feature	–
vdp (5)	140k	dense (abl.)	4	137480	Delta W in a fixed 4-dim linear subspace (trunk 16→64→4); heads 34050 x 4	4
vdp (5)	560k	HyperKKL r=4	170	560176	rank $\leq$ 4 per layer (max ranks [4, 4, 4, 2]), U(t)V(t) bilinear in a 170-dim trunk feature	170
vdp (5)	560k	dense (abl.)	16	546848	Delta W in a fixed 16-dim linear subspace (trunk 16→64→16); heads 34050 x 16	16
battery (7)	35k	HyperKKL r=4	11	36181	rank $\leq$ 4 per layer (max ranks [4, 4, 4, 4]), U(t)V(t) bilinear in a 11-dim trunk feature	11
battery (7)	35k	dense (abl.)	1	35781	Delta W in a fixed 1-dim linear subspace (trunk 16→64→1); heads 34693 x 1	1
battery (7)	140k	HyperKKL r=1	148	140103	rank $\leq$ 1 per layer (max ranks [1, 1, 1, 1]), U(t)V(t) bilinear in a 148-dim trunk feature	–
battery (7)	140k	HyperKKL r=4	44	141484	rank $\leq$ 4 per layer (max ranks [4, 4, 4, 4]), U(t)V(t) bilinear in a 44-dim trunk feature	44
battery (7)	140k	HyperKKL r=16	11	143833	rank $\leq$ 16 per layer (max ranks [7, 16, 16, 5]), U(t)V(t) bilinear in a 11-dim trunk feature	–
battery (7)	140k	dense (abl.)	4	140052	Delta W in a fixed 4-dim linear subspace (trunk 16→64→4); heads 34693 x 4	4
battery (7)	560k	HyperKKL r=4	169	560109	rank $\leq$ 4 per layer (max ranks [4, 4, 4, 4]), U(t)V(t) bilinear in a 169-dim trunk feature	169
battery (7)	560k	dense (abl.)	16	557136	Delta W in a fixed 16-dim linear subspace (trunk 16→64→16); heads 34693 x 16	16

F.3 C3: every scenario and plant

Table 9: When does a learned KKL observer win (C3)? id, dev systems. Counts of systems where HyperKKL ($r = 4$, the single learned method) beats the comparator with a paired 95% CI excluding 0; (kL) = comparator significantly better on k systems. (a) MODEL-BASED filters (ekf, ukf, pf, kkl_mb, Huber and condition-oracle variants); a filter is FAILED when its NRMSE exceeds climatology (constant mean-state predictor, same normalisation) and 'working' compares against the best filter that did not fail (*: no filter works on some systems). (b) DATA-DRIVEN same-data baselines (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle), trained on the same real-plant data; 'pend. m/N' = the new baselines are merged in m of N runs. Model error: median over dev plants. Per-system absolute numbers: Table 10.

		(a) vs model-based filters					(b) vs data-driven	
scenario	model err.	best	working	nominal	rob./orac.	failed	best	NLOX
exact	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	1/48	pend. 0/30	3/6
exact+outliers	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	13/48	pend. 0/30	1/6 (4L)
colored_t	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	1/48	pend. 0/18	4/6
mismatch10	0.09	5/6 (1L)	5/6 (1L)	5/6 (1L)	5/6 (1L)	13/48	0/5 (3L) [16/18]	3/6
mismatch20	0.17	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	15/48	0/1 (1L) [7/18]	4/6
drift30 (post-train)	0.24	0/6 (5L)	0/5 (5L)*	1/6 (3L)	1/6 (5L)	15/48	pend. 0/30	2/6 (1L)
mismatch30	0.24	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	15/48	0/6 (4L)	3/6
mismatch50	0.37	6/6	4/4*	6/6	6/6	23/48	pend. 1/18	3/6
partial	0.44	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	14/48	0/6 (4L)	3/6 (1L)
realistic	0.63	6/6	4/4*	6/6	6/6	24/48	1/6 (4L)	5/6 (1L)

Table 10: C3 per system, sequence OOD (dev). BL = HyperKKL ($r = 4$; HKKL), BF = best model-based filter, BW = best WORKING model-based filter (below climatology; $k/n^\dagger$ = failed / present), DD = best data-driven same-data baseline; NRMSE after each name. Changes: paired relative change of BL ± 95% CI half-width (Student-t over seeds; bold/underline: significant). FAILED filters are never used as the reference. Abbreviations: -or oracle, -hu Huber, koop koopman_kf, koop-bi bilinear, gb greybox.

scenario / system (clim.)	BL	BF	BW	(a) BL vs BW	DD	(b) BL vs DD
exact						
battery (0.78)	HKKL 0.0738	ekf 0.00521	ekf 0.00521 (1/8†)	+1317% ±363	pend. 0/5	–
bioreactor (0.928)	HKKL 0.0661	ekf 0.00619	ekf 0.00619 (0/8†)	+967% ±144	pend. 0/5	–
duffing (1.69)	HKKL 0.0547	ukf 0.00488	ukf 0.00488 (0/8†)	+1021% ±88	pend. 0/5	–
lorenz (1)	HKKL 0.0259	ukf 0.00645	ukf 0.00645 (0/8†)	+301% ±91	pend. 0/5	–
quad_tank (0.651)	HKKL 0.0712	ukf 0.0141	ukf 0.0141 (0/8†)	+407% ±31	pend. 0/5	–
vdp (1.12)	HKKL 0.0189	ekf 0.00578	ekf 0.00578 (0/8†)	+227% ±10	pend. 0/5	–
exact+outliers						
battery (0.78)	HKKL 2.13	pf 0.0293	pf 0.0293 (4/8†)	+7168% ±3883	pend. 0/5	–
bioreactor (0.928)	HKKL 0.104	ukf-hu 0.00619	ukf-hu 0.00619 (1/8†)	+1582% ±495	pend. 0/5	–
duffing (1.69)	HKKL 2.3	ukf-hu 0.00481	ukf-hu 0.00481 (1/8†)	+47783% ±6123	pend. 0/5	–
lorenz (1)	HKKL 0.599	ukf-hu 0.00693	ukf-hu 0.00693 (3/8†)	+8543% ±220	pend. 0/5	–
quad_tank (0.651)	HKKL 1.49	ekf-or 0.0252	ekf-or 0.0252 (5/8†)	+5807% ±237	pend. 0/5	–
vdp (1.12)	HKKL 0.84	ukf-hu 0.0059	ukf-hu 0.0059 (1/8†)	+14138% ±743	pend. 0/5	–
colored_t						
battery (0.78)	HKKL 0.0595	ekf-or 0.00989	ekf-or 0.00989 (1/8†)	+502% ±94	pend. 0/3	–
bioreactor (0.928)	HKKL 0.0645	ekf-or 0.00624	ekf-or 0.00624 (0/8†)	+933% ±252	pend. 0/3	–
duffing (1.69)	HKKL 0.0994	ekf-or 0.0104	ekf-or 0.0104 (0/8†)	+860% ±45	pend. 0/3	–
lorenz (1)	HKKL 0.0406	ukf-or 0.0177	ukf-or 0.0177 (0/8†)	+129% ±8	pend. 0/3	–
quad_tank (0.651)	HKKL 0.0696	ekf-or 0.0169	ekf-or 0.0169 (0/8†)	+313% ±21	pend. 0/3	–
vdp (1.12)	HKKL 0.035	pf 0.0134	pf 0.0134 (0/8†)	+161% ±18	pend. 0/3	–
mismatch10						
battery (0.776)	HKKL 0.0646	ekf 0.0605	ekf 0.0605 (1/8†)	+7% ±48	gb-ekf 0.00715	+800% ±316
bioreactor (0.841)	HKKL 0.0613	ekf-or 0.356	ekf-or 0.356 (5/8†)	**-83%** ±2	gb-ekf-or 0.0214	+187% ±25
duffing (1.7)	HKKL 0.0555	ukf 0.00797	ukf 0.00797 (0/8†)	+596% ±17	gb-ukf 0.005	+1011% ±40
lorenz (0.911)	HKKL 0.0219	ukf-or 0.157	ukf-or 0.157 (0/8†)	**-86%** ±3	gb-ekf-or 0.0201	+9% ±30
quad_tank (0.607)	HKKL 0.0513	kkl-mb 0.434	kkl-mb 0.434 (7/8†)	**-88%** ±2	pend. 1/3	–
vdp (1.11)	HKKL 0.02	ekf-or 0.0268	ekf-or 0.0268 (0/8†)	**-25%** ±9	gb-ekf 0.00634	+216% ±43
mismatch20						
battery (0.77)	HKKL 0.0569	ekf 0.12	ekf 0.12 (1/8†)	**-53%** ±6	gb-ekf 0.0111	+413% ±82
bioreactor (0.769)	HKKL 0.0783	ekf-or 0.401	ekf-or 0.401 (6/8†)	**-80%** ±10	pend. 2/3	–
duffing (1.7)	HKKL 0.0536	ukf 0.0142	ukf 0.0142 (0/8†)	+278% ±13	pend. 0/3	–
lorenz (0.616)	HKKL 0.00772	kkl-mb 0.0248	kkl-mb 0.0248 (0/8†)	**-69%** ±10	pend. 2/3	–
quad_tank (0.587)	HKKL 0.0421	kkl-mb 0.618†	none (8/8†)	–	pend. 0/3	–
vdp (1.09)	HKKL 0.0199	ekf-or 0.0429	ekf-or 0.0429 (0/8†)	**-54%** ±7	pend. 0/3	–
drift30 (post-train)						
battery (0.78)	HKKL 0.204	ekf-or 0.161	ekf-or 0.161 (1/8†)	+27% ±4	pend. 0/5	–
bioreactor (0.928)	HKKL 4.61	ekf-or 0.255	ekf-or 0.255 (6/8†)	+1708% ±21	pend. 0/5	–
duffing (1.69)	HKKL 0.0549	ekf-or 0.0184	ekf-or 0.0184 (0/8†)	+198% ±23	pend. 0/5	–

Table 10: C3 per system, sequence OOD (dev). BL = HyperKKL ($r = 4$; HKKL), BF = best model-based filter, BW = best WORKING model-based filter (below climatology; $k/n^{\dagger}$ = failed / present), DD = best data-driven same-data baseline; NRMSE after each name. Changes: paired relative change of BL ± 95% CI half-width (Student-t over seeds; bold/underline: significant). FAILED filters are never used as the reference. Abbreviations: -or oracle, -hu Huber, koop koopman_kf, koop-bi bilinear, gb greybox.

scenario / system (clim.)	BL	BF	BW	(a) BL vs BW	DD	(b) BL vs DD
lorenz (1)	HKKL 0.485	ukf-or 0.469	ukf-or 0.469 (0/8†)	+3% ±0	pend. 0/5	–
quad_tank (0.651)	HKKL 1.67	kkl-mb 1.58†	none (8/8†)	–	pend. 0/5	–
vdp (1.12)	HKKL 0.086	ekf-or 0.0711	ekf-or 0.0711 (0/8†)	+21% ±2	pend. 0/5	–
mismatch30						
battery (0.765)	HKKL 0.0515	ekf 0.177	ekf 0.177 (1/8†)	**-71%** ±7	gb-ekf 0.0151	+241% ±18
bioreactor (0.708)	HKKL 0.0977	ekf-or 0.247	ekf-or 0.247 (6/8†)	**-60%** ±35	nlox 0.0353	+194% ±317
duffing (1.71)	HKKL 0.0561	ukf 0.0216	ukf 0.0216 (0/8†)	+160% ±20	gb-ukf 0.00592	+848% ±22
lorenz (0.614)	HKKL 0.007	kkl-mb 0.0142	kkl-mb 0.0142 (1/8†)	**-51%** ±5	nlox 0.00682	+3% ±38
quad_tank (0.581)	HKKL 0.0422	kkl-mb 0.787†	none (8/8†)	–	gb-ekf 0.0129	+226% ±24
vdp (1.16)	HKKL 0.0205	ekf-or 0.0578	ekf-or 0.0578 (0/8†)	**-65%** ±7	gb-ekf 0.00976	+110% ±52
mismatch50						
battery (0.756)	HKKL 0.0503	ekf 0.284	ekf 0.284 (1/8†)	**-82%** ±9	pend. 0/3	–
bioreactor (0.614)	HKKL 0.135	ukf-or 0.353	ukf-or 0.353 (6/8†)	**-62%** ±23	pend. 1/3	–
duffing (1.73)	HKKL 0.0549	ekf-or 0.0361	ekf-or 0.0361 (0/8†)	+52% ±2	pend. 0/3	–
lorenz (0.814)	HKKL 0.0107	kkl-mb 0.0708	kkl-mb 0.0708 (1/8†)	**-85%** ±17	pend. 0/3	–
quad_tank (0.6)	HKKL 0.0442	kkl-mb 1.38†	none (8/8†)	–	pend. 0/3	–
vdp (1.23)	HKKL 0.0203	kkl-mb 0.0777	kkl-mb 0.0777 (0/8†)	**-74%** ±3	pend. 0/3	–
partial						
battery (0.78)	HKKL 0.0744	ukf-or 0.237	ukf-or 0.237 (1/8†)	**-69%** ±12	gb-ekf 0.0191	+295% ±190
bioreactor (0.928)	HKKL 0.0753	ukf 1.63†	none (8/8†)	–	koop 0.0271	+179% ±165
duffing (1.69)	HKKL 0.0536	ukf-or 0.0845	ukf-or 0.0845 (0/8†)	**-37%** ±14	gb-ukf 0.00775	+593% ±191
lorenz (1)	HKKL 0.0308	ukf-or 0.529	ukf-or 0.529 (5/8†)	**-94%** ±4	nlox 0.0287	+7% ±59
quad_tank (0.651)	HKKL 0.0717	ekf-or 0.0202	ekf-or 0.0202 (0/8†)	+255% ±10	gb-ekf 0.0142	+406% ±40
vdp (1.12)	HKKL 0.0179	kkl-mb 0.0819	kkl-mb 0.0819 (0/8†)	**-78%** ±3	gb-ekf 0.0257	**-30%** ±15
realistic						
battery (0.765)	HKKL 0.0642	ukf-or 0.29	ukf-or 0.29 (1/8†)	**-78%** ±1	gb-ekf-or 0.0214	+202% ±41
bioreactor (0.708)	HKKL 0.0987	ekf-or 2.44†	none (8/8†)	–	nlox 0.0389	+157% ±106
duffing (1.71)	HKKL 0.0997	ekf-or 0.127	ekf-or 0.127 (0/8†)	**-21%** ±2	gb-ekf 0.0144	+591% ±21
lorenz (0.614)	HKKL 0.0104	kkl-mb 0.0257	kkl-mb 0.0257 (7/8†)	**-58%** ±11	nlox 0.0129	**-19%** ±12
quad_tank (0.581)	HKKL 0.0441	kkl-mb 1.01†	none (8/8†)	–	gb-ekf 0.017	+160% ±16
vdp (1.16)	HKKL 0.0324	kkl-mb 0.125	kkl-mb 0.125 (0/8†)	**-74%** ±1	gb-ekf 0.0388	**-16%** ±5

† FAILED: NRMSE above climatology (in brackets after the system).

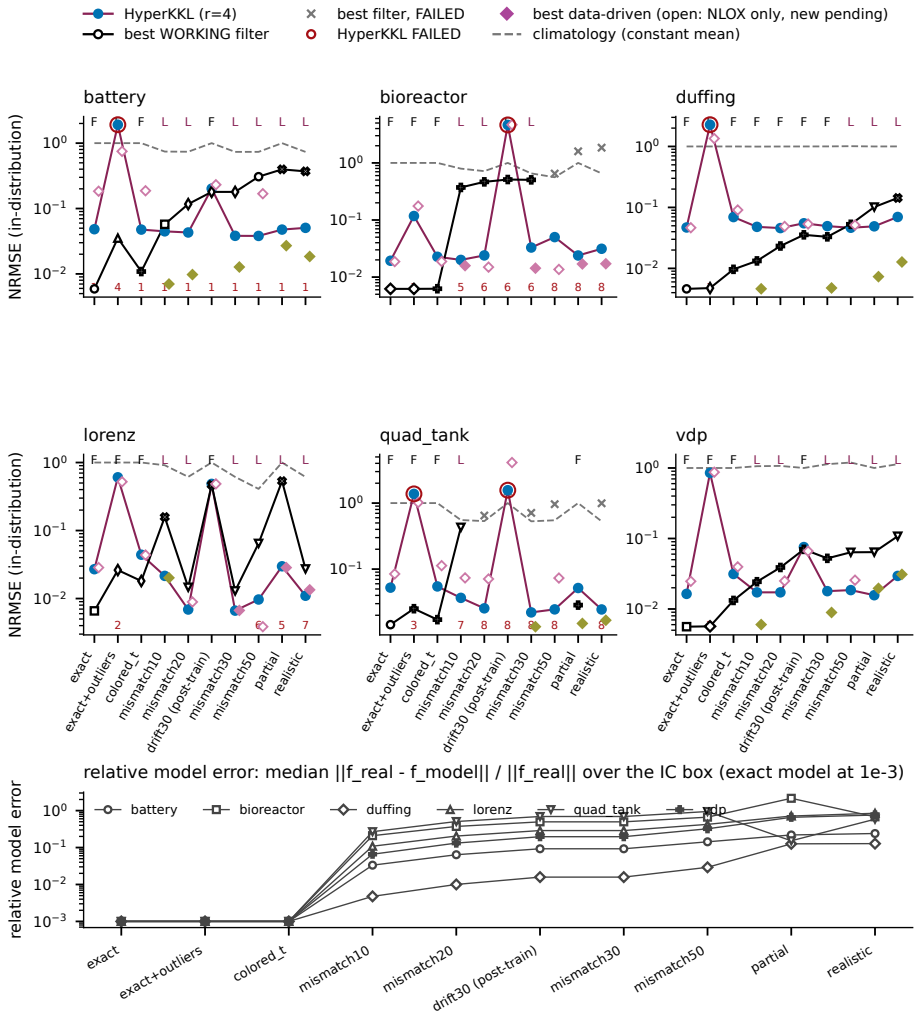
Table 11: C3 per system, id (dev). BL = HyperKKL ($r = 4$; HKKL), BF = best model-based filter, BW = best WORKING model-based filter (below climatology; $k/n^{\dagger}$ = failed / present), DD = best data-driven same-data baseline; NRMSE after each name. Changes: paired relative change of BL ± 95% CI half-width (Student-t over seeds; bold/underline: significant). FAILED filters are never used as the reference. Abbreviations: -or oracle, -hu Huber, koop koopman_kf, koop-bi bilinear, gb greybox.

scenario / system (clim.)	BL	BF	BW	(a) BL vs BW	DD	(b) BL vs DD
exact						
battery (1)	HKKL 0.0483	ekf 0.0059	ekf 0.0059 (1/8†)	+718% ±71	pend. 0/5	–
bioreactor (1)	HKKL 0.0193	ekf-hu 0.00622	ekf-hu 0.00622 (0/8†)	+211% ±46	pend. 0/5	–
duffing (1)	HKKL 0.0469	ekf 0.00461	ekf 0.00461 (0/8†)	+916% ±90	pend. 0/5	–
lorenz (1)	HKKL 0.027	ukf 0.00659	ukf 0.00659 (0/8†)	+310% ±101	pend. 0/5	–
quad_tank (1)	HKKL 0.0524	ekf 0.0144	ekf 0.0144 (0/8†)	+264% ±17	pend. 0/5	–
vdp (1)	HKKL 0.0164	ukf 0.00561	ukf 0.00561 (0/8†)	+192% ±5	pend. 0/5	–
exact+outliers						
battery (1)	HKKL 1.91	pf 0.0353	pf 0.0353 (4/8†)	+5328% ±1402	pend. 0/5	–
bioreactor (1)	HKKL 0.117	ekf-hu 0.00622	ekf-hu 0.00622 (1/8†)	+1790% ±858	pend. 0/5	–
duffing (1)	HKKL 2.27	ukf-hu 0.00475	ukf-hu 0.00475 (1/8†)	+47710% ±6069	pend. 0/5	–
lorenz (1)	HKKL 0.607	ukf-hu 0.0262	ukf-hu 0.0262 (2/8†)	+2215% ±57	pend. 0/5	–
quad_tank (1)	HKKL 1.38	ekf-or 0.0251	ekf-or 0.0251 (3/8†)	+5380% ±231	pend. 0/5	–
vdp (1)	HKKL 0.854	ekf-hu 0.00568	ekf-hu 0.00568 (2/8†)	+14939% ±740	pend. 0/5	–
colored_t						
battery (1)	HKKL 0.0476	ekf-or 0.0109	ekf-or 0.0109 (1/8†)	+338% ±14	pend. 0/3	–
bioreactor (1)	HKKL 0.0227	ekf-or 0.00625	ekf-or 0.00625 (0/8†)	+263% ±189	pend. 0/3	–
duffing (1)	HKKL 0.0694	ekf-or 0.00965	ekf-or 0.00965 (0/8†)	+619% ±37	pend. 0/3	–
lorenz (1)	HKKL 0.0441	ukf-hu 0.0181	ukf-hu 0.0181 (0/8†)	+143% ±4	pend. 0/3	–
quad_tank (1)	HKKL 0.0547	ekf-or 0.0172	ekf-or 0.0172 (0/8†)	+218% ±34	pend. 0/3	–
vdp (1)	HKKL 0.0314	ekf-or 0.0132	ekf-or 0.0132 (0/8†)	+137% ±20	pend. 0/3	–

Table 11: C3 per system, id (dev). BL = HyperKKL ($r = 4$; HKKL), BF = best model-based filter, BW = best WORKING model-based filter (below climatology; $k/n^{\dagger}$ = failed / present), DD = best data-driven same-data baseline; NRMSE after each name. Changes: paired relative change of BL ± 95% CI half-width (Student-t over seeds; bold/underline: significant). FAILED filters are never used as the reference. Abbreviations: -or oracle, -hu Huber, koop koopman_kf, koop-bi bilinear, gb greybox.

scenario / system (clim.)	BL	BF	BW	(a) BL vs BW	DD	(b) BL vs DD
mismatch10						
battery (0.743)	HKKL 0.0448	ukf 0.0574	ukf 0.0574 (1/8†)	**-22%** ±17	gb-ekf 0.007	+540% ±164
bioreactor (0.794)	HKKL 0.02	ekf-or 0.372	ekf-or 0.372 (5/8†)	**-95%** ±1	nlox 0.0159	+28% ±38
duffing (0.995)	HKKL 0.048	ekf-or 0.0133	ekf-or 0.0133 (0/8†)	+262% ±36	gb-ekf 0.00463	+937% ±104
lorenz (0.912)	HKKL 0.0215	ukf-or 0.158	ukf-or 0.158 (0/8†)	**-86%** ±3	gb-ekf-or 0.0202	+7% ±33
quad_tank (0.554)	HKKL 0.0367	kkl-mb 0.429	kkl-mb 0.429 (7/8†)	**-91%** ±2	pend. 1/3	–
vdp (1.06)	HKKL 0.0172	ekf-or 0.0244	ekf-or 0.0244 (0/8†)	**-29%** ±8	gb-ukf 0.00601	+187% ±37
mismatch20						
battery (0.739)	HKKL 0.043	ukf-hu 0.117	ukf-hu 0.117 (1/8†)	**-63%** ±5	gb-ekf 0.00977	+340% ±60
bioreactor (0.718)	HKKL 0.024	ekf-or 0.466	ekf-or 0.466 (6/8†)	**-95%** ±2	pend. 2/3	–
duffing (0.998)	HKKL 0.0458	ekf-or 0.0233	ekf-or 0.0233 (0/8†)	+97% ±25	pend. 0/3	–
lorenz (0.616)	HKKL 0.0069	kkl-mb 0.0146	kkl-mb 0.0146 (0/8†)	**-53%** ±9	pend. 2/3	–
quad_tank (0.534)	HKKL 0.0255	kkl-mb 0.647†	none (8/8†)	–	pend. 0/3	–
vdp (1.07)	HKKL 0.0172	ekf-or 0.0388	ekf-or 0.0388 (0/8†)	**-56%** ±5	pend. 0/3	–
drift30 (post-train)						
battery (1)	HKKL 0.202	ukf-hu 0.18	ukf-hu 0.18 (1/8†)	+12% ±2	pend. 0/5	–
bioreactor (1)	HKKL 4.63	ekf-or 0.512	ekf-or 0.512 (6/8†)	+805% ±8	pend. 0/5	–
duffing (1)	HKKL 0.055	ekf-or 0.0355	ekf-or 0.0355 (0/8†)	+55% ±10	pend. 0/5	–
lorenz (1)	HKKL 0.485	ukf-or 0.469	ukf-or 0.469 (0/8†)	+4% ±1	pend. 0/5	–
quad_tank (1)	HKKL 1.57	kkl-mb 1.5†	none (8/8†)	–	pend. 0/5	–
vdp (1)	HKKL 0.0756	ekf-or 0.0701	ekf-or 0.0701 (0/8†)	+8% ±1	pend. 0/5	–
mismatch30						
battery (0.735)	HKKL 0.0382	ukf-hu 0.18	ukf-hu 0.18 (1/8†)	**-79%** ±4	gb-ekf 0.0127	+200% ±35
bioreactor (0.655)	HKKL 0.0329	ekf-or 0.505	ekf-or 0.505 (6/8†)	**-93%** ±5	nlox 0.0143	+140% ±251
duffing (1)	HKKL 0.0493	ekf-or 0.0329	ekf-or 0.0329 (0/8†)	+50% ±32	gb-ekf 0.00477	+933% ±214
lorenz (0.614)	HKKL 0.00664	kkl-mb 0.0129	kkl-mb 0.0129 (0/8†)	**-49%** ±14	nlox 0.00669	-0% ±44
quad_tank (0.53)	HKKL 0.0221	kkl-mb 0.711†	none (8/8†)	–	gb-ekf 0.0134	+65% ±18
vdp (1.13)	HKKL 0.0179	ekf-or 0.0523	ekf-or 0.0523 (0/8†)	**-66%** ±7	gb-ekf 0.00893	+101% ±35
mismatch50						
battery (0.736)	HKKL 0.0381	ekf 0.307	ekf 0.307 (1/8†)	**-88%** ±5	pend. 0/3	–
bioreactor (0.561)	HKKL 0.0501	ukf-or 0.651†	none (8/8†)	–	pend. 1/3	–
duffing (1.01)	HKKL 0.0464	ekf-or 0.0529	ekf-or 0.0529 (0/8†)	**-12%** ±11	pend. 0/3	–
lorenz (0.409)	HKKL 0.00968	kkl-mb 0.0647	kkl-mb 0.0647 (6/8†)	**-85%** ±17	pend. 0/3	–
quad_tank (0.549)	HKKL 0.0246	kkl-mb 0.962†	none (8/8†)	–	pend. 0/3	–
vdp (1.2)	HKKL 0.0185	kkl-mb 0.0637	kkl-mb 0.0637 (0/8†)	**-71%** ±3	pend. 0/3	–
partial						
battery (1)	HKKL 0.0476	ukf-or 0.397	ukf-or 0.397 (1/8†)	**-88%** ±4	gb-ukf 0.027	+77% ±74
bioreactor (1)	HKKL 0.0239	ukf 1.6†	none (8/8†)	–	nlox 0.017	+42% ±37
duffing (1)	HKKL 0.049	kkl-mb 0.101	kkl-mb 0.101 (0/8†)	**-52%** ±15	gb-ukf 0.00732	+571% ±271
lorenz (1)	HKKL 0.0298	ukf-or 0.533	ukf-or 0.533 (5/8†)	**-94%** ±3	nlox 0.0285	+4% ±47
quad_tank (1)	HKKL 0.0517	ekf-or 0.0285	ekf-or 0.0285 (0/8†)	+82% ±4	gb-ekf 0.015	+244% ±16
vdp (1)	HKKL 0.0157	kkl-mb 0.064	kkl-mb 0.064 (0/8†)	**-75%** ±3	gb-ekf 0.0196	-18% ±36
realistic						
battery (0.735)	HKKL 0.0506	ukf-or 0.37	ukf-or 0.37 (1/8†)	**-86%** ±1	gb-ekf-or 0.0185	+174% ±11
bioreactor (0.655)	HKKL 0.0313	kkl-mb 1.85†	none (8/8†)	–	nlox 0.0171	+85% ±62
duffing (1)	HKKL 0.0697	ukf-or 0.143	ukf-or 0.143 (0/8†)	**-51%** ±2	gb-ekf 0.0127	+448% ±24
lorenz (0.614)	HKKL 0.011	kkl-mb 0.0273	kkl-mb 0.0273 (7/8†)	**-58%** ±10	nlox 0.0134	**-18%** ±12
quad_tank (0.53)	HKKL 0.0245	kkl-mb 0.994†	none (8/8†)	–	gb-ekf 0.0168	+46% ±10
vdp (1.13)	HKKL 0.0294	kkl-mb 0.108	kkl-mb 0.108 (0/8†)	**-73%** ±1	gb-ekf 0.031	-5% ±8

† FAILED: NRMSE above climatology (in brackets after the system).



Top marks: L = HyperKKL beats the best working model-based filter (paired 95% CI), F = filter wins, blank = n.s.; red numbers = filters above climatology (FAILED).
x ordered by median model error over dev plants; 'p' = pending. NRMSE on a log axis. FAILED = NRMSE above climatology.

Figure 4: As Figure 3, in-distribution tier (`id`). Climatology for `id` is analytic for the nominal plant and a proxy for a perturbed plant (Appendix E).

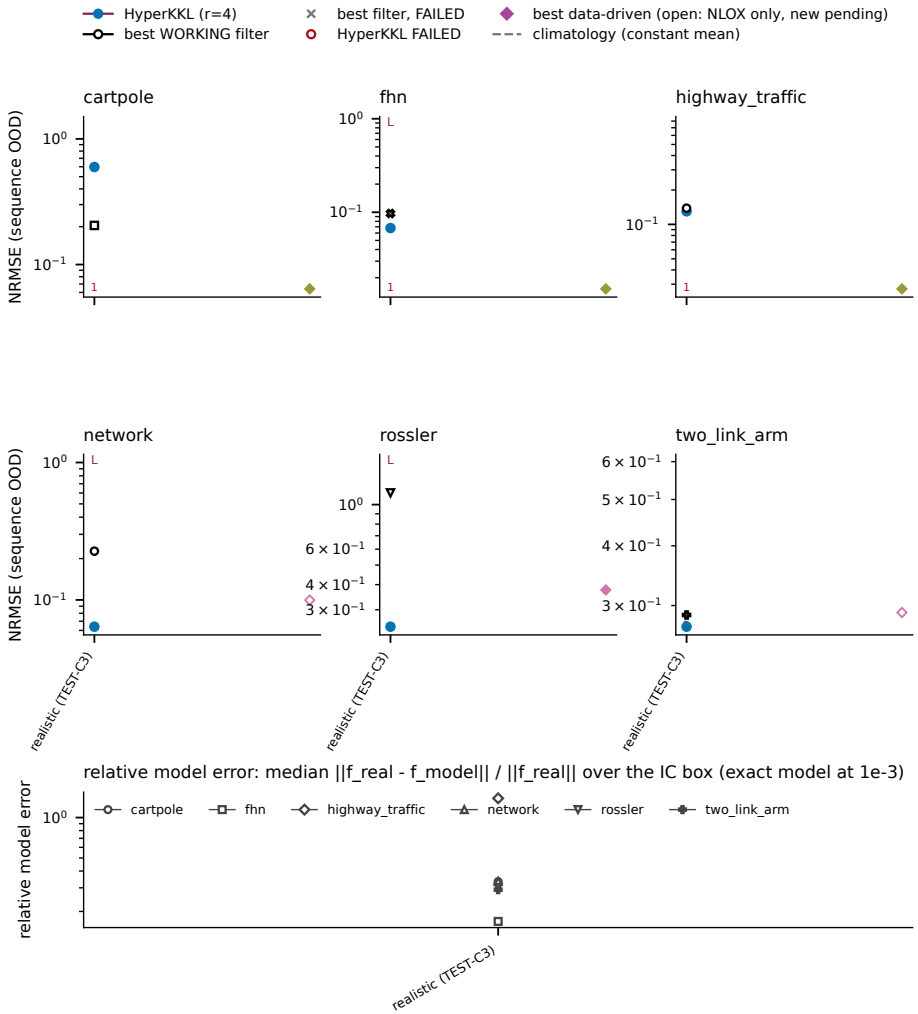
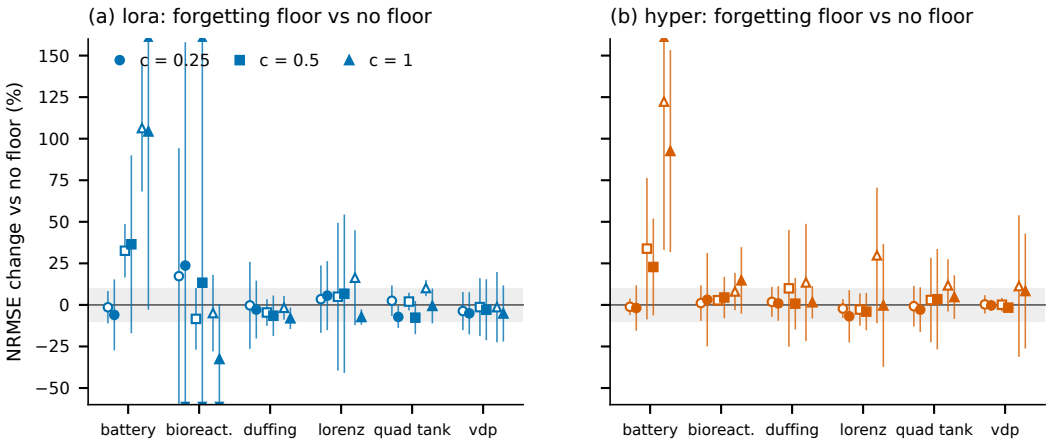


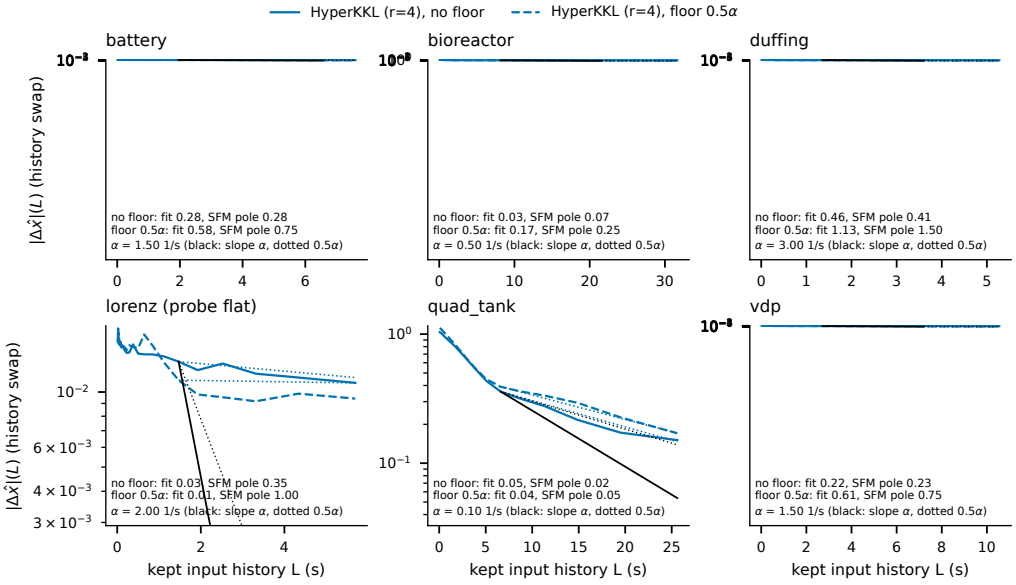
Figure 5: As Figure 3 for the held-out plants under the realistic deployment (`TEST-C3`, frozen recipe, run once): PASS* at the time of generation (17/18 jobs finished); plants without finished runs are marked PENDING.

F.4 C2: forgetting floor, memory probe and window sweeps



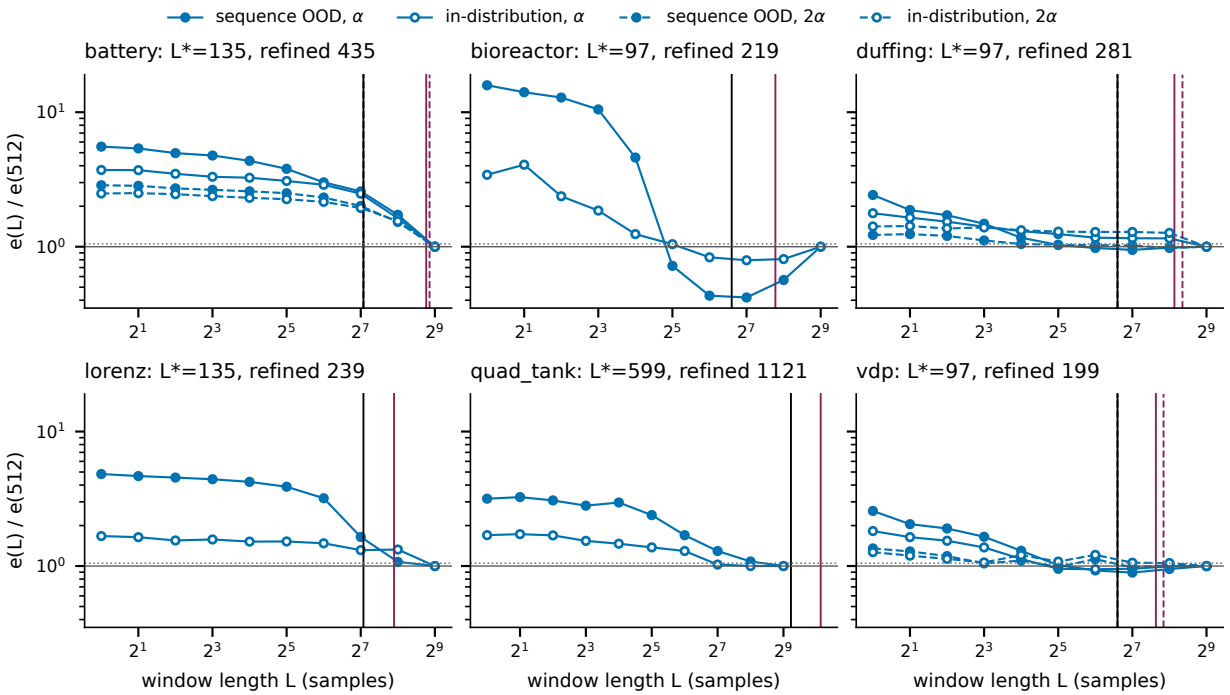
Context forced to forget at >= c alpha. Filled: sequence OOD; open: id (left/right of each pair). Bars: 95% CI over 3 paired seeds (C1-where seeds 0-2), arrows = clipped. Grey band: +-10% equivalence margin.
Pre-registered rule (c = 0.5): FAIL. no loss vs no floor (lora & hyper, id & seq): 4/6 (need 5/6); TOST(10%) non-inferior: 0/6; probe decays at >= floor rate: 0/6 systems.

Figure 6: C2 forgetting-floor intervention: change in NRMSE when the SFM context is forced to forget at $\geq c\alpha$, relative to no floor (paired, 3 seeds, 95% CI; grey band: $\pm 10\%$ equivalence margin). The pre-registered rule at $c = 0.5$ fails.



Memory probe (id tier, 16 episodes, geometric mean over seeds): keep the last L inputs, swap older history for another episode's. Coloured dotted: fitted tail rate (1/s).
Black: decay at the design rate alpha (Lemma 1) and at 0.5 alpha (the floor), anchored at the no-floor HyperKKL tail. Lemma 2 requires probe rate >= floor with a floor. Lorenz shown although flat.

Figure 7: C2 memory probe (`id`, 16 episodes, geometric mean over seeds): decoded change $|\Delta\hat{x}|$ when the last L seconds of input are kept and everything older is swapped for another episode's input, with and without a 0.5α forgetting floor. Black lines: decay at the design rate α (Lemma 11) and at 0.5α; coloured dotted lines: fitted tail rates. The `lorenz` probe is flat.



lora@window, mean over 3 seeds. Black: theory window $L^* = \ln(1/\eta)/\alpha$ ($\eta = 0.05$); wine: refined $L^* = (\ln(1/\eta) + \ln \ell)/\alpha$, ℓ = phase-1 $l_{dec,\,q95}$ (post hoc). Dotted: +5%.

Figure 8: C2 window sweeps (HyperKKL $r = 4$ with a raw window encoder, mean over 3 seeds): $e(L)/e(512)$ against the window length in samples at the default bandwidth and, on three plants, at 2α. Black: pre-registered window $L^* = \ln(1/\eta)/\alpha$ ($\eta = 0.05$); wine: refined $L^* = (\ln(1/\eta) + \ln \ell)/\alpha$ with the phase-1 proxy for ℓ (post hoc). Dotted: +5%.

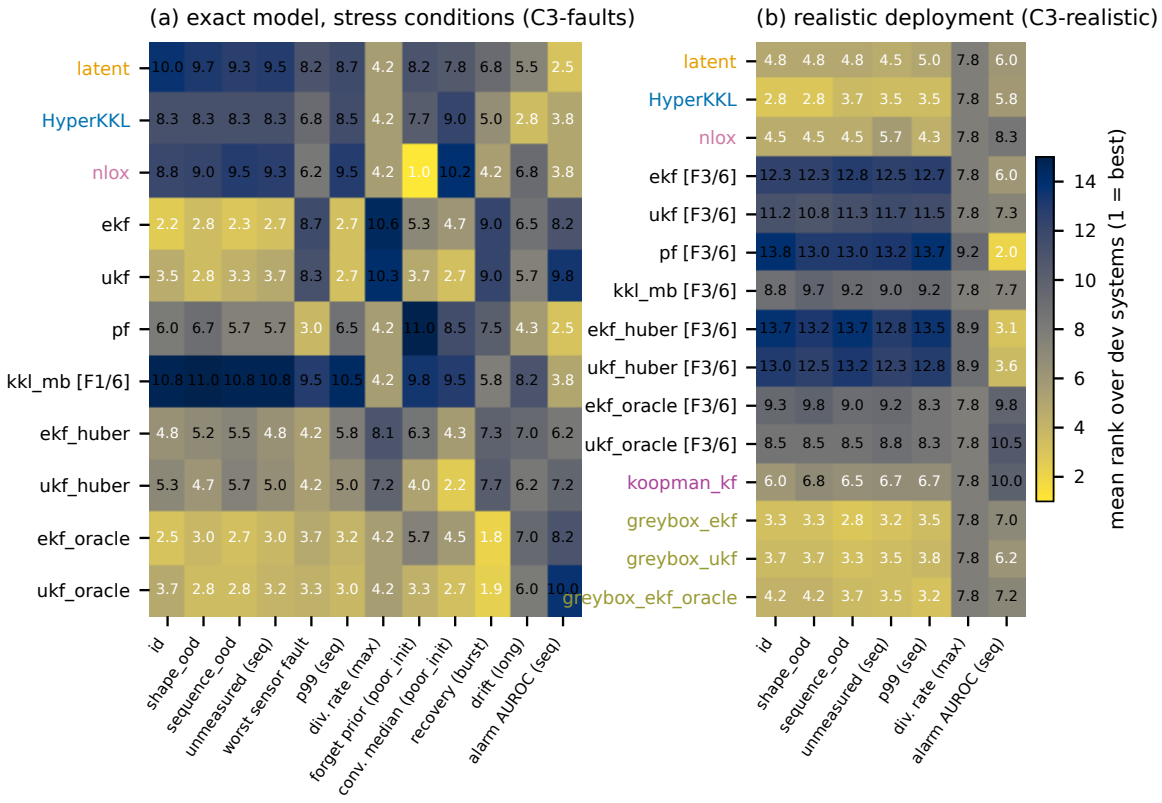
F.5 Deployment scorecard

Table 12: Deployment scorecard, realistic deployment (C3-realistic: partial model + 30% parameter error + AR(1) Student-t sensor noise), dev systems, mean over seeds. Columns as in hyperkkl/tree.py Groups: KKL (latent, lora, hyper), DATA-DRIVEN same-data baselines (nlox, koopman_kf, greybox_*), MODEL-BASED filters (the rest). (sequence OOD; div. = max divergence rate over all blocks; AUROC = output-residual alarm). FAILED: sequence-OOD NRMSE above climatology (value next to the system name).

system	method	id	shape	seq.	unmeas.	p99	div.	AUROC	μs/step	status
battery (0.765)	latent	0.1361	0.1325	0.1350	0.1432	0.2184	0.00	0.955	1.61e+03	
	lora	0.0506	0.0568	0.0642	0.0698	0.1495	0.00	0.940	1.66e+03	
	hyper	0.0478	0.0510	0.0618	0.0665	0.0954	0.00	0.945	1.17e+03	
	nlox	0.1589	0.1902	0.2837	0.3521	0.3630	0.00	0.780	1.04e+03	
	ekf	0.3725	0.3304	0.2959	0.3365	0.4376	0.00	0.878	1.14e+04	
	ukf	0.3709	0.2985	0.2925	0.3321	0.4353	0.00	0.824	6.56e+03	
	pf	0.3765	0.3042	0.2940	0.3288	0.4374	0.00	0.943	5.9e+03	
	kkl_mb	1.3581	1.2096	1.4683	1.5779	2.1881	0.00	0.842	1.69e+04	FAILED
	ekf_huber	0.3730	0.3211	0.2951	0.3203	0.4358	0.00	0.882	1.18e+04	
	ukf_huber	0.3720	0.2997	0.2942	0.3201	0.4352	0.00	0.871	7.42e+03	
	ekf_oracle	0.3715	0.3235	0.2929	0.3310	0.4347	0.00	0.862	1.25e+04	
	ukf_oracle	0.3702	0.2980	0.2901	0.3275	0.4329	0.00	0.819	5.07e+03	
	koopman_kf	0.1022	0.1078	0.0961	0.0815	0.3155	0.00	0.736	2.36e+04	
	greybox_ekf	0.0185	0.0206	0.0217	0.0224	0.0392	0.00	0.928	2.22e+04	
	greybox_ukf	0.0191	0.0216	0.0222	0.0233	0.0401	0.00	0.930	8.06e+03	
	greybox_ekf_oracle	0.0185	0.0206	0.0214	0.0221	0.0390	0.00	0.925	2.14e+04	
bioreactor (0.708)	latent	0.0614	0.0561	0.0702	0.0859	0.1097	0.00	0.889	1.86e+03	
	lora	0.0313	0.0350	0.0987	0.1023	0.2843	0.00	0.890	1.61e+03	
	hyper	0.0328	0.0313	0.0922	0.0991	0.2431	0.00	0.911	1.16e+03	
	nlox	0.0171	0.0211	0.0389	0.0334	0.0484	0.00	0.885	1e+03	
	ekf	5.7330	5.7218	5.7313	8.0282	5.7950	0.00	–	8.68e+03	FAILED
	ukf	5.7103	5.6993	5.7076	7.9949	5.7692	0.00	–	3.34e+03	FAILED
	pf	5.7978	5.7874	5.8001	8.1239	5.8636	0.00	–	3.12e+03	FAILED
	kkl_mb	1.8498	1.9388	2.9636	3.8966	3.7956	0.00	0.863	1.2e+04	FAILED
	ekf_huber	5.7229	5.7118	5.7243	8.0155	5.7882	0.00	–	6.61e+03	FAILED
	ukf_huber	5.7220	5.7110	5.7234	8.0143	5.7874	0.00	–	3.29e+03	FAILED
	ekf_oracle	2.5540	2.5247	2.4441	3.4564	2.6467	0.00	0.994	6.77e+03	FAILED
	ukf_oracle	3.0024	2.9752	2.8975	4.0976	3.1013	0.00	0.995	2.29e+03	FAILED
	koopman_kf	0.0402	0.0399	0.0933	0.1287	0.1274	0.00	0.967	3.22e+03	
	greybox_ekf	0.0640	0.0615	0.0649	0.0889	0.0772	0.00	0.994	1.12e+04	
	greybox_ukf	0.0630	0.0607	0.0642	0.0878	0.0760	0.00	0.995	4.63e+03	
	greybox_ekf_oracle	0.0529	0.0528	0.0521	0.0677	0.0618	0.00	0.986	1.21e+04	
duffing (1.71)	latent	0.0554	0.0569	0.0824	0.1117	0.1580	0.00	0.699	1.67e+03	
	lora	0.0697	0.0760	0.0997	0.1380	0.1196	0.00	0.686	1.31e+03	
	hyper	0.0707	0.0768	0.1072	0.1478	0.1264	0.00	0.699	1.12e+03	
	nlox	0.0901	0.0881	0.1104	0.1521	0.1529	0.00	0.773	873	
	ekf	0.3347	0.1204	0.2441	0.3319	0.4500	0.00	0.752	5.37e+03	
	ukf	0.3343	0.1203	0.2438	0.3318	0.4494	0.00	0.753	1.66e+03	
	pf	0.3988	0.1213	0.2438	0.3322	0.4572	0.00	0.776	1.51e+03	
	kkl_mb	0.1973	0.1858	0.1957	0.2753	0.2377	0.00	0.703	1.11e+04	
	ekf_huber	0.4071	0.1224	0.2763	0.3485	0.5616	0.00	0.826	7.21e+03	
	ukf_huber	0.4048	0.1221	0.2754	0.3476	0.5595	0.00	0.825	2.7e+03	
	ekf_oracle	0.1430	0.0958	0.1266	0.1773	0.1948	0.00	0.673	3.69e+03	
	ukf_oracle	0.1430	0.0958	0.1266	0.1773	0.1948	0.00	0.673	2.01e+03	
	koopman_kf	0.1595	0.1584	0.1649	0.2316	0.1967	0.00	0.678	2.94e+03	
	greybox_ekf	0.0127	0.0127	0.0144	0.0183	0.0194	0.00	0.684	7.78e+03	
	greybox_ukf	0.0128	0.0128	0.0145	0.0184	0.0195	0.00	0.684	2.98e+03	
	greybox_ekf_oracle	0.0128	0.0128	0.0144	0.0175	0.0194	0.00	0.679	7.88e+03	
lorenz (0.614)	latent	0.0191	0.0176	0.0175	0.0190	0.0309	0.00	0.941	2.09e+03	
	lora	0.0110	0.0101	0.0104	0.0109	0.0156	0.00	0.926	1.47e+03	
	hyper	0.0165	0.0159	0.0150	0.0160	0.0223	0.00	0.930	1.42e+03	
	nlox	0.0134	0.0125	0.0129	0.0131	0.0182	0.00	0.947	1.06e+03	
	ekf	3.1657	3.1673	3.1705	2.7913	3.1959	0.00	–	7.01e+03	FAILED
	ukf	3.5217	3.5237	3.5275	3.1037	3.5576	0.00	–	1.84e+03	FAILED
	pf	100.00	100.00	100.00	100.00	100.00	1.00	–	1.92e+03	FAILED
	kkl_mb	0.0273	0.0254	0.0257	0.0281	0.0397	0.00	0.937	1.22e+04	
	ekf_huber	3.2244	3.2236	3.2337	2.8441	3.4158	0.03	0.996	6.37e+03	FAILED
	ukf_huber	3.5837	3.5828	3.5947	3.1600	3.8092	0.03	0.996	2.71e+03	FAILED
	ekf_oracle	0.6663	0.6691	0.6722	0.6107	0.7029	0.00	–	5.11e+03	FAILED
	ukf_oracle	0.6478	0.6505	0.6535	0.5945	0.6839	0.00	–	1.92e+03	FAILED
	koopman_kf	0.0252	0.0236	0.0242	0.0206	0.0366	0.00	0.921	4.24e+03	
	greybox_ekf	0.1112	0.0763	0.1067	0.0512	1.1442	0.00	0.995	1.18e+04	
	greybox_ukf	0.0863	0.0713	0.1017	0.0478	0.8615	0.00	0.995	4.77e+03	
	greybox_ekf_oracle	0.2184	0.2156	0.2160	0.2217	0.2254	0.00	–	1.26e+04	
quad_tank (0.581)	latent	0.3311	0.3315	0.3421	0.4799	0.4528	0.00	0.960	2.12e+03	
	lora	0.0245	0.0257	0.0441	0.0538	0.0628	0.00	0.963	1.56e+03	
	hyper	0.0229	0.0244	0.0436	0.0536	0.0625	0.00	0.965	1.46e+03	

Table 12: Deployment scorecard, realistic deployment (C3-realistic: partial model + 30% parameter error + AR(1) Student-t sensor noise), dev systems, mean over seeds. Columns as in hyperkkl/tree.py Groups: KKL (latent, lora, hyper), DATA-DRIVEN same-data baselines (nlox, koopman_kf, greybox_*), MODEL-BASED filters (the rest). (sequence OOD; div. = max divergence rate over all blocks; AUROC = output-residual alarm). FAILED: sequence-OOD NRMSE above climatology (value next to the system name).

system	method	id	shape	seq.	unmeas.	p99	div.	AUROC	μs/step	status
	nlox	0.0890	0.0968	0.1540	0.1835	0.1938	0.00	0.940	746	
	ekf	4.7397	4.6491	4.6806	5.4427	4.8801	0.00	–	1.14e+04	FAILED
	ukf	4.7362	4.6462	4.6781	5.4403	4.8771	0.00	–	4.42e+03	FAILED
	pf	5.2010	5.1278	5.1540	5.9683	5.4029	0.00	–	4.55e+03	FAILED
	kkl_mb	0.9938	0.8785	1.0089	1.4042	1.3148	0.00	0.991	1.83e+04	FAILED
	ekf_huber	4.8023	4.7098	4.7429	5.5171	4.9454	0.00	–	1.6e+04	FAILED
	ukf_huber	4.7984	4.7066	4.7402	5.5145	4.9423	0.00	–	5.68e+03	FAILED
	ekf_oracle	3.1109	3.0471	3.0997	3.9109	3.2578	0.00	–	1.12e+04	FAILED
	ukf_oracle	3.1093	3.0456	3.0983	3.9090	3.2562	0.00	–	5.09e+03	FAILED
	koopman_kf	0.0428	0.0356	0.1051	0.1303	0.1559	0.00	0.894	2.27e+04	
	greybox_ekf	0.0168	0.0170	0.0170	0.0178	0.0220	0.00	0.962	1.62e+04	
	greybox_ukf	0.0168	0.0170	0.0170	0.0178	0.0220	0.00	0.962	5.17e+03	
	greybox_ekf_oracle	0.0193	0.0196	0.0195	0.0194	0.0250	0.00	0.952	1.83e+04	
vdp (1.16)	latent	0.0387	0.0360	0.0408	0.0513	0.0614	0.00	0.685	1.81e+03	
	lora	0.0294	0.0265	0.0324	0.0390	0.0469	0.00	0.700	1.52e+03	
	hyper	0.0315	0.0287	0.0334	0.0405	0.0506	0.00	0.697	1.13e+03	
	nlox	0.0412	0.0392	0.0472	0.0586	0.0657	0.00	0.662	902	
	ekf	0.4279	0.4479	0.4617	0.6008	0.5003	0.00	0.861	4.74e+03	
	ukf	0.4279	0.4478	0.4617	0.6008	0.5003	0.00	0.861	2.04e+03	
	pf	0.3613	0.3790	0.3924	0.5514	0.4191	0.00	0.997	1.8e+03	
	kkl_mb	0.1077	0.1009	0.1247	0.1732	0.1597	0.00	0.670	1.57e+04	
	ekf_huber	0.7247	0.7666	0.7901	0.8661	0.9804	0.00	0.996	6.54e+03	
	ukf_huber	0.7244	0.7663	0.7899	0.8659	0.9801	0.00	0.996	3.36e+03	
	ekf_oracle	0.1477	0.1483	0.1560	0.2184	0.2109	0.00	0.626	5.35e+03	
	ukf_oracle	0.1477	0.1483	0.1560	0.2184	0.2109	0.00	0.626	2.17e+03	
	koopman_kf	0.2150	0.2115	0.2170	0.3052	0.3364	0.00	0.682	2.7e+03	
	greybox_ekf	0.0310	0.0295	0.0388	0.0492	0.0510	0.00	0.484	6.6e+03	
	greybox_ukf	0.0310	0.0296	0.0389	0.0493	0.0511	0.00	0.484	2.72e+03	
	greybox_ekf_oracle	0.0858	0.0833	0.0875	0.1202	0.1401	0.00	0.691	5.35e+03	



Rank of each method per scorecard column (hyperkkl/tree.py SC_COLS) per dev system (ties share the mean rank), averaged over systems.
[Fk/n]: sequence-OOD NRMSE above climatology (FAILED) on k of n systems.

Figure 9: Deployment scorecard as mean rank per column over the dev plants (1 = best). (a) exact model under the stress conditions of `C3-faults`; (b) realistic deployment (`C3-realistic`). [Fk/6]: the method's `sequence_ood` NRMSE is above climatology on k of six plants.

F.6 Every decision rule

Table 13: Every pre-registered decision rule (node.yaml, written before the runs), applied mechanically (analysis/evidence.py). PASS/FAIL: the rule's condition holds / fails and cannot change with the pending jobs; *: provisional (decided on the finished jobs, but further seeds of counted systems are still running and can move the CIs); INCONCL.: pending systems decide; PENDING: no data; REPORT.: descriptive rule (reported, no pass/fail). Vote counts k/n are not corrected for multiplicity (6 systems, several tiers and nodes).

node	question	rule (verbatim, abridged)	outcome	evidence
dev systems (design decisions)				
R0	Does the v2 input protocol (one amplitude box, per-plant t_ref, id / shape_ood / sequence_ood tiers) train and evaluate cleanly, with no mass...	Pass if no job fails, rejection < 5% per tier, and every conditioned variant beats autonomous on id/clean on every dev system.	**PASS**	no failed jobs: True; max rejection 0.0% (< 5%: True); every conditioned variant beats autonomous on id: 6/6 systems
C1-where	With identical frozen base maps, context encoder (SFM) and a matched 140k-parameter conditioner, which location generalises best to unseen input...	A location is preferred if it beats every other location with a paired 95% CI excluding 0 on the all-families column on ≥ 4/6 dev systems, on shape_ood and...	**FAIL**	preferred location: NONE. Weaker claim (not the pre-registered rule), weight-space beats latent and concat: shape_ood 6/6; sequence_ood 5/6
C1-ctx	Does the C1 ranking survive when the context encoder's parameters count toward the budget?	Ranking preserved if the C1-where winner stays within CI of the best here on ≥ 4/6 systems.	**PASS**	C1-where winner best-or-within-CI: shape_ood 6/6; sequence_ood 6/6 (need 4/6); TOST(±10%) version: shape_ood 5/6; sequence_ood 5/6
C1-budget-s	Is the C1 ranking an artefact of one parameter budget?	Stable if the C1-where winner is best-or-within-CI at both budgets on ≥ 4/6 systems.	**PASS**	C1-where winner best-or-within-CI: shape_ood 4/6; sequence_ood 4/6 (need 4/6); TOST(±10%) version: shape_ood 4/6; sequence_ood 3/6
C1-budget-l	Is the C1 ranking an artefact of one parameter budget?	Stable if the C1-where winner is best-or-within-CI at both budgets on ≥ 4/6 systems.	**PASS**	C1-where winner best-or-within-CI: shape_ood 6/6; sequence_ood 6/6 (need 4/6); TOST(±10%) version: shape_ood 5/6; sequence_ood 5/6
C1-budget	Is the C1 ranking an artefact of one parameter budget?	Stable if the C1-where winner is best-or-within-CI at both budgets on ≥ 4/6 systems.	**PASS**	stable at BOTH budgets: shape_ood 4/6; sequence_ood 4/6
C1-rank1	Is the LoRA ~ hypernetwork tie an artefact of the low-rank choice?	Stable if lora at this rank stays within an equivalence margin of ±10% of lora rank 4 (C1-where) or better, on ≥ 4/6 dev systems (sequence_ood).	**FAIL**	lora at this rank within ±10% of rank 4 or better (TOST non-inferiority, 90% CI): 1/6 (need 4/6)
C1-rank16	Is the LoRA ~ hypernetwork tie an artefact of the low-rank choice?	Stable if lora at this rank stays within an equivalence margin of ±10% of lora rank 4 (C1-where) or better, on ≥ 4/6 dev systems (sequence_ood).	**FAIL**	lora at this rank within ±10% of rank 4 or better (TOST non-inferiority, 90% CI): 3/6 (need 4/6)
C1-basewidth-64	Does a stronger or weaker base decoder change where the input should enter?	Report the C1 ranking per width; 'base-dependent' if the winner changes on ≥ 3/6.	**PASS**	winner changes on shape_ood 1/6; sequence_ood 0/6 ('base-dependent' if ≥ 3/6; PASS = not base-dependent)
C1-basewidth-256	Does a stronger or weaker base decoder change where the input should enter?	Report the C1 ranking per width; 'base-dependent' if the winner changes on ≥ 3/6.	**PASS**	winner changes on shape_ood 2/6; sequence_ood 2/6 ('base-dependent' if ≥ 3/6; PASS = not base-dependent)
C1-modelfree	How much of the accuracy needs the model in phase 1?	Model needed if C1-where beats this node with CI excluding 0 on ≥ 4/6.	**FAIL**	C1-where beats model-free phase 1 (same method, CI): shape 0/6; seq 0/6 (need 4/6)
C2-floor-025	Corollary 1: input history older than ~ln(1/eta)/alpha barely affects the map. If the context is forced to forget at rate ≥ c alpha (Lemma 2 then...	Memory beyond the theory window is unnecessary if C2-floor-05 is within CI of C1-where (no floor) for lora and hyper on ≥ 5/6 dev systems, on id and sequence_ood....	**FAIL**	no loss vs no floor (lora & hyper, id & seq): 6/6 (need 5/6); TOST(10%) non-inferior: 3/6; probe decays at ≥ floor rate: 1/6 systems
C2-floor-05	Corollary 1: input history older than ~ln(1/eta)/alpha barely affects the map. If the context is forced to forget at rate ≥ c alpha (Lemma 2 then...	Memory beyond the theory window is unnecessary if C2-floor-05 is within CI of C1-where (no floor) for lora and hyper on ≥ 5/6 dev systems, on id and sequence_ood....	**FAIL**	no loss vs no floor (lora & hyper, id & seq): 4/6 (need 5/6); TOST(10%) non-inferior: 0/6; probe decays at ≥ floor rate: 0/6 systems

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node	question	rule (verbatim, abridged)	outcome	evidence
C2-floor-1	Corollary 1: input history older than ~ln(1/eta)/alpha barely affects the map. If the context is forced to forget at rate ≥ c alpha (Lemma 2 then...	Memory beyond the theory window is unnecessary if C2-floor-05 is within CI of C1-where (no floor) for lora and hyper on ≥ 5/6 dev systems, on id and sequence_ood....	**FAIL**	no loss vs no floor (lora & hyper, id & seq): 4/6 (need 5/6); TOST(10%) non-inferior: 0/6; probe decays at ≥ floor rate: 0/6 systems
C2-window-a1	With a raw input window of L samples instead of the full-history SFM context, how long must the window be?	Supported if e(L ≥ L*) is within 5% of e(512) on ≥ 5/6 dev systems, on id and sequence_ood.	**FAIL**	e(L ≥ L*) within 5% of e(512): id 2/5; seq 3/5 (need 5/6). Refined L* = (ln(1/eta)+ln l)/alpha (post hoc, not pre-registered): id 2/3; seq 2/3
C2-window-a2	Does the window knee move as 1/alpha (Theorem: L* ~ (ln(1/eta) + ln l)/alpha)?	Compare knees with C2-window-a1 on vdp, duffing, battery; supported if the knee in samples changes by ≤ 2x on ≥ 2/3 systems.	**FAIL**	knee in samples changes by ≤ 2x: 1/2 (need 2/3)
C3-faults	Against EKF, UKF, PF, model-based KKL and NLOX (nominal model, oracle noise for clean), and against robust (Huber) and condition-oracle filters,...	Per condition, count dev systems where the best learned observer beats the best filter (nominal, and separately the best of robust/oracle) with a paired CI excluding...	REPORT.	learned-wins regime (≥ 4/6 vs the best filter of either kind, shape and seq): none of 15 conditions
C3-noise-ct	The sensor's noise is colored and heavy-tailed in training and deployment. Filters assume white Gaussian noise (nominal R; oracle filters use its...	Best learned vs best filter of each kind, paired CI on id and sequence_ood, ≥ 4/6.	**FAIL**	id: vs best filter 0/6, vs best working 0/6; seq: vs best filter 0/6, vs best working 0/6; failed filters per system 0-1
C3-noise-ct (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	PENDING	pending: new data-driven baselines merged in 0/18 runs
C3-mismatch10	Filters use the nominal model; learned observers are trained on data from the real plant. When does learning from data beat a wrong model?	Learned wins at a level if the best learned observer beats the best filter with CI excluding 0 on ≥ 4/6 systems, on id and sequence_ood.	**PASS**	id: vs best filter 5/6, vs best working 5/6; seq: vs best filter 4/6, vs best working 4/6; failed filters per system 0-7
C3-mismatch10 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL***	best KKL vs best data-driven: id 0/4; seq 1/4 (merged runs 15/18)
C3-mismatch20	Filters use the nominal model; learned observers are trained on data from the real plant. When does learning from data beat a wrong model?	Learned wins at a level if the best learned observer beats the best filter with CI excluding 0 on ≥ 4/6 systems, on id and sequence_ood.	**PASS**	id: vs best filter 5/6, vs best working 5/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-8
C3-mismatch20 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	INCONCL.	best KKL vs best data-driven: id 0/1; seq 0/1 (merged runs 6/18)
C3-mismatch30	Filters use the nominal model; learned observers are trained on data from the real plant. When does learning from data beat a wrong model?	Learned wins at a level if the best learned observer beats the best filter with CI excluding 0 on ≥ 4/6 systems, on id and sequence_ood.	**PASS**	id: vs best filter 5/6, vs best working 5/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-8
C3-mismatch30 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL**	best KKL vs best data-driven: id 0/6; seq 0/6 (merged runs 18/18)

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node	question	rule (verbatim, abridged)	outcome	evidence
C3-mismatch50	Filters use the nominal model; learned observers are trained on data from the real plant. When does learning from data beat a wrong model?	Learned wins at a level if the best learned observer beats the best filter with CI excluding 0 on $\geq$ 4/6 systems, on id and sequence_ood.	**PASS**	id: vs best filter 6/6, vs best working 6/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-8
C3-mismatch50 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	PENDING	pending: merged in 1/18 runs, but no system has the new baselines in all its seeds yet
C3-mismatch (crossover)	Where does the filter-vs-learned crossover sit?	Hypothesis of C3-mismatch*: crossover between 10% and 30% mismatch (learned wins at a level per that node's rule).	**FAIL**	level outcomes: 10% PASS, 20% PASS, 30% PASS, 50% PASS; first winning level 10%
C3-partial	When the modeller knows only part of the dynamics (a missing or simplified term, not a wrong parameter), do learned observers trained on...	Per system, best learned vs best filter (nominal and robust/oracle), paired CI on id and sequence_ood; report against the relative model error median |f - f_known| / |f|.	REPORT.	id: vs best filter 5/6, vs best working 5/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-8; hypothesis pattern (learned wins on bioreactor/vdp/lorenz, filters on duffing/quad_tank) holds on 4/5...
C3-partial (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL**	best KKL vs best data-driven: id 1/6; seq 2/6 (merged runs 18/18)
C3-partial-modelfree	Is a partial model in the phase-1 PDE term still worth using, or does it hurt?	Model helps if C3-partial beats this node with CI excluding 0 on $\geq$ 4/6 systems.	**FAIL**	C3-partial (partial model in the PDE term) beats model-free phase 1: id 0/6; seq 0/6 (need 4/6; decided on the compare tier sequence_ood)
C3-realistic	All three kinds of incomplete knowledge at once, as on a real plant.	Best learned vs best filter (any variant), paired CI, id and sequence_ood, $\geq$ 4/6.	**PASS**	id: vs best filter 6/6, vs best working 6/6; seq: vs best filter 6/6, vs best working 6/6; failed filters per system 0-8
C3-realistic (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL**	best KKL vs best data-driven: id 1/6; seq 2/6 (merged runs 30/30)
C3-regimes	Under partial knowledge, do learned KKL observers beat the grey-box EKF/UKF (partial model + learned residual, same data) under a poor initial...	Per condition, best KKL (lora/hyper) vs best grey-box filter (greybox_ekf/ukf/ekf_oracle): paired 95% CI excluding 0 on $\geq$ 4/6 dev systems on sequence_ood = 'KKL-wins...	PENDING	pending (0/18 jobs done, 18 running)
C3-data50	How much data does a learned observer need before it pays off?	Report error vs n_train (50/150/600) per method; crossover with the filters under 30% mismatch read from C3-mismatch30.	REPORT.	learning curve reported (Figure F3 companion); no pass/fail rule
C3-data150	How much data does a learned observer need before it pays off?	Report error vs n_train (50/150/600) per method; crossover with the filters under 30% mismatch read from C3-mismatch30.	REPORT.	learning curve reported (Figure F3 companion); no pass/fail rule
TEST systems (frozen recipe, run once)				
TEST-C1	Does the dev-system C1 conclusion (weight-space conditioning beats latent injection and the additive 'concat' variant) hold on six plants never...	Supported if the better of lora/hyper beats both latent and concat with a paired CI excluding 0 on $\geq$ 4/6 test systems on shape_ood and on sequence_ood. Run once; no...	**PASS**	weight-space beats latent and concat: shape_ood 5/6; sequence_ood 5/6 (need 4/6 on each)

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node	question	rule (verbatim, abridged)	outcome	evidence
TEST-C3	Partial model + 30% parameter error + colored Student-t noise on six held-out plants: does the best learned observer beat every filter variant?	Best learned vs best WORKING filter (NRMSE below climatology), paired CI, id and sequence_ood, $\geq 4/6$; report against the relative model error of each plant. Run once.	**PASS***	[PARTIAL DATA: 17/18 jobs done, 1 running] id: vs best filter 4/6, vs best working 4/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-2
TEST-C3 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL***	best KKL vs best data-driven: id 1/4; seq 1/4 (merged runs 14/18)

G Real-data validation: details

All numbers here are macros generated by `analysis/real_numbers.py` from `experiments/real/results.json`; the pre-registration is `experiments/real/DESIGN.md` (written before any training or evaluation) and the data provenance `DATA.md`.

Data. Panasonic NCR18650PF cell, Kollmeyer, Mendeley Data v1, DOI 10.17632/wykht8y7tg.1, CC BY 4.0 (Kollmeyer, 2018); every downloaded file's SHA-256 matches the repository's. Resampled to $dt = 1\,\mathrm{s}$; input u = mean current over the second (discharge positive); outputs $y = (\Delta T, V_m)$: case-temperature rise and the terminal voltage behind the plant's 2-s acquisition filter, so that every observer uses the unchanged $y = x[\text{sensors}]$ contract of the simulated battery plant. States $(\mathrm{SOC}, V_1, V_2, \Delta T, V_m)$; labelled on real data: SOC (coulomb count), ΔT, V_m; V_1, V_2 have no labels and are excluded from the phase-2 loss. Split by drive cycle at 25°C: train Cycle 1–3, UDDS, LA92; validation Cycle 4; test US06, HWFET (HWFTa and HWFTb pooled), NN; disjoint files and absolute time intervals. Disclosed overlap of shapes: the Cycle 1–4 mixes contain US06/HWFET/NN segments and NN combines US06 and LA92 portions, so the test cycles are new *sequences* on real hardware, not wholly new waveform shapes.

Model identification. 2-RC equivalent circuit with constant parameters identified by least squares on the training cycles: $R_0 = 32.9\,\mathrm{m\Omega}$, $\tau_1 = 25\,\mathrm{s}$, $\tau_2 = 1166\,\mathrm{s}$, $Q = 2.995\,\mathrm{Ah}$ (C/20 capacity), OCV polynomial, lumped thermal model. Open-loop terminal-voltage RMSE 24–37 mV on the training cycles, 23–72 mV on validation/test (US06 70 mV, errors up to 0.5 V near empty), 58–221 mV at 0°C with Arrhenius-scaled resistances (factor 2.5): a realistic but incomplete model, as intended.

Observers. Learned KKL: the paper's default recipe ($n_z = 7$, $\alpha = 1.3/\mathrm{min}$ so that $dt = 1\,\mathrm{s}$ passes the design gate, SFM context, budget 140k (139.6k–141.5k parameters per conditioner), 3000 phase-2 steps, 3 seeds). Phase 1 on the identified circuit (zero-input episodes, collocation states, and the training currents replayed on the model); phase 2 is the unchanged `train.phase2` on real training windows with the loss restricted to the labelled states. WS-best: per seed, whichever of HyperKKL ($r = 4$) and its dense-subspace ablation has the lower validation SOC RMSE (seed 0: lora, seed 1: lora, seed 2: hyper); the single-method reading uses the $r = 4$ rows (HyperKKL vs EKF: 3 of 3 cycles decided for HyperKKL; vs latent: 1 of 3, differences US06 -0.28 [-0.42, -0.17], HWFET -0.18 [-0.19, +0.02], NN -0.45 [-0.66, -0.21]; per-seed means -1.13, -0.64, +0.97; -0.21, -0.36, +0.04; -0.63, -0.30, -0.61). Filters: the unchanged EKF/UKF on the identified circuit, Q, R from a 72-point grid scored on the validation cycle with the same wrong-prior protocol (oracle-tuned, as `ekf_oracle`); the EKF's chosen $r_{V_m} = 0.10\,\mathrm{V}$ sits at the top of the pre-registered grid (score flat there), the UKF's (0.03 V) is interior. Reference: coulomb counting from the wrong prior (error 20% by construction). Episodes: 30-min windows every 5 min through each discharge (US06 11 windows, HWFET 40, NN 34; validation 35), each run with prior $\mathrm{SOC}_0 \pm 0.2$ where that stays in $[0, 1]$.

Table 14: Real-data validation: settled SOC RMSE [% SOC] ($t \geq 15$ min of 30-min windows, prior SOC 20% off) on the three held-out 25°C drive cycles and on the validation cycle (Cycle 4, used for model selection and filter tuning only). Learned: mean ± sd over 3 seeds. WS-best = HyperKKL ($r = 4$) or its dense-subspace ablation, whichever has the lower validation RMSE per seed (the pre-registered selection). Same-data baselines (Koopman KF, grey-box EKF/UKF): pending.

method	US06	HWFET	NN	validatio
WS-best (HyperKKL $r = 4$ or dense ablation, chosen per seed on validation)	**1.76 ± 0.15**	**0.86 ± 0.05**	**0.69 ± 0.04**	0.7
HyperKKL ($r = 4$)	2.18 ± 0.59	0.91 ± 0.11	0.73 ± 0.04	0.7
dense-subspace ablation	1.75 ± 0.25	0.93 ± 0.08	0.69 ± 0.03	0.7
latent injection	2.44 ± 0.58	1.09 ± 0.10	1.25 ± 0.22	1.3
concat (additive)	7.37 ± 1.22	2.80 ± 0.14	4.51 ± 0.11	4.1
autonomous (no input)	31.52 ± 2.65	15.54 ± 0.60	31.08 ± 1.55	23.0
EKF (identified 2-RC ECM, Q, R tuned on validation)	8.85	1.19	1.95	3.2
UKF (same model, tuned)	9.13	1.39	1.55	3.1
coulomb counting from the wrong prior	20.00	20.00	20.00	20.0
hyperkkl	2.18 ± 0.59	0.91 ± 0.11	0.73 ± 0.04	0.7

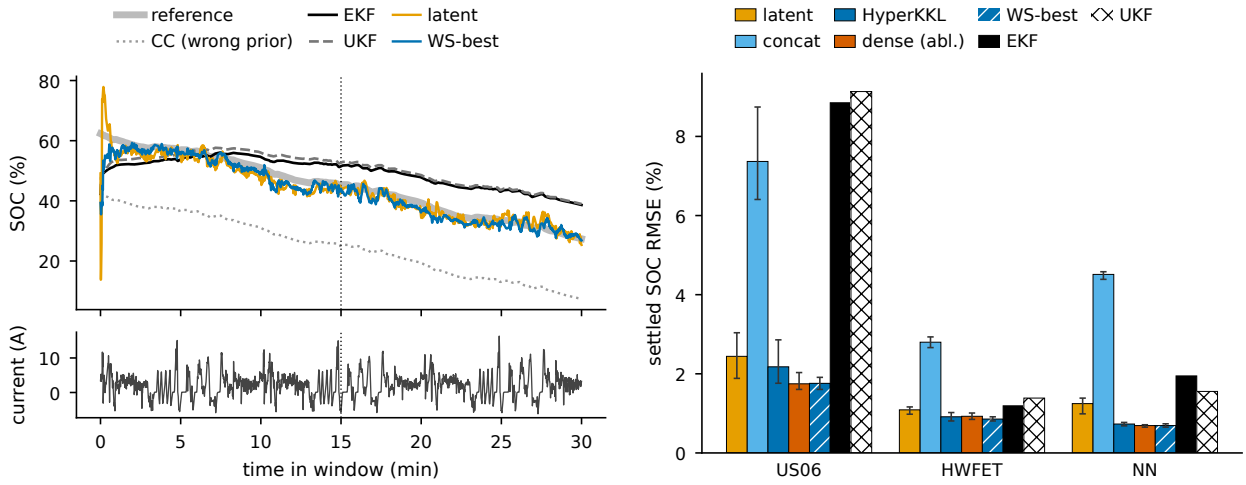


Figure 10: Real data, 25°C. Left: one US06 window from a prior 20% off (the EKF settles on a biased SOC; the learned observer corrects it from data with a visible ripple). Dotted line: start of the settled window ($t \geq 15$ min); the current is positive on discharge. Right: settled SOC RMSE per method and test cycle, prior SOC ±20%; learned observers: mean over 3 seeds, whiskers min–max (`experiments/real/figures/`).

Statistics and rules. Per-window paired differences averaged over prior signs and seeds; 95% moving-block bootstrap over window start times (block 30 min, 10 000 resamples) and per-seed means; a cycle counts only if the CI excludes 0 *and* all seeds agree. H1: WS-best beats the tuned EKF on ≥ 2 of 3 test cycles; H2: WS-best beats latent injection, same rule. Everything else is descriptive.

Results. H1: learned better on 3, EKF better on 0 of 3 cycles. Paired settled-RMSE differences (WS-best − EKF, % SOC; CI; per-seed means): US06 -6.50 [-8.31, -4.04] (-6.94, -7.09, -7.25); HWFET -0.41 [-0.66, -0.29] (-0.28, -0.38, -0.34); NN -1.27 [-1.76, -0.83] (-1.21, -1.26, -1.29). Against the UKF: US06 -6.81, HWFET -0.48, NN -0.91 (all CIs below 0, all seeds negative). H2: weight-space better on 3, latent better on 0: US06 -0.63 [-0.72, -0.21] (-1.13, -0.64, -0.28); HWFET -0.24 [-0.26, -0.03] (-0.21, -0.36, -0.13); NN -0.49 [-0.71, -0.23] (-0.63, -0.30, -0.73). Maximum settled error: US06 WS-best 6.1 % vs EKF 10.3 %; HWFET 1.9 vs 1.9; NN 2.4 vs 3.1. Median time to stay within 5% SOC (30-min windows): WS-best US06 16.2 min, HWFET 0.4, NN 0.8; EKF 30.0 (never within 5% on US06 windows), 6.8, 6.7. The learned estimate has a sample-level ripple of 1–2% SOC (Figure 10); no output smoothing was applied to any method.

Whole discharges (secondary). From full charge with prior SOC 0.8, RMSE over $t \geq 15$ min: US06 WS-best 1.94 vs EKF 6.98 / UKF 7.01; HWFTa 0.74 vs 0.80 / 1.03; HWFTb 0.98 vs 0.75 / 0.93; NN 0.69 vs 0.66 / 0.70. With hours to converge the filters match the learned observers on the mild cycles; the advantage on those cycles is convergence speed from a wrong prior, and on the aggressive cycle it is accuracy.

Temperature shift (secondary, 0°C). Models trained and identified at 25°C only, evaluated on every 0°C drive cycle. The claim does *not* transfer: the EKF (Arrhenius-scaled resistances) is better on 8 of 9 cycles (learned 9.9–16.0 %, EKF 3.6–22.2 %); the learned observers win only on US06. The decoder has never seen the 0°C voltage–SOC–current relationship, whereas the filter carries an explicit temperature model. Weight-space vs latent is decided for neither side on 9 of 9 cycles. Training on several temperatures (the dataset covers −20 to 25°C) is the obvious next step and was not part of the pre-registration.

Limitations (as recorded in the experiment report). (1) The reference SOC is coulomb counting, the standard in this literature but itself an estimate (tester current error; capacity measured once at C/20 after the drive cycles, i.e. after some ageing; SOC = 1 at the end of each CC–CV charge by definition); using the nominal 2.9 Ah instead would shift the reference by up to 2.8% SOC at the end of a discharge; the learned methods are trained on this reference and can absorb its convention. (2) One cell, one test campaign: windows within a cycle are dependent (handled by the block bootstrap), cell-to-cell variation is not sampled, and US06 has only 11 windows. (3) The advantage holds at the training temperature only. (4) The filters use a constant-parameter 2-RC circuit without hysteresis or SOC-dependent resistance, i.e. what a quick least-squares identification gives; a richer model (lookup tables from the HPPC tests, a hysteresis state) would narrow the gap and was not built. This is the incomplete-model regime of C3 by design. (5) The short-window protocol favours fast convergence. (6) The learned estimate has sample-level ripple. Deviations from the pre-registration (implementation only): filters run batched on GPU over the whole Q, R grid (tested equal to individual filters); phase-1 simulations use clean outputs and no process noise; not run: the strict shape split, the residual alarm, NLOX, and the same-data baselines (pending).

H History of the study: what was tried, superseded or abandoned

This appendix is the part of the record that a reader needs to judge how much freedom the authors had. It is condensed from the generated history of the experiment tree (`experiments/HISTORY.md`, built from the node files, the journal and the mechanical verdicts).

Protocol v1 (archived). A first tree used an *amplitude* OOD tier (inputs scaled beyond the training range) next to a waveform tier, five dev plants and nominal filters only. Its single-winner C1 rule also failed mechanically (preferred location: none, at the $\geq 4/5$ bar), its exact-model C3 rule failed (filters won on every plant), and its mismatch nodes gave mixed verdicts at 10–20% and passes at 30–50%. The protocol was archived because the amplitude tier drove inputs beyond the physical input bounds of bounded plants (the bioreactor's dilution rate, the traffic ramp), so that "OOD" amplitudes were compressed by the actuator and the tier did not test what it claimed. Protocol v2 replaced it by the shape and sequence tiers inside one amplitude box, per-plant t_{ref}, partial-knowledge models, coloured Student-t noise, robust and oracle filters, the deployment scorecard and a sixth dev plant (`battery`). Every v2 node was re-registered with its rule before its runs; no v1 number appears in this paper.

Dead ends. A 5% mismatch node failed most of its jobs twice in v1 and was dropped (the v2 sweep starts at 10%, which is why the crossover below 10% is not located). A control that trained the latent variant with a one-step rollout failed jobs twice and was not carried to v2; the latent variant's trainability confound (Section 4) is therefore unresolved. Two v1 bugs in the plant-perturbation rule (the bioreactor's operating point and the quad-tank's gravity and valve constants were scaled with the "physical parameters", rejecting every episode) were fixed before v2 and the affected runs moved out of the tree.

Verbal readings corrected by the mechanical verdicts. Before the results layer existed, interim digests read the forgetting-floor node as "passes on 5/6" and the partial-model-helps comparison as "8× worse without the model on the bioreactor"; the mechanical application of the rules gives FAIL (no-loss 4/6,

TOST 0/6, probe 0/6) and FAIL (the bioreactor difference is not significant at three seeds). The digest's attribution of the battery's memory need to the theory's lower bound was withdrawn in the theory revision; the sufficient-side explanation through $\ln \ell$ replaced it (Corollary 13). The C1 rank controls were expected to pass and **FAIL**.

Data problems recorded by the results layer. `TEST-C1`'s eighteen runs come from four snapshots of the code (per plant: `network` from one, `cartpole`, `fhn` (two seeds) and `highway_traffic` from a second, `fhn` (third seed) and `two_link_arm` from a third, `rossler` from a fourth); `C1-budget-s`'s runs come from two snapshots; the node files and recipes were frozen throughout, and the verdicts are reported as the layer prints them. The same-data comparison (b) was added after the simulated reviews and is labelled not pre-registered wherever it appears. The decisions table overflows its width by up to 4 pt.

Pending at the time of writing. `C1-modelfree` (FAIL), the learning-curve nodes `C3-data50` and `C3-data150` (REPORTED, REPORTED), the base-width controls (PASS, PASS), the third seeds of the budget controls (18/18, 18/18 jobs finished), and `TEST-C3` (PASS*, 17/18). The Koopman Kalman filter and grey-box EKF baselines and the real-data battery experiment are implemented and pre-registered and are not part of this version.

Simulated reviews. Five simulated reviews written from the perspective of likely action editors (control theory, Koopman methods, dynamical-systems reconstruction, deep learning for dynamics, ML theory) were commissioned on the theory and the interim results before this draft; their must-fix items (a circular contraction hypothesis, a hidden reachability hypothesis in the window lower bound, the chaotic-plant caveats, the missing model-incompleteness proposition, the single-winner C1 rule, the capacity description of the hypernetwork, the climatology reference, the label requirement, the same-data baselines) are addressed in the theory revision and in the framing of this paper, or listed above as open.